Results: figures and tables

Main text

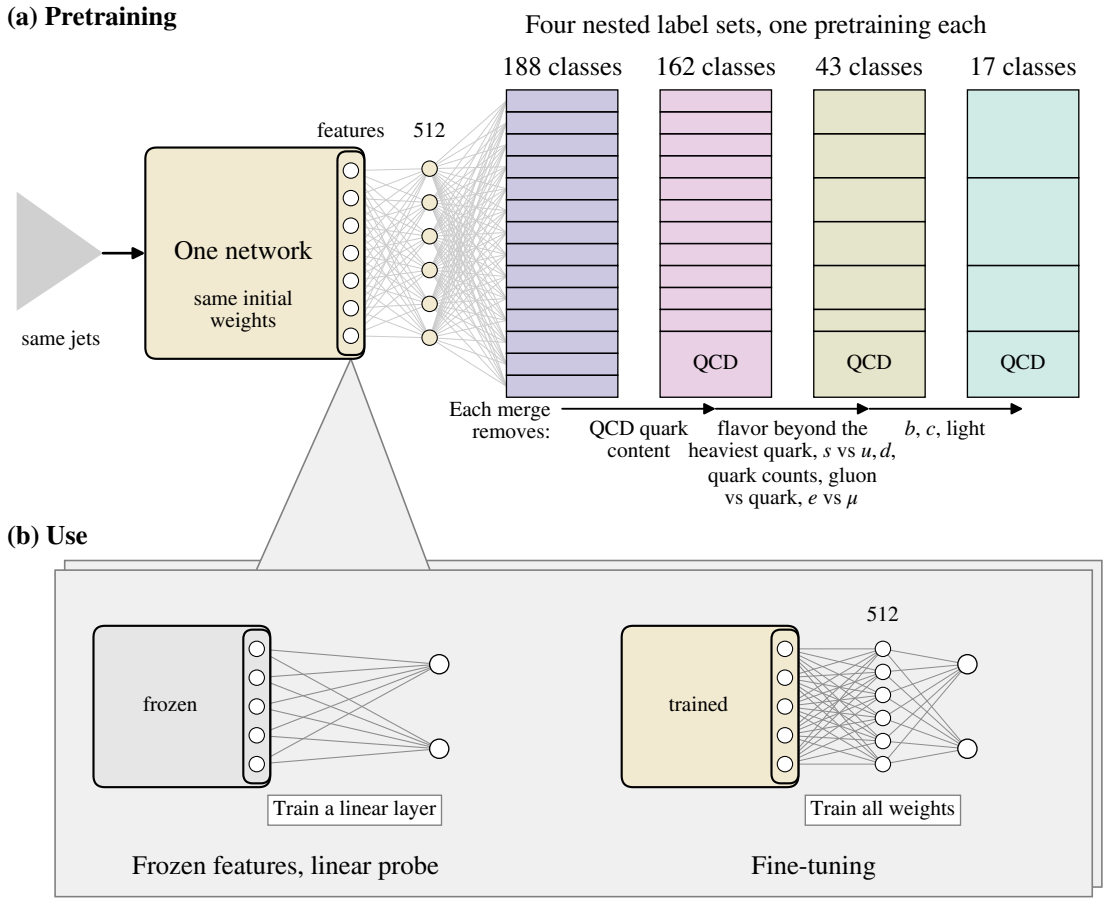


Figure 1: Design of the study. (a) One network is pretrained on the same jets four times, once for each of four nested label sets with 188, 162, 43 and 17 classes, through a hidden layer of 512 units; below each merge is the distinction it removes. (b) The pretrained network is then used in two ways: its frozen features feed a linear probe, or a new head with its own 512-unit hidden layer is added and all weights are fine-tuned.

Table 1: Discriminants analyses build by summing the output scores of a pretrained tagger, against the number of classes in the label set (columns, finest first): Sophon's and the boosted $|V_{cb}|$ measurement's, and a discriminant that weights each QCD subclass separately, as GN3X evaluates its rejection per QCD subclass. •: the discriminant can be built exactly from that label set's outputs, because a merged class's score is the sum of the scores of the classes it merges; —: it cannot. This is a property of the label set, not of a trained model.

discriminant	source	188	162	43	17
$X \to b\bar{b}$ vs QCD	[**?**], Eq. (4)	•	•	—	—
$X \to c\bar{s}$ vs QCD (Sophon's A_1)	[**?**], Eqs. (5)–(6)	•	•	—	—
three quarks, exactly one b, vs QCD (Sophon's A_3)	[**?**], Eqs. (5)–(6)	•	•	—	—
$W' \to W\phi \to WWW$ event discriminant	[**?**], Eqs. (7)–(9)	•	•	—	—
D_{bc}, boosted $\|V_{cb}\|$ measurement	[**?**], Eq. (1)	•	•	—	—
QCD subclasses weighted separately (GN3X-style)	[**?**], Sec. 6	•	—	—	—

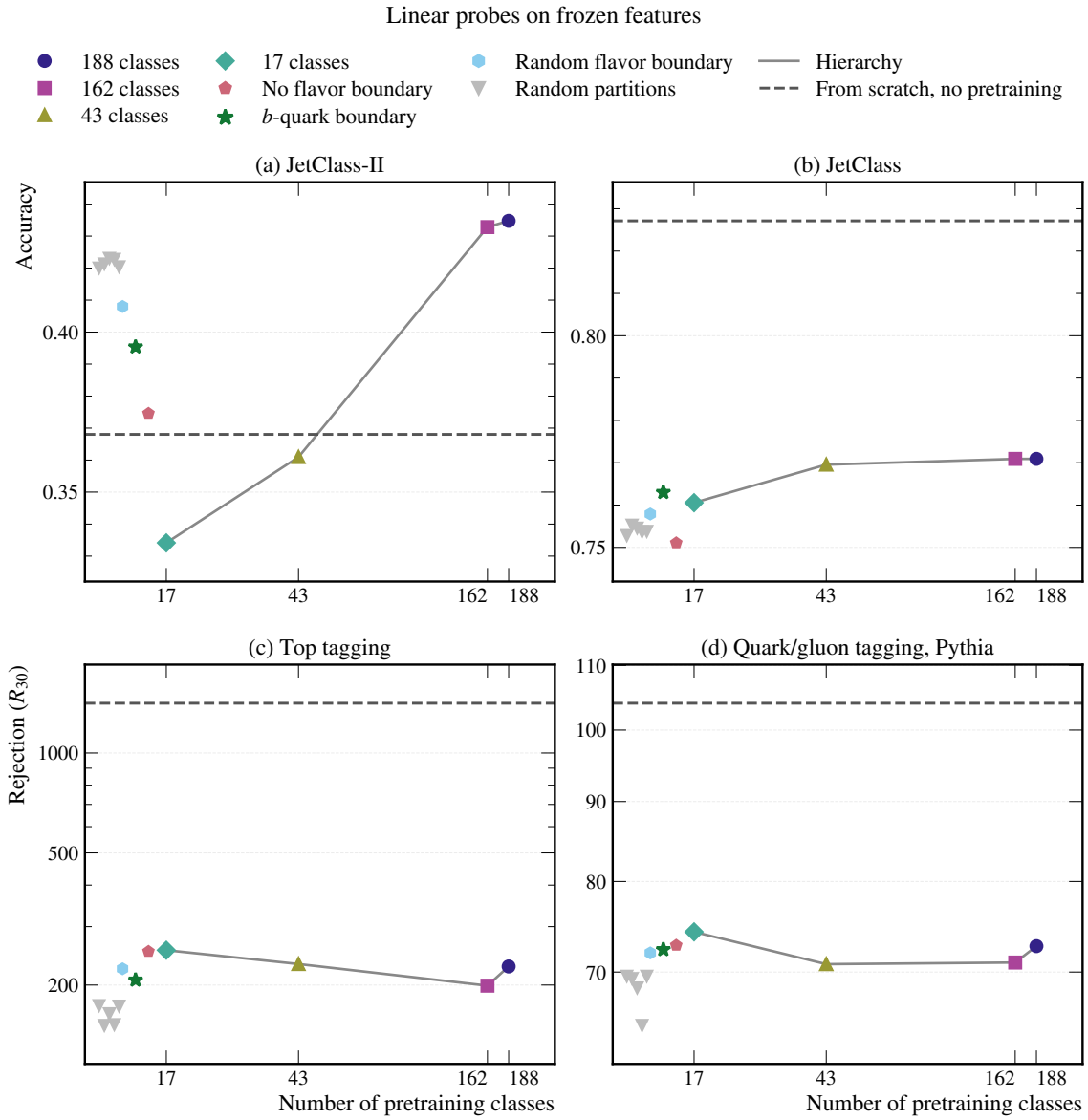


Figure 2: Linear probes on the frozen features against the number of pretraining classes: accuracy on class-balanced JetClass-II and on JetClass, and background rejection R_{30} at 30% signal efficiency on top and quark/gluon (Pythia) tagging, log axis. Markers are means over runs and the grey line (hierarchy) joins the four label sets. The 17-class random partitions, three with controlled flavor boundaries and five plain, are drawn just left of 17. The dashed line is the network trained from scratch.

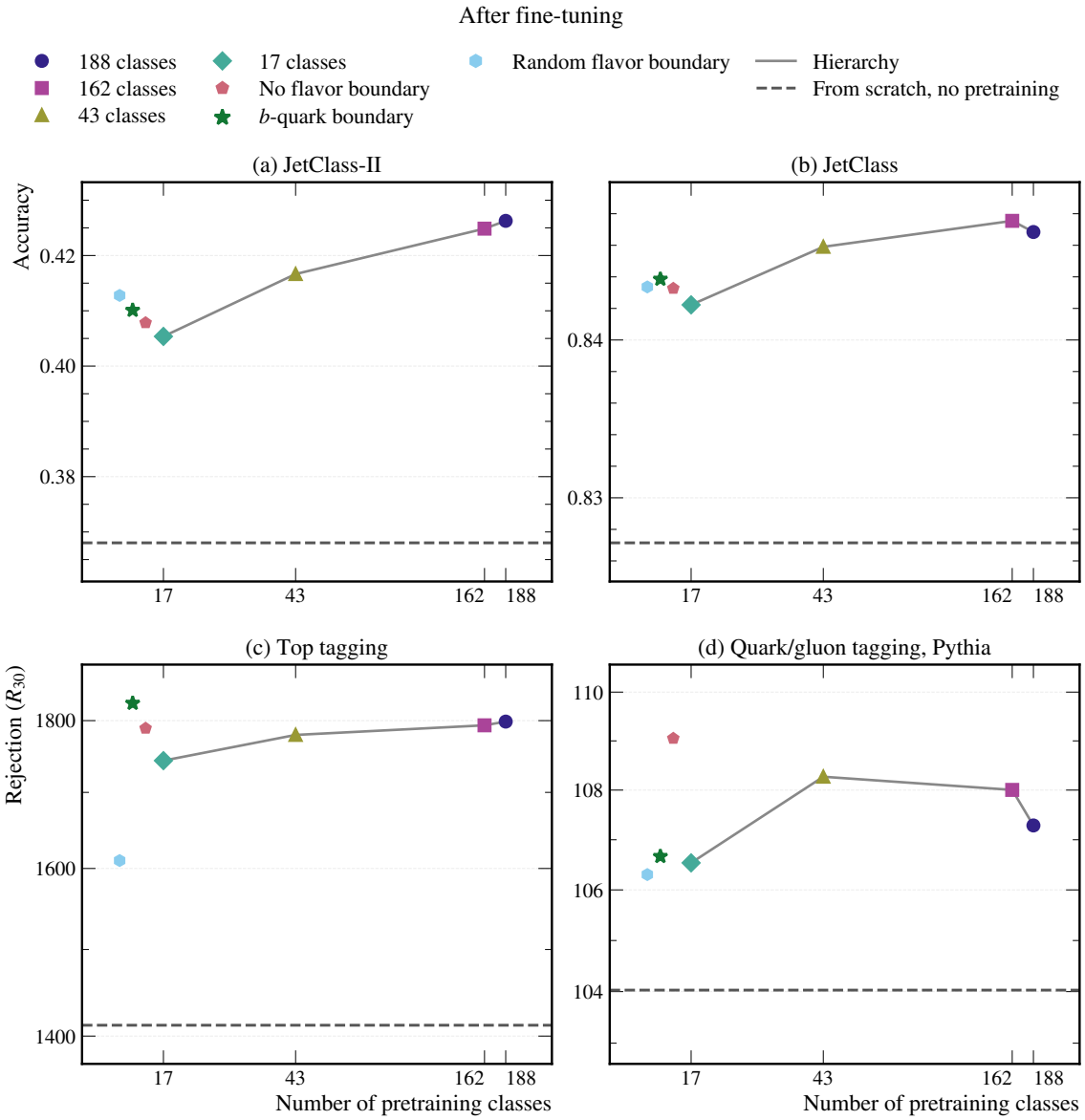


Figure 3: Fine-tuned models against the number of pretraining classes: accuracy on class-balanced JetClass-II and on JetClass, and background rejection R_{30} at 30% signal efficiency on top and quark/gluon tagging. Log axes on (c, d). Markers are means over runs and the grey line (hierarchy) joins the four label sets. The three 17-class partitions with controlled flavor boundaries are drawn just left of 17. The dashed line is the network trained from scratch.

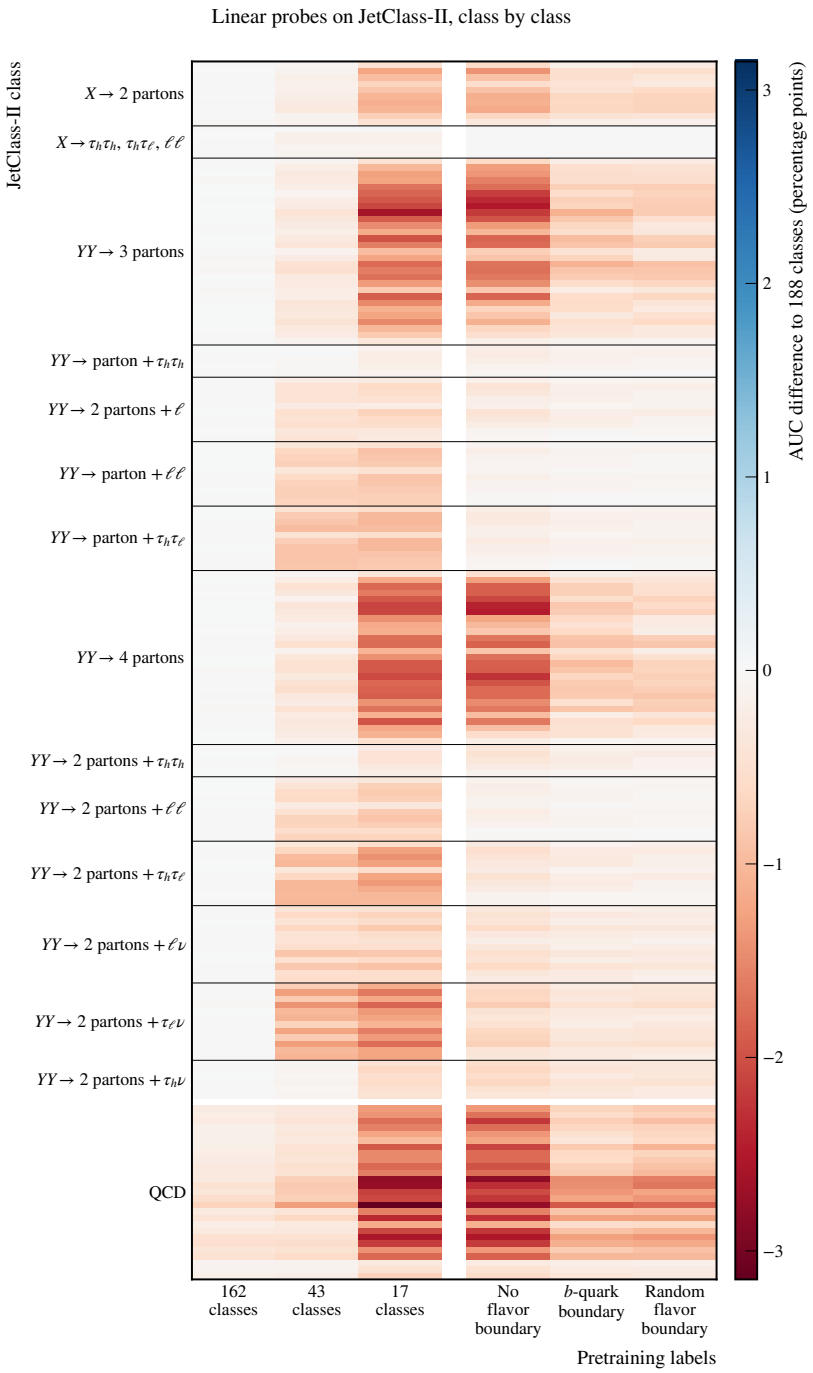


Figure 4: Difference in per-class AUC between each model and the 188-class model, in percentage points, for the 188 classes of class-balanced JetClass-II, from linear probes. Columns are the 162-, 43- and 17-class label sets and the three random partitions with controlled flavor boundaries. Rows are grouped by decay topology, with the QCD classes at the bottom. Values are means over runs; red marks an AUC below that of the 188-class model. Every class's values are in the data release.

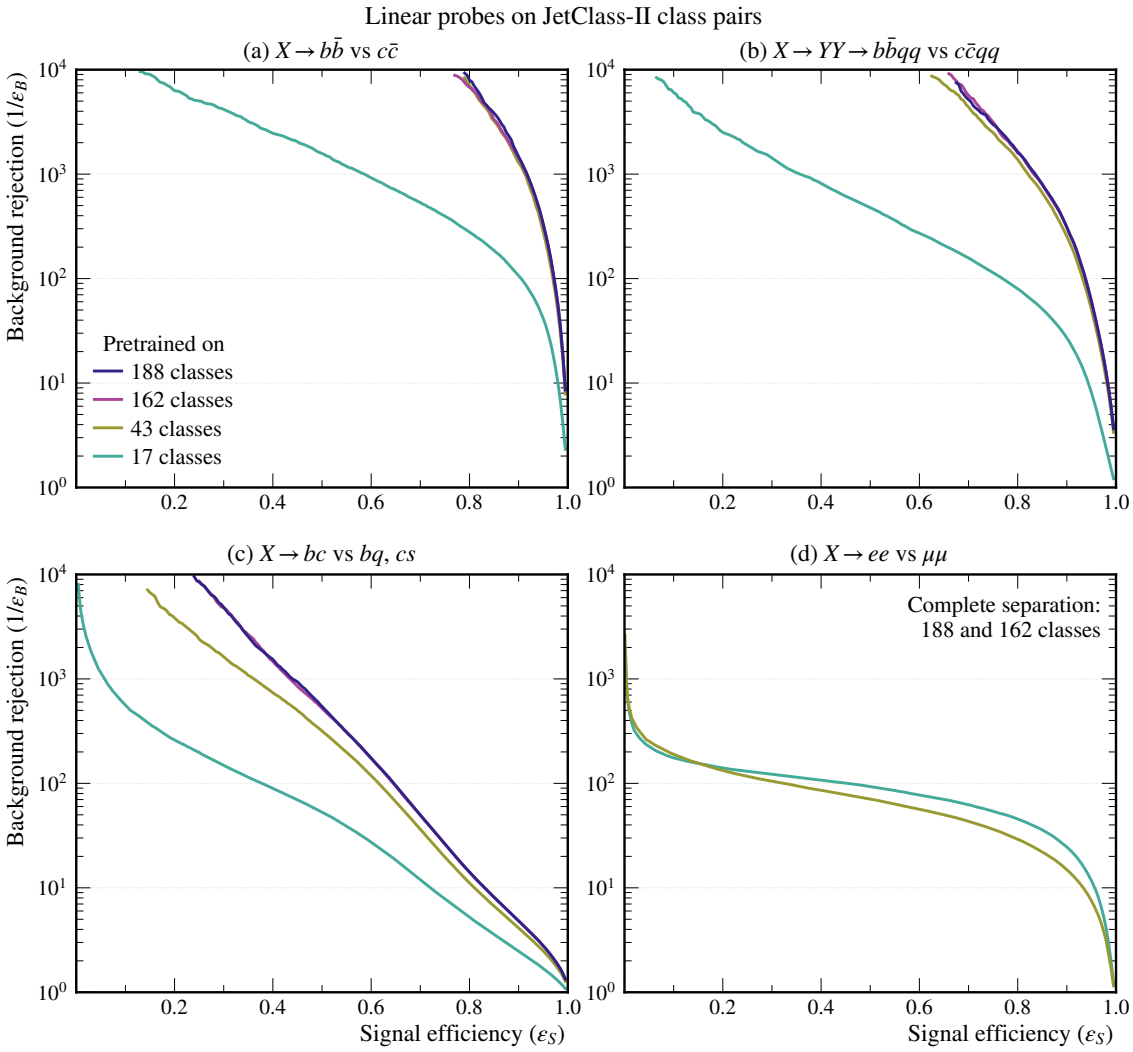


Figure 5: Background rejection against signal efficiency for four pairs of JetClass-II classes that differ only in flavor, from linear probes on networks pretrained with 188, 162, 43 and 17 classes. The first class named is the signal. Curves are means over three runs and end where fewer than 20 background jets remain; models with fewer at every efficiency are named in the panel. In (a, b) the three finest label sets nearly coincide.

Table 2: JetClass, in the form of ParT's Table 1: accuracy, AUC averaged over all pairs of classes (one against one, ParT's convention; on JetClass's equal class sizes it equals the one-against-rest average, checked), and for each signal class the rejection of QCD jets at 50% signal efficiency (99% for $H \to \ell\nu qq'$, 99.5% for $t \to b\ell\nu$) on $D_S = g_S/(g_S + g_{\mathrm{QCD}})$ of equation (**??**). Published rows: trained from scratch on 2 million jets and scored on ParT's 20-million-jet test set (ParT on 100 million jets reaches an accuracy of 0.861 and an AUC of 0.9877); ours fine-tuned on one million jets and scored on two million (the AUC on 500,000 of them). Mean ± standard deviation over the three pretraining runs. From scratch: three seeds. Bold: the best value of each column, and every value within one standard deviation of it. †: fewer than $n_{\min}$ QCD jets pass in at least one run (equation (**??**)), so that run's rejection has a counting uncertainty above 22%; these columns carry no claim.

				rejection of QCD								
	jets	accuracy	AUC	$H \to b\bar{b}$ R_{50}	$H \to c\bar{c}$ R_{50}	$H \to gg$ R_{50}	$H \to 4q$ R_{50}	$H \to \ell\nu qq'$ R_{99}	$t \to bqq'$ R_{50}	$t \to b\ell\nu$ $R_{99.5}$	$W \to qq'$ R_{50}	$Z \to q\bar{q}$ R_{50}
ParticleNet [**?**]	2 M	0.828	0.9820	5,540	1,681	90	662	1,654	4,049	4,673	260	215
ParT [**?**]	2 M	0.836	0.9834	5,587	1,982	93	761	1,609	6,061	4,474	307	236
this work (Sophon's network, fine-tuned)												
from scratch	1 M	0.8271 ± 0.0012	0.9819 ± 0.0002	$4{,}920 \pm 190$	$1{,}580 \pm 60$	88.3 ± 3.2	670 ± 40	$1{,}270 \pm 100$	$6{,}100 \pm 600$	$3{,}940 \pm 320$	272 ± 6	215 ± 8
188 classes	1 M	0.8468 ± 0.0006	0.9854 ± 0.0001	$\mathbf{6{,}600 \pm 330}$	$\mathbf{2{,}650 \pm 260}$	$\mathbf{104 \pm 4}$	$1{,}091 \pm 15$	$\mathbf{3{,}100 \pm 400}$	$\mathbf{12{,}900 \pm 1{,}600}^{\dagger}$	$\mathbf{9{,}400 \pm 1{,}500}^{\dagger}$	374 ± 8	$\mathbf{312 \pm 6}$
162 classes	1 M	$\mathbf{0.8475 \pm 0.0009}$	$\mathbf{0.9856 \pm <0.0001}$	6,060	$\mathbf{2{,}820 \pm 40}$	101.8 ± 1.5	$\mathbf{1{,}150 \pm 70}$	$\mathbf{3{,}070 \pm 330}$	$\mathbf{14{,}100 \pm 3{,}500}^{\dagger}$	$7{,}800 \pm 340$	$\mathbf{389 \pm 9}$	307.7 ± 1.2
43 classes	1 M	0.8459 ± 0.0002	0.9853 ± 0.0001	$\mathbf{6{,}500 \pm 500}$	$2{,}610 \pm 160$	$\mathbf{103.5 \pm 1.3}$	$1{,}030 \pm 40$	$\mathbf{3{,}160 \pm 280}$	$\mathbf{14{,}200 \pm 2{,}100}^{\dagger}$	$8{,}450 \pm 210$	376 ± 6	$\mathbf{307 \pm 7}$
17 classes	1 M	0.8422 ± 0.0005	0.9847 ± 0.0001	$\mathbf{6{,}670 \pm 220}$	$2{,}390 \pm 40$	$\mathbf{101 \pm 6}$	$1{,}049 \pm 17$	$2{,}800 \pm 200$	$10{,}600 \pm 2{,}700^{\dagger}$	$\mathbf{8{,}600 \pm 1{,}300}$	$\mathbf{378 \pm 22}$	285 ± 11

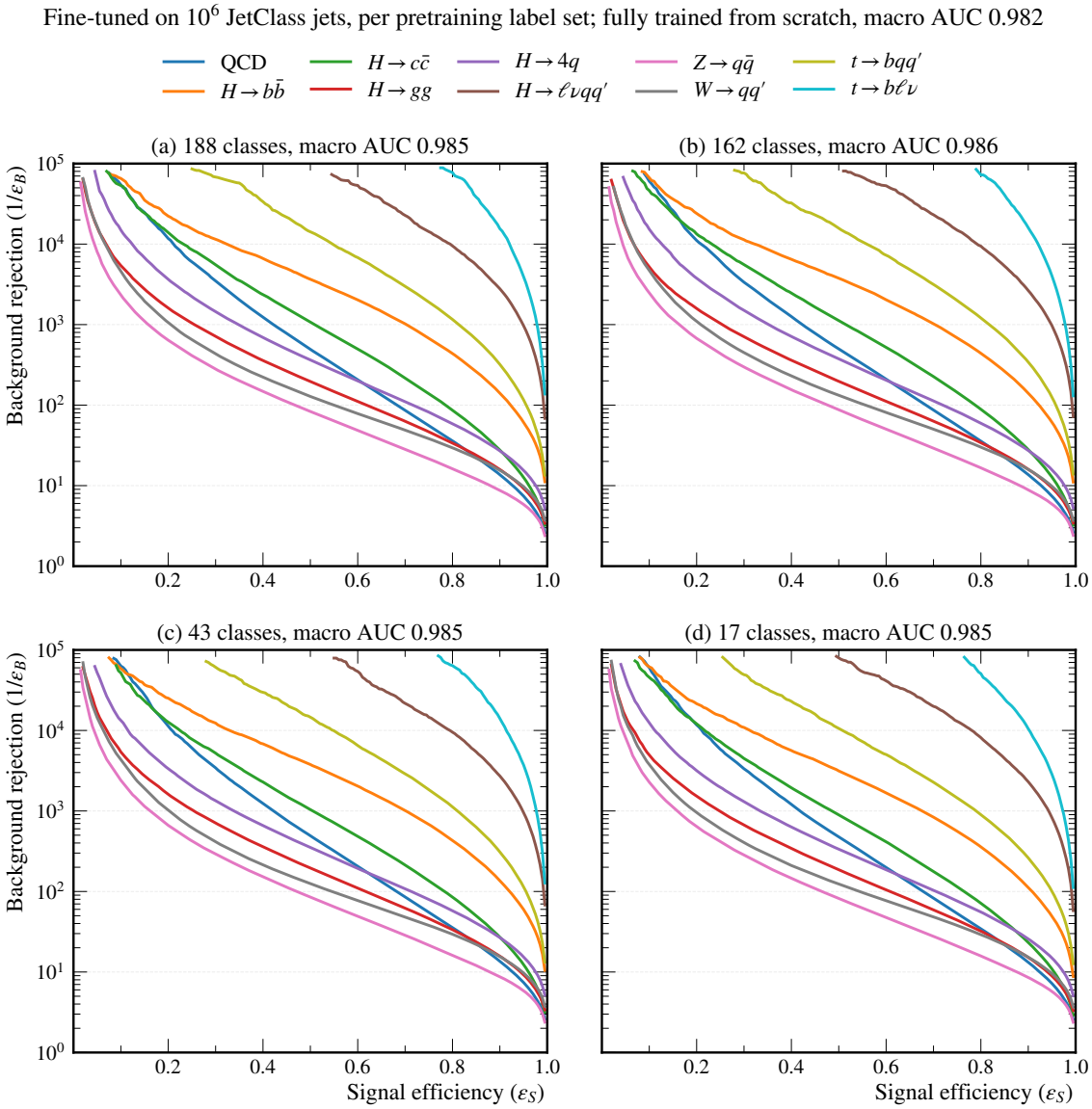


Figure 6: Background rejection against signal efficiency on JetClass after fine-tuning on one million jets. Each curve takes one of the ten JetClass classes as signal and the other nine as background, and is the mean over three pretraining runs. Curves begin where 20 background jets remain. One panel per pretraining label set; its title gives the mean area under the curve over the ten classes, and the heading that of the network trained from scratch.

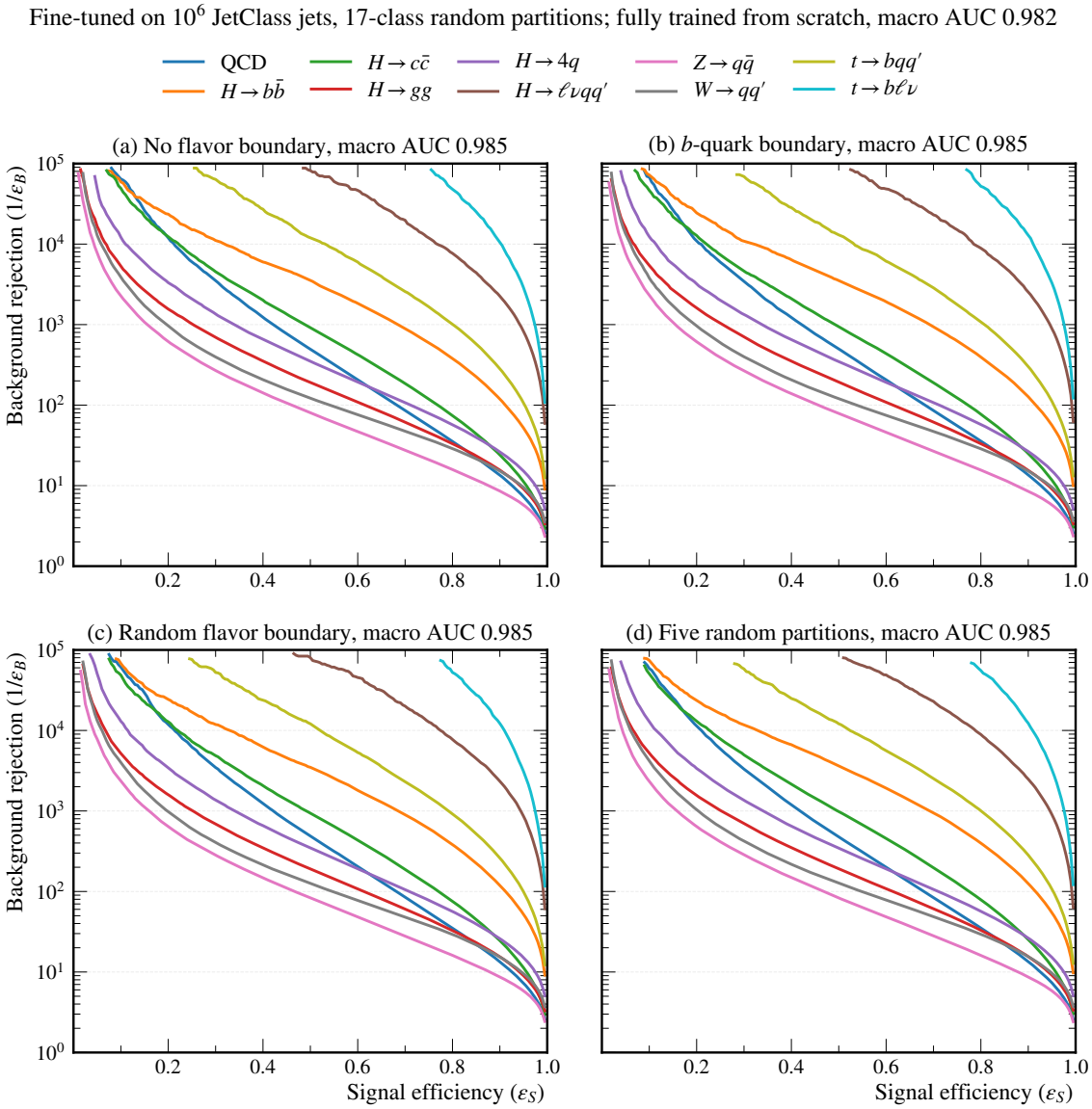


Figure 7: Background rejection against signal efficiency on JetClass after fine-tuning on one million jets, for networks pretrained on 17-class random partitions: with no flavor boundary, a b-quark boundary or a random flavor boundary (two runs each), and five plain random partitions (averaged). Each curve takes one class as signal against the other nine and begins where 20 background jets remain. Titles give the mean area under the curve; the heading, from scratch.

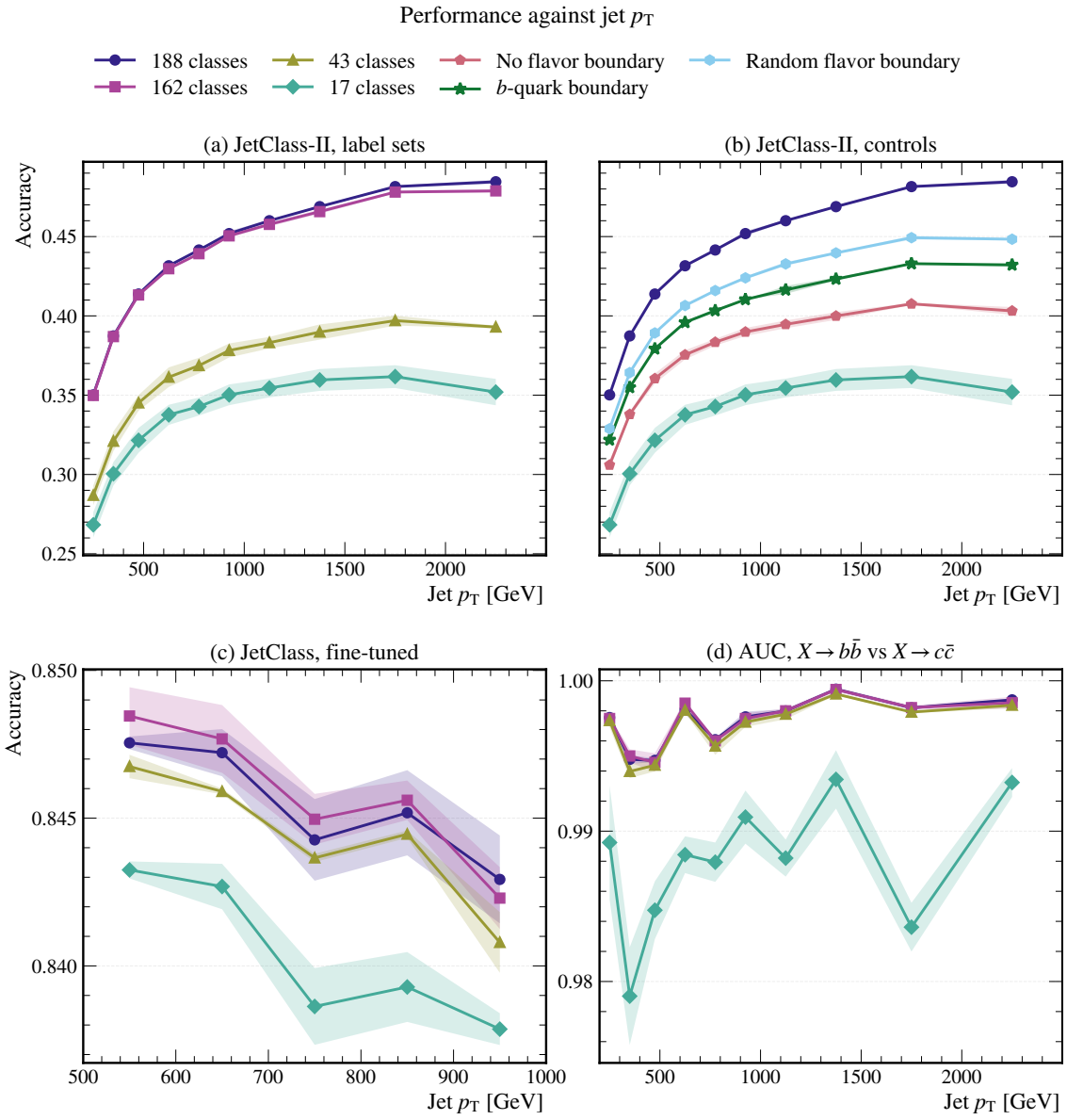


Figure 8: Performance in bins of jet transverse momentum: (a) accuracy of linear probes on class-balanced JetClass-II test jets per pretraining label set; (b) the same for the 17-class random partitions with controlled flavor boundaries, with the 188- and 17-class models for reference; (c) accuracy on JetClass after fine-tuning on one million jets; (d) the probes' area under the curve of $X \to b\bar{b}$ against $X \to c\bar{c}$. Points are means over three runs (two per partition), bands the standard deviation.

Table 3: Top tagging, fine-tuned on the full training set (1.2 million jets): accuracy, AUC and background rejection $1/\varepsilon_B$ at 50% and 30% signal efficiency. Published rows from the tables cited; f.t. marks a model pretrained on JetClass and fine-tuned. Our pretrained models see the 10 inputs the top data lack set to zero; the model with kinematic inputs only, a diagnostic of that mismatch, has none missing (one run). Uncertainties: ours, the standard deviation over the pretraining runs, one fine-tuning each, and over three seeds from scratch; ParT, the spread over 9 fine-tunings of one pretrained model with different output-layer initializations, quoted for the run of median accuracy; L-GATr and OmniLearned, the standard deviation of five trainings. Bold: the best value of each column, and every value within one standard deviation of it.

	inputs set to zero	accuracy	AUC	R_{50}	R_{30}
ParticleNet [**?**]	–	0.940	0.9858	397 ± 7	$1{,}615 \pm 93$
ParT [**?**]	–	0.940	0.9858	413 ± 16	$1{,}602 \pm 81$
ParT-f.t. [**?**]	–	0.944	0.9877	$\mathbf{691 \pm 15}$	$2{,}766 \pm 130$
MIParT-L-f.t. [**?**]	–	0.944	0.9878	640 ± 10	$2{,}789 \pm 133$
L-GATr-f.t. [**?**]	–	**0.9446**	0.9879	651 ± 11	$2{,}894 \pm 84$
OmniLearned-l [**?**]	–	0.944	**0.9880**	$\mathbf{688 \pm 9}$	$\mathbf{3{,}486 \pm 157}$
this work (Sophon's network)					
from scratch	10	0.9387 ± 0.0002	0.9852 ± 0.0001	358 ± 11	$1{,}412 \pm 20$
188 classes	10	0.9400 ± 0.0002	0.9861 ± 0.0001	461 ± 4	$1{,}800 \pm 60$
162 classes	10	0.9401 ± 0.0002	0.9861 ± 0.0001	443 ± 17	$1{,}790 \pm 60$
43 classes	10	0.9402 ± 0.0004	0.9861 ± 0.0001	450 ± 13	$1{,}780 \pm 150$
17 classes	10	0.9398 ± 0.0001	0.9859 ± 0.0001	429 ± 10	$1{,}740 \pm 140$
188 classes, kinematic inputs only (diagnostic)	none	0.941	0.9862	443	1,803

ParT's top-tagging training script, unchanged, on our cluster gives R_{30} of 1,711 (published $1{,}602 \pm 81$); with Sophon's output head, 1,577; with the head and Sophon's jet-p_{T} rescaling, 1,224 (one run each; our network from scratch, which uses the rescaling, $1{,}412 \pm 20$); table 13.

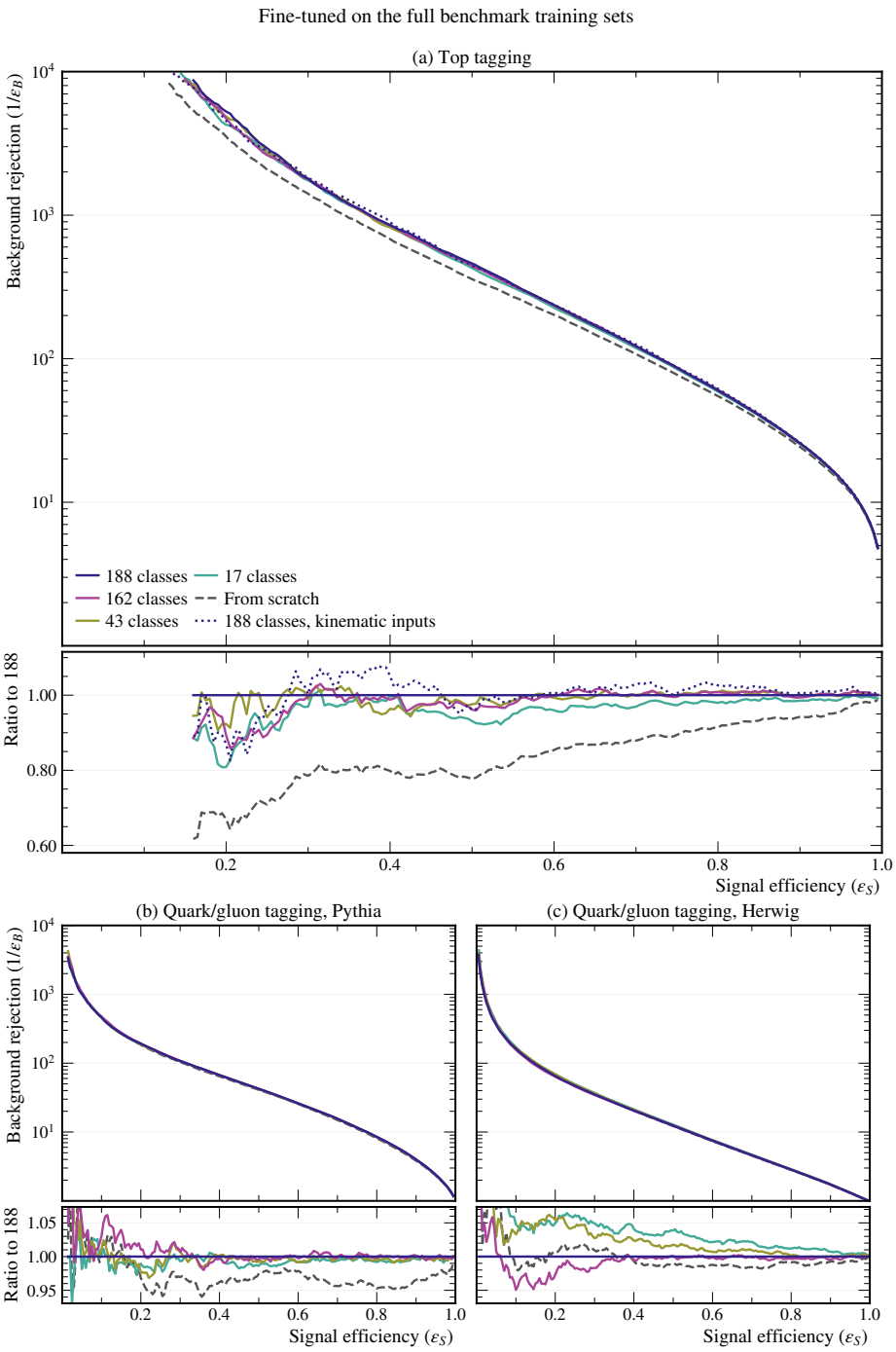


Figure 9: Background rejection against signal efficiency after fine-tuning on the full training sets: top tagging, and quark/gluon separation tested on Pythia and Herwig jets. Curves are networks pretrained with 188, 162, 43 and 17 classes and the network trained from scratch, each the mean over three runs; in (a), dotted, one run of the 188-class model with kinematic inputs only. Curves end where fewer than 20 background jets remain. Lower panels: ratio to 188 classes.

Table 4: Linear probes of the four label sets with the missing inputs set to zero, to the training mean or to random draws of a real particle, at the largest training set. Upper part: relative drop in JetClass-II accuracy (per cent) when the slots the top or the quark/gluon data lack are substituted, against the same model with every input. Lower part: R_{30} on top and quark/gluon tagging. Mean and standard deviation over three runs. The best substitution of each row and data set is in bold. Every configuration is in table 8.

	JetClass-II, top slots			JetClass-II, quark/gluon slots		
pretraining	zero	mean	draws	zero	mean	draws
188 classes	$88.9 \pm 1.0\%$	$87.9 \pm 1.1\%$	$\mathbf{85.98 \pm 0.07\%}$	$61.2 \pm 3.0\%$	$\mathbf{48.8 \pm 1.1\%}$	$54.30 \pm 0.21\%$
162 classes	$89.0 \pm 0.9\%$	$87.8 \pm 0.7\%$	$\mathbf{85.89 \pm 0.11\%}$	$64 \pm 5\%$	$\mathbf{49.5 \pm 0.5\%}$	$54.6 \pm 0.6\%$
43 classes	$87.81 \pm 0.28\%$	$86.7 \pm 0.9\%$	$\mathbf{83.17 \pm 0.29\%}$	$68 \pm 5\%$	$\mathbf{47.6 \pm 1.6\%}$	$53.5 \pm 1.4\%$
17 classes	$86.7 \pm 0.7\%$	$86.1 \pm 0.8\%$	$\mathbf{81.4 \pm 0.5\%}$	$70.9 \pm 2.3\%$	$\mathbf{50.0 \pm 1.0\%}$	$51.7 \pm 0.4\%$

	top, R_{30}			quark/gluon, R_{30}		
pretraining	zero	mean	draws	zero	mean	draws
188 classes	60 ± 12	73 ± 15	$\mathbf{228 \pm 25}$	66 ± 7	$\mathbf{72.7 \pm 2.4}$	68.1 ± 1.8
162 classes	76 ± 17	64 ± 11	$\mathbf{199 \pm 25}$	58 ± 9	$\mathbf{70.99 \pm 0.31}$	66.8 ± 0.4
43 classes	57 ± 13	58 ± 14	$\mathbf{231 \pm 8}$	61 ± 9	$\mathbf{70.8 \pm 1.8}$	69.0 ± 1.1
17 classes	55 ± 5	55 ± 5	$\mathbf{255 \pm 17}$	62 ± 7	$\mathbf{74.3 \pm 1.9}$	71.6 ± 2.8
188 classes, kinematic inputs only	231	nothing missing		–	–	–

Table 5: Quark/gluon discrimination with the particle type as five flags and the charge (ParT's exp inputs), fine-tuned on the full training set (1.6 million Pythia jets) and scored on Pythia and on Herwig jets: accuracy, AUC and $1/\varepsilon_B$ at 50% and 30% quark efficiency. Our models see the 4 displacement inputs set to zero. Published rows (exp inputs) from the tables cited; PFN-Ex's uncertainty is half the interquartile range of ten trainings, the others as in table 3, with nine trainings for OmniLearned. Bold as in table 3.

		Pythia				Herwig			
	inputs set to zero	accuracy	AUC	R_{50}	R_{30}	accuracy	AUC	R_{50}	R_{30}
PFN-Ex [?]	–	–	0.9005	34.7 ± 0.4	–	–	–	–	–
ParticleNet [?]	–	0.840	0.9116	39.8 ± 0.2	98.6 ± 1.3	–	–	–	–
ParT [?]	–	0.840	0.9121	41.3 ± 0.3	101.2 ± 1.1	–	–	–	–
ParticleNet-f.t. [?]	–	0.839	0.9115	40.1 ± 0.2	100.3 ± 1.0	–	–	–	–
ParT-f.t. [?]	–	0.843	0.9151	42.4 ± 0.2	107.9 ± 0.5	–	–	–	–
OmniLearned-m [?]	–	**0.845**	**0.9162**	**43.2 ± 0.1**	**111.2 ± 1.5**	–	–	–	–
this work (Sophon's network)									
from scratch	4	0.8397 ± 0.0007	0.9121 ± 0.0002	41.47 ± 0.29	104.0 ± 1.0	**0.7153 ± 0.0024**	0.8061 ± 0.0007	12.18 ± 0.11	35.5 ± 1.2
188 classes	4	0.8433 ± 0.0002	0.9148 ± 0.0001	42.56 ± 0.10	107.3 ± 2.2	0.712 ± 0.005	0.8078 ± 0.0023	12.3 ± 0.4	**35.2 ± 2.1**
162 classes	4	$0.8432 \pm <0.0001$	0.9148 ± 0.0001	42.39 ± 0.22	108.0 ± 1.4	0.7100 ± 0.0017	0.8074 ± 0.0010	12.33 ± 0.28	34.6 ± 2.0
43 classes	4	0.8430 ± 0.0005	0.9146 ± 0.0003	42.2 ± 0.8	108.3 ± 1.0	**0.713 ± 0.006**	**0.809 ± 0.005**	**12.5 ± 0.6**	**36.1 ± 2.7**
17 classes	4	0.8427 ± 0.0003	$0.9144 \pm <0.0001$	41.97 ± 0.25	106.5 ± 0.8	**0.7175 ± 0.0033**	**0.8103 ± 0.0018**	**12.79 ± 0.20**	**36.9 ± 0.9**

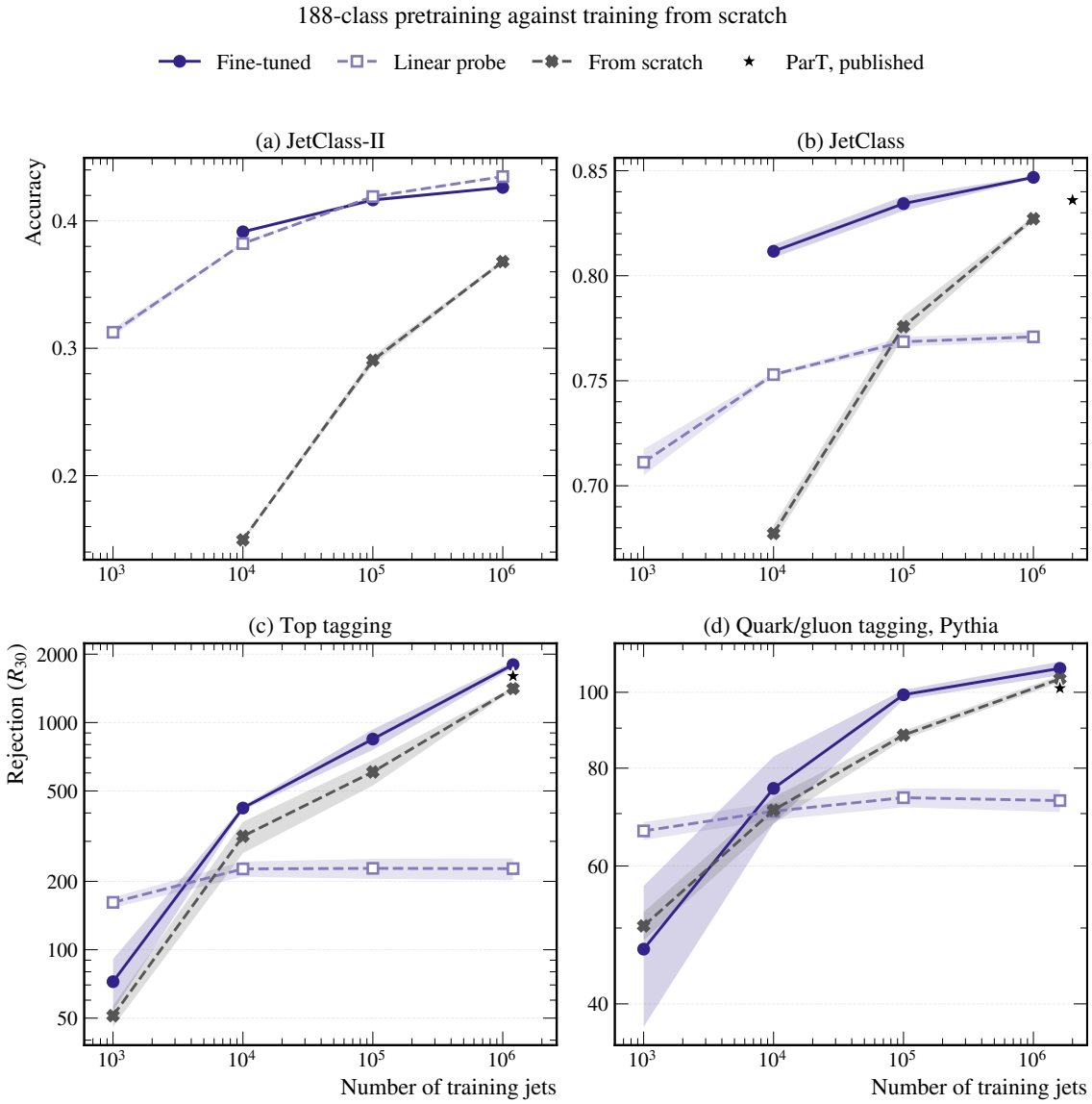


Figure 10: Accuracy on class-balanced JetClass-II and on JetClass, and rejection R_{30} at 30% signal efficiency on top and quark/gluon tagging, against the number of training jets: the 188-class model fine-tuned and with a linear probe, and a network trained from scratch. Points are means; bands the standard deviation over three runs. Fine-tuning on JetClass-II and JetClass starts at 10^4 jets. Stars are published ParT: same inputs on JetClass, kinematic only on top, kinematic and particle type on quark/gluon.

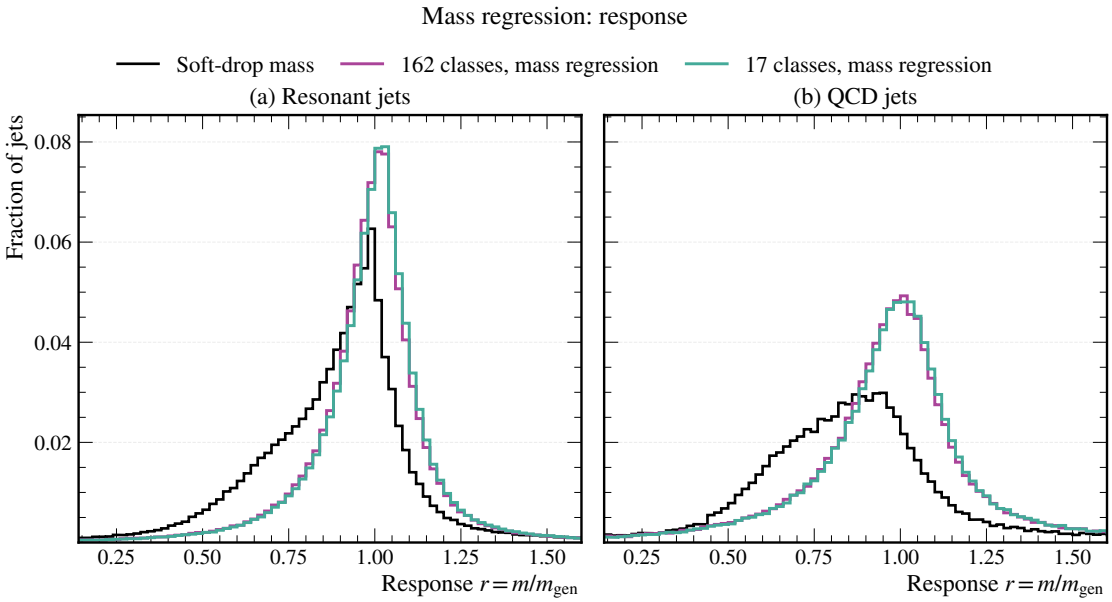


Figure 11: Distribution of the jet mass response $r = m/m_{\text{gen}}$, a jet's mass over its generator-level mass, for (a) resonant and (b) QCD jets of the JetClass-II test set: the soft-drop mass (solid black) and the mass regression outputs of the models pretrained on 162 and 17 classes, each the mean of two runs, which nearly coincide. Both panels share their axes.

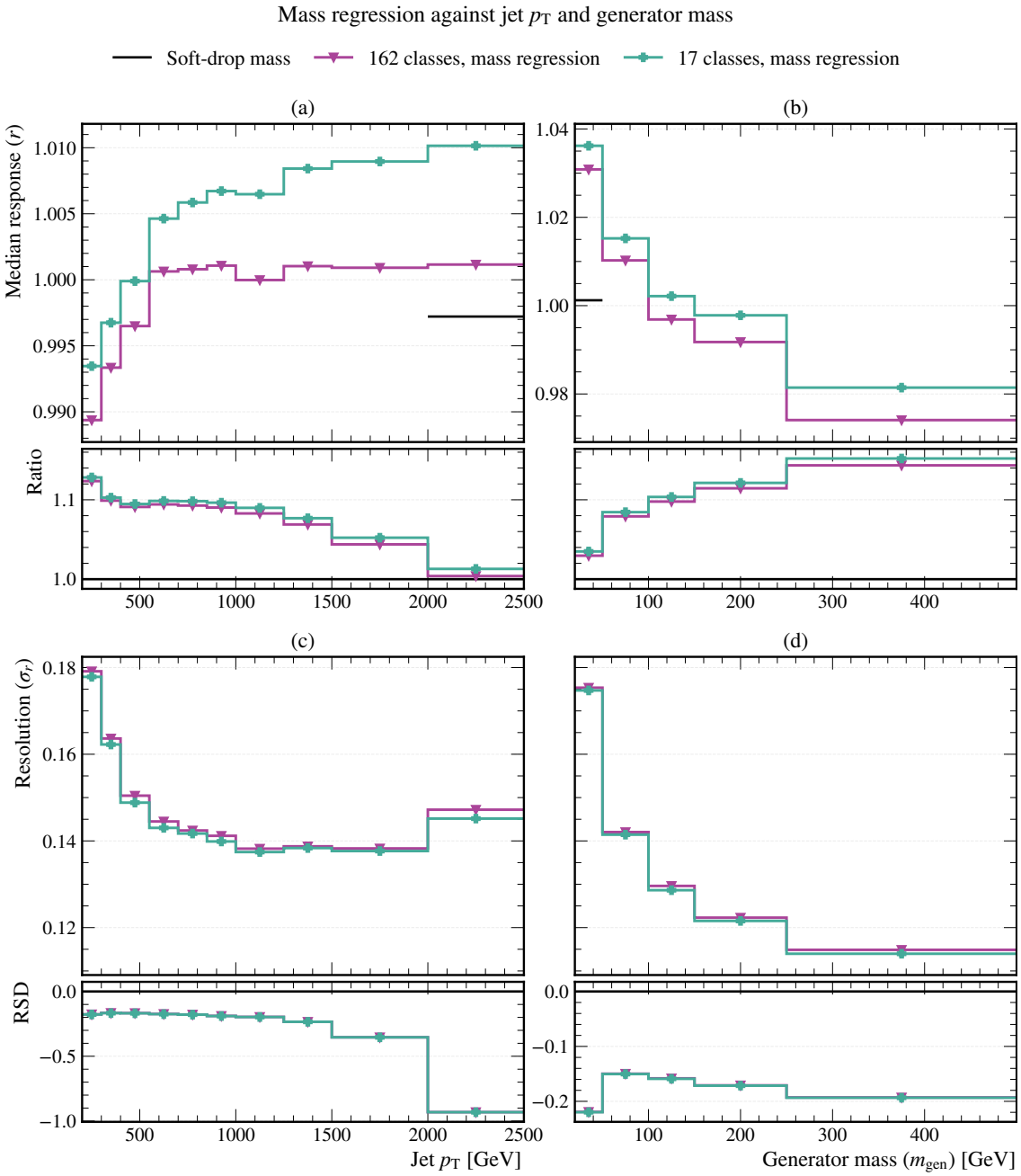


Figure 12: Median mass response $r = m/m_{\text{gen}}$ (a, b) and resolution σ_r, half the 16–84% quantile range of r over its median (c, d), of JetClass-II test jets against transverse momentum (a, c; all jets) and generator mass (b, d; 20–500 GeV): mass regression of the models pretrained on 162 and 17 classes (two runs). Lower panels: ratio to the soft-drop mass (black where in range) and signed root square difference (RSD), negative where better. Unlabelled axes share the left ones.

Table 6: Jet mass from frozen features: a multilayer perceptron regresses the mass from the frozen features of each configuration; the groomed (soft-drop) mass is the reference. Resolution: σ_r of equation (**??**), half the 15.9–84.1% inter-quantile range of the response $r = m_{\text{est}}/m_{\text{SD}}^{\text{gen}}$ divided by its median; and the median response, over all JetClass-II test jets with a generator match (resonant and QCD jets separately in table 25); lower resolution is better. The regression target is centred on the training median of each of the 188 classes, a truth-label input the groomed-mass row does not have. A multilayer perceptron is used because the mass is not linear in the features (ridge regression in table 25). The randomly initialized network shows what a network that never trained exposes. The mass regression in pretraining has the weight $\lambda = 5$ at both label sets, a share of the training loss of 13.78 and 13.87% (162 classes) and 28.65 and 28.67% (17 classes). Mean ± standard deviation over the three pretraining runs; the models with the mass regression have two runs, and both values are given.

	resolution	median response
groomed mass	0.245	0.929
MLP regression on frozen features		
randomly initialized network	0.2421 ± 0.0018	0.983 ± 0.005
188 classes	0.2200 ± 0.0014	0.991 ± 0.005
162 classes	0.2195 ± 0.0017	0.9876 ± 0.0014
43 classes	0.2158 ± 0.0029	0.991 ± 0.005
17 classes	0.2098 ± 0.0031	0.991 ± 0.004
162 classes, mass regression	0.1591, 0.1586	1.004, 1.006
17 classes, mass regression	0.1583, 0.1588	1.009, 1.006

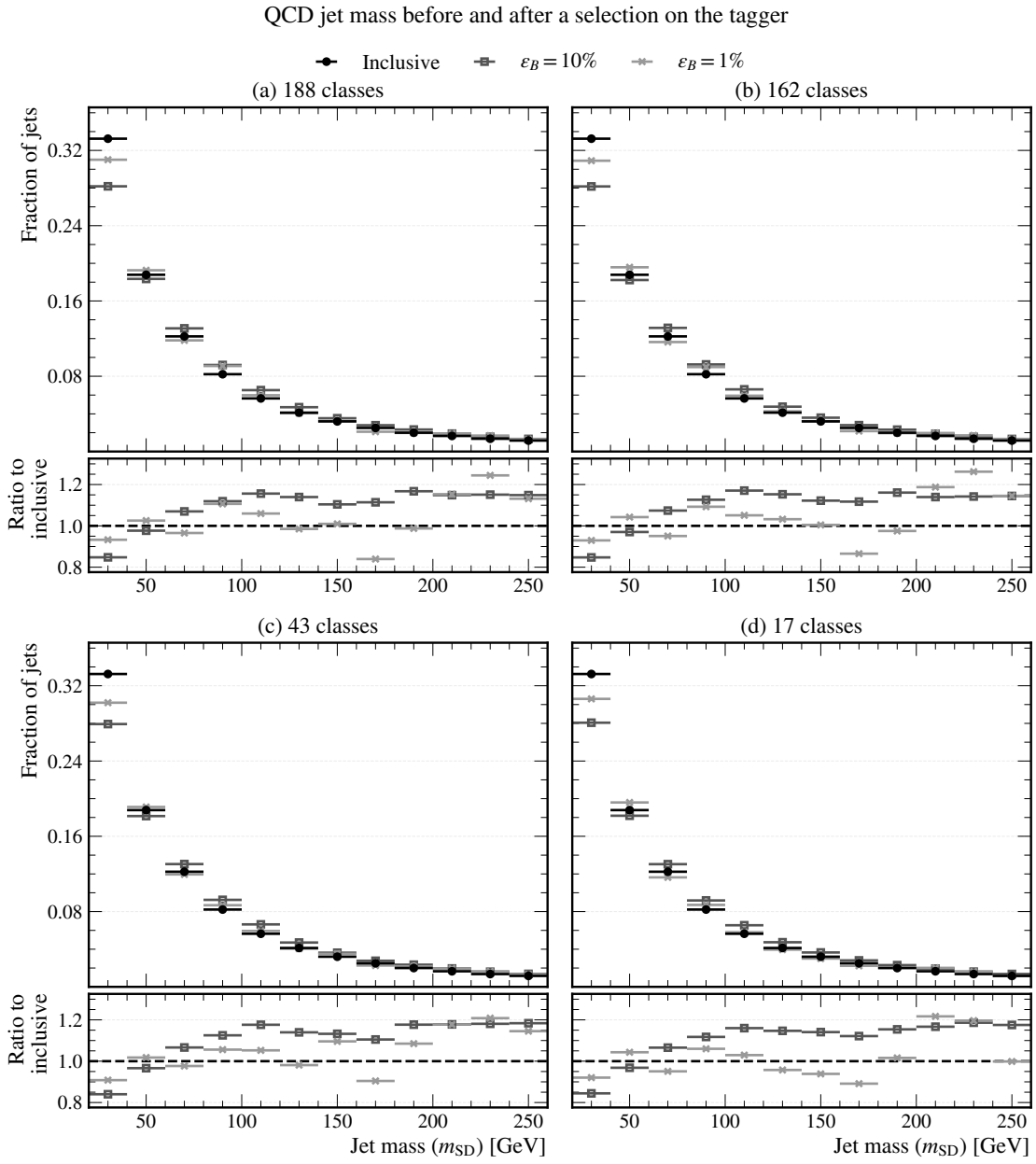


Figure 13: Soft-drop mass distribution of QCD test jets of JetClass-II before any selection (inclusive) and after selections on $1 - g_{\mathrm{QCD}}$, one minus the summed QCD score, at QCD efficiencies ε_B of 10% and 1%, for the models pretrained on 188, 162, 43 and 17 classes (mean of three runs). Bars show the bin width; the axis stops at 260 GeV. Below each panel, the ratio to the inclusive distribution.

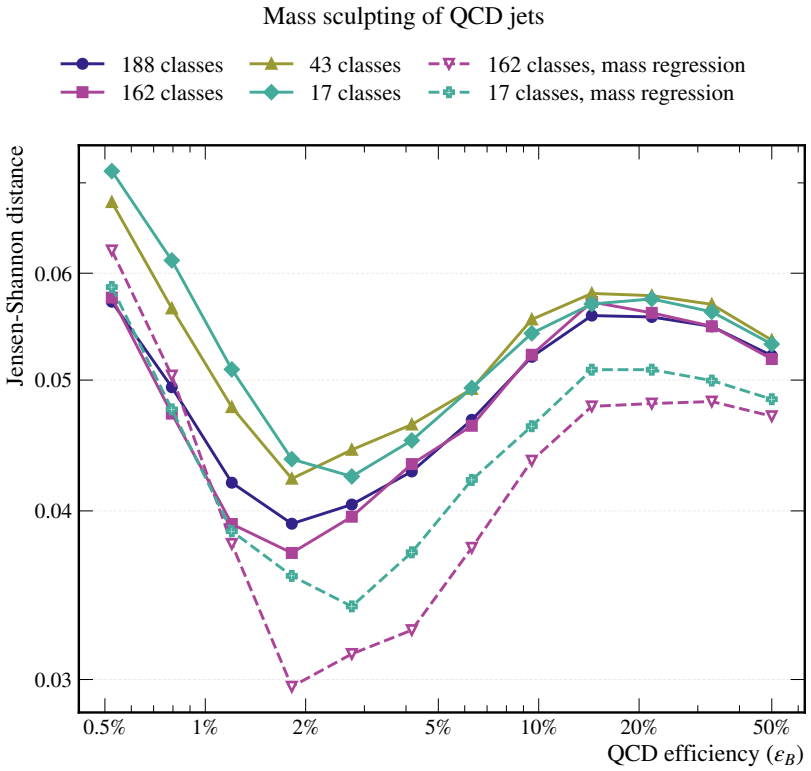


Figure 14: Jensen-Shannon distance (base 2) between the soft-drop mass distributions of QCD test jets of JetClass-II before and after a selection on $1 - g_{\mathrm{QCD}}$, one minus the summed QCD score, against the QCD efficiency ε_B of the selection (tighter to the left), for the models pretrained on 188, 162, 43 and 17 classes (solid, mean of three runs) and on 162 and 17 classes with mass regression (dashed, mean of two).

Appendix

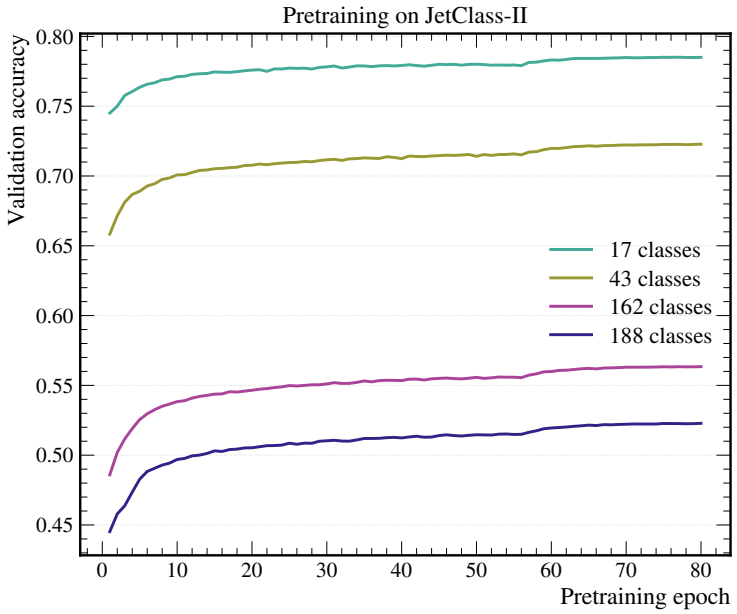


Figure 15: Validation accuracy of the pretraining against epoch for the 188-, 162-, 43- and 17-class label sets, each on its own classes, as the mean over three runs. A coarser label set has fewer classes to tell apart, so its accuracy is not comparable with a finer set's. The step near epoch 56 is where the learning rate starts to decay.

Table 7: Learning-rate choice at 10^4 training jets: the best validation accuracy of each rate on top and quark/gluon and their sum. *: the rate chosen, on the sum for the pretrained weights and on each task's accuracy for the networks trained from scratch; the test jets play no part.

network	learning rate	top	quark/gluon	sum
pretrained, backbone	10^{-4}	0.9204	0.8204	1.741
pretrained, backbone	3×10^{-4}	0.9192	0.8202	1.739
pretrained, backbone	10^{-3}	0.9212	0.8211	1.742*
from scratch	3×10^{-4}	0.9152	0.8052	1.720
from scratch	10^{-3}	0.9188*	0.8029	1.722
from scratch	3×10^{-3}	0.9177	0.8059*	1.724

Table 8: The three substitutions of the missing inputs (zero, the training mean, random draws of a real particle) for every configuration, at the largest training set. Upper part: relative drop in JetClass-II accuracy (per cent) when the slots the top (10) or the quark/gluon (4) data lack are substituted, against the same model with every input. Lower part: R_{30} of the linear probe on top and quark/gluon jets under each substitution; the model with kinematic inputs only has nothing missing. The substitution chosen on validation is random draws for top and the training mean for quark/gluon. The best substitution of each row and data set is in bold. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

	JetClass-II, top slots			JetClass-II, quark/gluon slots		
pretraining	zero	mean	draws	zero	mean	draws
188 classes	88.9 ± 1.0%	87.9 ± 1.1%	**85.98 ± 0.07%**	61.2 ± 3.0%	**48.8 ± 1.1%**	54.30 ± 0.21%
162 classes	89.0 ± 0.9%	87.8 ± 0.7%	**85.89 ± 0.11%**	64 ± 5%	**49.5 ± 0.5%**	54.6 ± 0.6%
43 classes	87.81 ± 0.28%	86.7 ± 0.9%	**83.17 ± 0.29%**	68 ± 5%	**47.6 ± 1.6%**	53.5 ± 1.4%
17 classes	86.7 ± 0.7%	86.1 ± 0.8%	**81.4 ± 0.5%**	70.9 ± 2.3%	**50.0 ± 1.0%**	51.7 ± 0.4%
162 classes, mass regression	89.20, 88.29%	88.47, 87.78%	**85.35, 85.37%**	61.8, 67.9%	**49.7, 52.2%**	54.34, 55.13%
17 classes, mass regression	84.8, 87.4%	85.93, 86.89%	**79.1, 81.7%**	66.7, 65.4%	**49.04, 49.64%**	50.77, 51.07%
random partition 1, seed 1	90.2%	88.6%	**85.7%**	66.5%	**51.6%**	56.3%
random partition 2, seed 2	90.4%	88.8%	**85.9%**	69.5%	**52.4%**	56.3%
random partition 3, seed 1	90.8%	89.5%	**86.2%**	63.7%	**52.8%**	57.8%
random partition 4, seed 2	89.9%	88.7%	**87.1%**	72.9%	**53.4%**	55.5%
random partition 5, seed 2	89.5%	89.7%	**86.1%**	69.5%	**53.9%**	56.3%
random partition, no flavor boundary	88.88, 88.46%	88.3, 87.0%	**83.26, 83.89%**	62.72, 62.24%	**51.87, 51.28%**	53.92, 53.44%
random partition, b-quark boundary	87.80, 87.65%	88.9, 86.9%	**84.44, 85.21%**	66.2, 62.8%	**52.67, 52.80%**	55.58, 56.49%
random partition, random flavor boundary	89.58, 90.28%	89.44, 89.66%	**84.94, 85.58%**	67.2, 69.0%	**52.9, 51.8%**	57.2, 55.4%
randomly initialized network, BatchNorm recomputed	**56.4 ± 1.4%**	59.8 ± 0.8%	59.8 ± 1.2%	**8.7 ± 2.6%**	10.4 ± 1.4%	11.4 ± 2.1%

	top, R_{30}			quark/gluon, R_{30}		
pretraining	zero	mean	draws	zero	mean	draws
188 classes	60 ± 12	73 ± 15	**228 ± 25**	66 ± 7	**72.7 ± 2.4**	68.1 ± 1.8
162 classes	76 ± 17	64 ± 11	**199 ± 25**	58 ± 9	**70.99 ± 0.31**	66.8 ± 0.4
43 classes	57 ± 13	58 ± 14	**231 ± 8**	61 ± 9	**70.8 ± 1.8**	69.0 ± 1.1
17 classes	55 ± 5	55 ± 5	**255 ± 17**	62 ± 7	**74.3 ± 1.9**	71.6 ± 2.8
162 classes, mass regression	77, 95	57, 79	**260, 306**	65.4, 61.1	**73.9, 70.7**	70.4, 72.1
17 classes, mass regression	103, 69	55, 81	**306, 378**	66.01, 65.57	68.4, 72.2	**71.3, 75.3**
random partition 1, seed 1	44.2	55.5	**174**	65.9	**69.6**	66.9
random partition 2, seed 2	60.5	53.6	**151**	59.0	**69.3**	64.6
random partition 3, seed 1	50.0	42.9	**164**	65.2	**68.4**	62.3
random partition 4, seed 2	30.2	56.5	**152**	51.0	**64.7**	61.6
random partition 5, seed 2	73.9	45.5	**173**	54.9	**69.6**	68.2
random partition, no flavor boundary	69.65, 69.34	44, 65	**262, 243**	63.1, 66.4	**73.9, 71.7**	71.07, 71.48
random partition, b-quark boundary	73.7, 80.7	45, 68	**202.5, 211.6**	54.6, 59.4	**72.52, 72.20**	70.87, 70.47
random partition, random flavor boundary	65, 41	40.9, 33.2	**214, 233**	66, 54	**71.2, 72.8**	71.2, 69.8
randomly initialized network, BatchNorm recomputed	58 ± 13	**60.4 ± 2.4**	48.6 ± 3.0	**54.2 ± 2.4**	52.2 ± 0.8	52.4 ± 0.4
188 classes, kinematic inputs only	231	nothing missing		–	–	–

Table 9: Nonlinear probes (a multilayer perceptron with one hidden layer of 256 units, three seeds averaged) at the largest training set: class-balanced JetClass-II AUC (all, resonant and QCD classes) with the ratio ρ of $1 - \text{AUC}$, equation (**??**), to the 188-class run, JetClass AUC, and AUC and R_{30} on top, quark/gluon and Herwig jets (missing inputs set to zero); then the class pairs of table 24, class-balanced, except $X \to ee$ vs $\mu\mu$, which has no nonlinear probe. The models with the mass regression and the numbered random partitions have no nonlinear probe on the class-balanced test set, so their cells there are dashes and they leave the pair blocks; the next block gives them on the natural-composition test set beside the levels of the hierarchy on that set, and no ρ crosses the two test sets. Last, the linear probe on the 512-unit hidden layer of the 188-class models' pretraining head beside the one on the class token. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. The 188-class model with kinematic inputs only, a diagnostic of the missing inputs, has one run and no input missing, and is evaluated on top tagging only.

	JetClass-II				JetClass	top		quark/gluon		Herwig	
pretraining	AUC	resonant	QCD	ρ	AUC	AUC	R_{30}	AUC	R_{30}	AUC	R_{30}
188 classes	$0.9806 \pm <0.0001$	$0.9834 \pm <0.0001$	0.9643 ± 0.0001	1.00	0.9786 ± 0.0004	0.9713 ± 0.0034	290 ± 60	0.894 ± 0.004	82.6 ± 3.5	0.8077 ± 0.0006	36.64 ± 0.25
162 classes	0.9802 ± 0.0001	0.9833 ± 0.0001	0.9618 ± 0.0003	1.019 ± 0.007	0.9786 ± 0.0002	0.971 ± 0.004	280 ± 90	0.889 ± 0.005	77.0 ± 2.2	0.8060 ± 0.0014	35.6 ± 0.6
43 classes	0.9773 ± 0.0001	0.9802 ± 0.0002	0.9605 ± 0.0001	1.170 ± 0.006	0.9782 ± 0.0003	0.9688 ± 0.0029	250 ± 60	0.8904 ± 0.0031	81 ± 4	0.8071 ± 0.0007	35.67 ± 0.28
17 classes	0.9731 ± 0.0004	0.9766 ± 0.0004	0.9524 ± 0.0004	1.388 ± 0.021	0.9776 ± 0.0005	0.9688 ± 0.0018	224 ± 24	0.887 ± 0.005	78.9 ± 1.7	0.8062 ± 0.0024	36.2 ± 0.5
162 classes, mass regression	–	–	–	–	0.9794, 0.9799	0.9731, 0.9754	317, 376	0.8932, 0.8914	81.8, 78.4	0.8094, 0.8071	37.64, 36.83
17 classes, mass regression	–	–	–	–	0.9787, 0.9785	0.9747, 0.9704	356, 261	0.8933, 0.8927	80.91, 81.90	0.8079, 0.8084	35.95, 36.47
random partition 1, seed 1	–	–	–	–	0.974	0.964	161	0.893	80.9	0.809	36.6
random partition 2, seed 2	–	–	–	–	0.975	0.968	210	0.888	79.0	0.806	35.2
random partition 3, seed 1	–	–	–	–	0.974	0.967	190	0.894	80.5	0.806	35.2
random partition 4, seed 2	–	–	–	–	0.974	0.959	125	0.881	72.0	0.806	35.5
random partition 5, seed 2	–	–	–	–	0.975	0.970	251	0.886	75.5	0.805	35.4
random partition, no flavor boundary	0.9749, 0.9746	0.9786, 0.9783	0.9528, 0.9522	1.294, 1.310	0.9764, 0.9763	0.9668, 0.9682	210.1, 218.3	0.8912, 0.8921	77.5, 83.1	0.8073, 0.8071	35.27, 34.98
random partition, b-quark boundary	0.9774, 0.9775	0.9806, 0.9807	0.9582, 0.9584	1.168, 1.159	0.9767, 0.9776	0.9718, 0.9724	304.1, 313.5	0.8887, 0.8911	80.8, 78.6	0.8069, 0.8067	36.08, 35.57
random partition, random flavor boundary	0.9776, 0.9777	0.9810, 0.9811	0.9571, 0.9576	1.158, 1.150	0.9767, 0.9763	0.9670, 0.9626	215, 160	0.8906, 0.8870	80.6, 77.9	0.8074, 0.8052	35.31, 34.53
randomly initialized network	0.8648 ± 0.0014	0.8694 ± 0.0017	0.8372 ± 0.0015	–	0.8859 ± 0.0021	0.9667 ± 0.0006	210 ± 4	0.8844 ± 0.0001	76.6 ± 0.8	0.8074 ± 0.0007	38.2 ± 0.4
randomly initialized network, BatchNorm recomputed	0.8974 ± 0.0018	0.9035 ± 0.0020	0.8610 ± 0.0008	–	0.9146 ± 0.0012	0.9712 ± 0.0024	300 ± 50	0.8831 ± 0.0003	76.3 ± 1.9	0.8087 ± 0.0009	37.8 ± 1.5
188 classes, kinematic inputs only (diagnostic)	–	–	–	–	–	0.981	716	–	–	–	–

pretraining	$X \to bc$ vs bq, cs		$X \to b\bar{b}$ vs $c\bar{c}$		$X \to YY \to b\bar{b}qq$ vs $c\bar{c}qq$	
	AUC	ρ	AUC	ρ	AUC	ρ
188 classes	0.9434 ± 0.0001	1.00	$0.9978 \pm <0.0001$	1.00	0.9941 ± 0.0001	1.00
162 classes	0.9432 ± 0.0002	1.002 ± 0.006	$0.9977 \pm <0.0001$	1.023 ± 0.021	0.9941 ± 0.0001	1.011 ± 0.030
43 classes	0.9375 ± 0.0008	1.104 ± 0.017	$0.9975 \pm <0.0001$	1.125 ± 0.011	$0.9935 \pm <0.0001$	1.112 ± 0.006
17 classes	0.9080 ± 0.0013	1.624 ± 0.022	0.9932 ± 0.0002	3.04 ± 0.09	0.9828 ± 0.0011	2.93 ± 0.17
random partition, no flavor boundary	0.9077, 0.9046	1.633, 1.679	0.9926, 0.9902	3.3, 4.3	0.9813, 0.9776	3.20, 3.78
random partition, b-quark boundary	0.9216, 0.9227	1.385, 1.360	0.9961, 0.9963	1.74, 1.64	0.9916, 0.9919	1.442, 1.372
random partition, random flavor boundary	0.9231, 0.9242	1.359, 1.335	0.9951, 0.9946	2.21, 2.37	0.9889, 0.9891	1.890, 1.843
randomly initialized network	0.630 ± 0.004	–	0.759 ± 0.009	–	0.747 ± 0.006	–
randomly initialized network, BatchNorm recomputed	0.6865 ± 0.0030	–	0.866 ± 0.008	–	0.837 ± 0.006	–

pretraining	QCD $b\bar{b}$ vs $c\bar{c}$		QCD with b vs light		QCD with c, no b, vs light	
	AUC	ρ	AUC	ρ	AUC	ρ
188 classes	0.9871 ± 0.0001	1.00	$0.9882 \pm <0.0001$	1.00	0.9329 ± 0.0001	1.00
162 classes	0.9860 ± 0.0003	1.085 ± 0.030	0.9876 ± 0.0001	1.050 ± 0.004	0.9303 ± 0.0002	1.039 ± 0.005
43 classes	0.9855 ± 0.0001	1.121 ± 0.017	0.9875 ± 0.0001	1.055 ± 0.005	0.9292 ± 0.0003	1.055 ± 0.006
17 classes	0.9701 ± 0.0008	2.31 ± 0.04	0.9842 ± 0.0002	1.337 ± 0.021	0.9145 ± 0.0004	1.274 ± 0.008
random partition, no flavor boundary	0.9680, 0.9647	2.50, 2.72	0.9845, 0.9847	1.315, 1.288	0.9174, 0.9186	1.234, 1.211
random partition, b-quark boundary	0.9817, 0.9821	1.432, 1.372	0.9863, 0.9865	1.158, 1.140	0.9224, 0.9229	1.159, 1.147
random partition, random flavor boundary	0.9776, 0.9775	1.745, 1.725	0.9858, 0.9859	1.201, 1.187	0.9236, 0.9247	1.141, 1.120
randomly initialized network	0.7228 ± 0.0031	–	0.8185 ± 0.0019	–	0.682 ± 0.005	–
randomly initialized network, BatchNorm recomputed	0.822 ± 0.005	–	0.8879 ± 0.0033	–	0.7143 ± 0.0013	–

natural-composition	JetClass-II		ρ on the class pairs					
test set	AUC	ρ	$X \to bc$ vs bq, cs	$X \to b\bar{b}$ vs $c\bar{c}$	$X \to YY \to b\bar{b}qq$ vs $c\bar{c}qq$	QCD $b\bar{b}$ vs $c\bar{c}$	QCD with b vs light	QCD with c, no b, vs light
188 classes	0.9767 ± 0.0002	1.00	1.00	1.00	1.00	1.00	1.00	1.00
162 classes	0.9759 ± 0.0003	1.033 ± 0.016	1.003 ± 0.004	1.00 ± 0.09	0.97 ± 0.08	1.08 ± 0.09	1.06 ± 0.06	1.041 ± 0.021
43 classes	0.9736 ± 0.0005	1.135 ± 0.018	1.135 ± 0.017	1.112 ± 0.009	1.02 ± 0.06	1.13 ± 0.05	1.05 ± 0.09	1.057 ± 0.016
17 classes	0.9682 ± 0.0003	1.364 ± 0.008	1.76 ± 0.04	3.26 ± 0.10	3.34 ± 0.29	2.71 ± 0.13	1.32 ± 0.18	1.303 ± 0.032
162 classes, mass regression	0.9752, 0.9757	1.061, 1.051	1.057, 1.018	1.21, 1.09	1.17, 1.04	1.30, 1.16	1.027, 0.938	1.071, 1.045
17 classes, mass regression	0.9674, 0.9679	1.393, 1.391	1.839, 1.797	3.65, 3.11	3.74, 3.34	3.26, 2.88	1.38, 1.26	1.326, 1.323
random partition 1, seed 1	0.974	1.13	1.20	1.44	1.35	1.53	1.05	1.13
random partition 2, seed 2	0.974	1.14	1.17	1.35	1.30	1.47	1.00	1.10
random partition 3, seed 1	0.974	1.12	1.10	1.44	1.32	1.45	1.16	1.12
random partition 4, seed 2	0.974	1.13	1.09	1.25	1.25	1.37	1.16	1.07
random partition 5, seed 2	0.974	1.15	1.21	1.56	1.30	1.53	1.16	1.10

188 classes, linear probe on	JetClass-II AUC	quark/gluon AUC	quark/gluon R_{30}
class token	$0.9801 \pm <0.0001$	0.878 ± 0.006	66 ± 7
512-unit hidden layer	$0.9802 \pm <0.0001$	0.880 ± 0.007	70 ± 6

Table 10: Linear probes on JetClass-II (natural composition, largest training set) with the BatchNorm statistics as accumulated in pretraining and recomputed on training jets: the AUC averaged over the classes and the ratio ρ of $1 - \text{AUC}$, equation (**??**), to the 188-class run. The class-balanced probes use the checkpoints as trained, so this check uses the natural-composition test set. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

	AUC		ρ	
pretraining	as trained	recomputed	as trained	recomputed
188 classes	$0.9760 \pm <0.0001$	$0.9760 \pm <0.0001$	1.00	1.00
162 classes	0.9748 ± 0.0003	0.9748 ± 0.0003	1.049 ± 0.014	1.050 ± 0.014
43 classes	0.9719 ± 0.0006	0.9719 ± 0.0006	1.171 ± 0.024	1.172 ± 0.025
17 classes	0.9631 ± 0.0006	0.9631 ± 0.0006	1.538 ± 0.025	1.537 ± 0.026
162 classes, mass regression	0.9735, 0.9740	0.9735, 0.9740	1.106, 1.082	1.105, 1.083
17 classes, mass regression	0.9623, 0.9632	0.9623, 0.9632	1.571, 1.530	1.572, 1.530
random partition 1, seed 1	0.972	0.972	1.17	1.17
random partition 2, seed 2	0.972	0.972	1.17	1.17
random partition 3, seed 1	0.972	0.972	1.15	1.15
random partition 4, seed 2	0.972	0.972	1.15	1.15
random partition 5, seed 2	0.972	0.972	1.16	1.16
random partition, no flavor boundary	0.9657, 0.9645	0.9657, 0.9645	1.432, 1.477	1.432, 1.477
random partition, b-quark boundary	0.9698, 0.9701	0.9698, 0.9701	1.260, 1.245	1.260, 1.245
random partition, random flavor boundary	0.9698, 0.9698	0.9698, 0.9698	1.259, 1.254	1.260, 1.254
randomly initialized network	0.8058 ± 0.0018	0.8525 ± 0.0019	–	–

Table 11: Uncertainty of the label-set comparisons at the largest training set, in per cent: the standard deviation of the logarithm of the ratio over the pretraining runs, then that from the finite test sample. JetClass-II: the ratio of $1 - \mathrm{AUC}$, the test-sample part from a bootstrap of the class-balanced test jets (200 resamples within each class). Top and quark/gluon: the ratio of R_{30}, the test-sample part from the number of background jets passing the cut.

	linear probe			fine-tuning		
ratio	JetClass-II, $1 - \mathrm{AUC}$	top, R_{30}	q/g, R_{30}	JetClass-II, $1 - \mathrm{AUC}$	top, R_{30}	q/g, R_{30}
162 classes / 188 classes	0.45 / 0.15	25 / 2.6	24 / 3.5	1.1 / 0.13	3.2 / 13	2.6 / 4.6
43 classes / 188 classes	0.70 / 0.16	29 / 2.4	20 / 3.6	0.20 / 0.13	7.3 / 13	2.8 / 4.6
17 classes / 188 classes	1.3 / 0.21	16 / 2.4	9.8 / 3.6	0.29 / 0.15	9.9 / 13	2.6 / 4.6

Table 12: JetClass-II linear probes with 10^6 training jets on the natural-composition test set and on the class-balanced one: AUC averaged over the classes and the ratio ρ of $1 - \mathrm{AUC}$, equation (**??**), to the 188-class run. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

	natural composition		class-balanced	
pretraining	AUC	ρ	AUC	ρ
188 classes	$0.9760 \pm <0.0001$	1.00	$0.9801 \pm <0.0001$	1.00
162 classes	0.9748 ± 0.0003	1.049 ± 0.014	0.9796 ± 0.0001	1.026 ± 0.005
43 classes	0.9719 ± 0.0006	1.171 ± 0.024	0.9756 ± 0.0002	1.227 ± 0.009
17 classes	0.9631 ± 0.0006	1.538 ± 0.025	0.9691 ± 0.0004	1.557 ± 0.019
162 classes, mass regression	0.9735, 0.9740	1.106, 1.082	0.9791, 0.9794	1.053, 1.037
17 classes, mass regression	0.9623, 0.9632	1.571, 1.530	0.9679, 0.9696	1.617, 1.528
random partition 1, seed 1	0.972	1.17	0.978	1.13
random partition 2, seed 2	0.972	1.17	0.978	1.13
random partition 3, seed 1	0.972	1.15	0.978	1.11
random partition 4, seed 2	0.972	1.15	0.978	1.12
random partition 5, seed 2	0.972	1.16	0.977	1.13
random partition, no flavor boundary	0.9657, 0.9645	1.432, 1.477	0.9717, 0.9710	1.424, 1.462
random partition, b-quark boundary	0.9698, 0.9701	1.260, 1.245	0.9758, 0.9759	1.218, 1.212
random partition, random flavor boundary	0.9698, 0.9698	1.259, 1.254	0.9760, 0.9760	1.206, 1.208
randomly initialized network, BatchNorm recomputed	0.8525 ± 0.0019	–	0.866 ± 0.004	–

Table 13: ParT's top-tagging training script on our cluster, then with Sophon's output head, then also with Sophon's rescaling of every jet's four-momenta to $p_{\mathrm{T}} = 500$ GeV (cumulative, one run each), beside ParT's published row and our network trained from scratch, which uses the rescaling.

	accuracy	AUC	R_{50}	R_{30}	background jets passing at R_{30}
ParT's script, unchanged	0.9398	0.9858	424	1,711	118
+ Sophon's output head	0.9403	0.9859	416	1,577	128
+ Sophon's jet-p_{T} rescaling	0.9385	0.9850	357	1,224	165
ParT, published [?]	0.940	0.9858	413 ± 16	1602 ± 81	–
our network from scratch	0.9387 ± 0.0002	0.9852 ± 0.0001	358 ± 11	$1{,}412 \pm 20$	–

Table 14: Linear probes on the frozen features at the largest training set of each task. Upper part: JetClass-II (class-balanced test set) accuracy and AUC averaged over its 188 classes each against the rest, over the classes with the full 5,000 test jets, over the resonant and over the QCD classes; JetClass accuracy and AUC. Lower part: top tagging and quark/gluon discrimination (also on Herwig jets) with the missing inputs substituted as chosen on validation for every model (random draws for top, the training mean for quark/gluon; column headings): accuracy, AUC and the background rejection $1/\varepsilon_B$ at 50% and 30% signal efficiency. The ratio column is ρ of equation (**??**), $(1 - \text{AUC})$ over that of the 188-class run of the same seed (a random partition: of its seed). Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. The 188-class model with kinematic inputs only, a diagnostic of the missing inputs, has one run and no input missing, and is evaluated on top tagging only. AUC: area under the curve of signal efficiency against background efficiency. *: a probe that did not converge.

	JetClass-II						JetClass	
pretraining	accuracy	AUC	AUC, full count	AUC, resonant	AUC, QCD	ρ	accuracy	AUC
188 classes	0.4348 ± 0.0003	$0.9801 \pm <0.0001$	$0.9813 \pm <0.0001$	$0.9830 \pm <0.0001$	$0.9632 \pm <0.0001$	1.00	0.7709 ± 0.0023	0.9682 ± 0.0006
162 classes	0.4328 ± 0.0008	0.9796 ± 0.0001	0.9810 ± 0.0001	0.9829 ± 0.0001	0.9600 ± 0.0002	1.026 ± 0.005	0.7709 ± 0.0005	0.9683 ± 0.0002
43 classes	0.361 ± 0.005	0.9756 ± 0.0002	0.9769 ± 0.0002	0.9785 ± 0.0002	0.9583 ± 0.0003	1.227 ± 0.009	0.7695 ± 0.0014	0.9680 ± 0.0004
17 classes	0.334 ± 0.007	0.9691 ± 0.0004	0.9708 ± 0.0005	0.9729 ± 0.0005	0.9460 ± 0.0006	1.557 ± 0.019	0.7605 ± 0.0027	0.9650 ± 0.0008
162 classes, mass regression	0.4271, 0.4305	0.9791, 0.9794	0.9805, 0.9808	0.9824, 0.9827	0.9592, 0.9598	1.053, 1.037	0.7718, 0.7760	0.9684, 0.9697
17 classes, mass regression	0.312, 0.350	0.9679, 0.9696	0.9695, 0.9714	0.9716, 0.9737	0.9458, 0.9457	1.617, 1.528	0.7624, 0.7612	0.9655, 0.9651
random partition 1, seed 1	0.420	0.978	0.979	0.981	0.956	1.13	0.753	0.964
random partition 2, seed 2	0.421	0.978	0.979	0.981	0.956	1.13	0.755	0.964
random partition 3, seed 1	0.423	0.978	0.979	0.982	0.956	1.11	0.755	0.964
random partition 4, seed 2	0.423	0.978	0.979	0.981	0.956	1.12	0.754	0.964
random partition 5, seed 2	0.420	0.977	0.979	0.981	0.955	1.13	0.754	0.964
random partition, no flavor boundary	0.3769, 0.3723	0.9717, 0.9710	0.9736, 0.9729	0.9760, 0.9753	0.9460, 0.9449	1.424, 1.462	0.7535, 0.7486	0.9633, 0.9622
random partition, b-quark boundary	0.3945, 0.3963	0.9758, 0.9759	0.9774, 0.9775	0.9794, 0.9795	0.9543, 0.9545	1.218, 1.212	0.7615, 0.7644	0.9661, 0.9665
random partition, random flavor boundary	0.4071, 0.4089	0.9760, 0.9760	0.9777, 0.9777	0.9798, 0.9798	0.9536, 0.9535	1.206, 1.208	0.7593, 0.7564	0.9654, 0.9645
randomly initialized network	0.0724 ± 0.0015	0.8121 ± 0.0019	0.8123 ± 0.0021	0.8152 ± 0.0024	0.794 ± 0.005	–	0.4077 ± 0.0013	0.8280 ± 0.0011
randomly initialized network, BatchNorm recomputed	0.1017 ± 0.0017	0.866 ± 0.004	0.869 ± 0.004	0.874 ± 0.004	0.8232 ± 0.0012	–	0.475 ± 0.005	0.866 ± 0.004

	top missing inputs: random draws				quark/gluon missing inputs: the training mean				quark/gluon, Herwig missing inputs: the training mean			
pretraining	accuracy	AUC	R_{50}	R_{30}	accuracy	AUC	R_{50}	R_{30}	accuracy	AUC	R_{50}	R_{30}
188 classes	0.9112 ± 0.0015	0.9686 ± 0.0007	87 ± 4	228 ± 25	0.8076 ± 0.0004	0.8835 ± 0.0002	26.17 ± 0.35	72.7 ± 2.4	0.7229 ± 0.0033	0.8075 ± 0.0006	11.81 ± 0.10	35.0 ± 2.6
162 classes	0.9098 ± 0.0031	0.9671 ± 0.0020	79 ± 9	199 ± 25	0.8051 ± 0.0015	0.8818 ± 0.0018	25.6 ± 0.4	70.99 ± 0.31	0.7234 ± 0.0022	0.8068 ± 0.0022	11.6 ± 0.4	35.9 ± 1.7
43 classes	0.9101 ± 0.0019	0.9681 ± 0.0013	87.7 ± 3.5	231 ± 8	0.8055 ± 0.0009	0.8821 ± 0.0011	25.7 ± 0.6	70.8 ± 1.8	0.7206 ± 0.0005	0.8058 ± 0.0011	11.52 ± 0.23	34.4 ± 1.7
17 classes	0.9152 ± 0.0010	0.9710 ± 0.0006	97 ± 4	255 ± 17	0.8064 ± 0.0014	0.8821 ± 0.0015	25.8 ± 0.7	74.3 ± 1.9	0.7201 ± 0.0028	0.8053 ± 0.0020	11.50 ± 0.16	35.5 ± 1.0
162 classes, mass regression	0.9166, 0.9123	0.9715, 0.9702	96.8, 103.2	260, 306	0.8075, 0.8073	0.8837, 0.8831	25.69, 25.53	73.9, 70.7	0.7227, 0.7218	0.8064, 0.8078	11.71, 11.94	34.93, 35.60
17 classes, mass regression	0.9186, 0.9206	0.9734, 0.9748	112, 124	306, 378	0.8062, 0.8040	0.8805, 0.8798	25.82, 25.85	68.4, 72.2	0.7256, 0.7240	0.8079, 0.8078	11.93, 11.91	34.7, 36.3
random partition 1, seed 1	0.907	0.965	69.4	174	0.805	0.881	25.0	69.6	0.726	0.809	11.8	36.2
random partition 2, seed 2	0.901	0.962	62.3	151	0.803	0.879	24.5	69.3	0.717	0.803	11.2	34.0
random partition 3, seed 1	0.905	0.963	66.0	164	0.805	0.880	24.0	68.4	0.719	0.804	10.9	32.1
random partition 4, seed 2	0.898	0.960	58.7	152	0.798	0.875	22.8	64.7	0.722	0.805	11.5	33.8
random partition 5, seed 2	0.904	0.963	65.6	173	0.803	0.879	24.1	69.6	0.721	0.802	11.2	34.1
random partition, no flavor boundary	0.9145, 0.9103	0.9708, 0.9683	98, 87	262, 243	0.8056, 0.8006	0.8823, 0.8774	25.55, 24.64	73.9, 71.7	0.7200, 0.7213	0.8058, 0.8055	11.36, 11.59	34.32, 34.93
random partition, b-quark boundary	0.9088, 0.9063	0.9673, 0.9657	84.0, 75.1	202.5, 211.6	0.8043, 0.8025	0.8813, 0.8797	24.97, 24.58	72.52, 72.20	0.7218, 0.7209	0.8057, 0.8060	11.66, 11.47	35.27, 34.60
random partition, random flavor boundary	0.9091, 0.9086	0.9671, 0.9674	83.1, 84.9	214, 233	0.8058, 0.8043	0.8829, 0.8807	25.19, 25.50	71.2, 72.8	0.7201, 0.7232	0.8065, 0.8060	11.60, 11.72	34.99, 35.51
randomly initialized network	0.8867 ± 0.0013	0.9408 ± 0.0012	25.0 ± 0.8	56.8 ± 2.7	0.7851 ± 0.0018	0.8612 ± 0.0016	19.40 ± 0.28	56.4 ± 1.0	0.7205 ± 0.0021	0.7965 ± 0.0009	10.58 ± 0.11	31.6 ± 0.8
randomly initialized network, BatchNorm recomputed	0.8783 ± 0.0021	0.9354 ± 0.0017	22.6 ± 1.0	48.6 ± 3.0	0.7845 ± 0.0013	0.8600 ± 0.0009	18.70 ± 0.18	52.2 ± 0.8	0.7194 ± 0.0034	0.7972 ± 0.0019	10.48 ± 0.21	30.7 ± 0.5
188 classes, kinematic inputs only (diagnostic)	0.916	0.971	94.2	231	–	–	–	–	–	–	–	–

Table 15: Linear probes against the number of training jets (column headers): JetClass-II AUC averaged over the classes, JetClass accuracy, and R_{30} on top, quark/gluon and Herwig jets with the missing inputs substituted as chosen on validation (random draws for top, the training mean for quark/gluon). Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. Here the five random partitions are pooled into one row, mean ± standard deviation over their runs. The 188-class model with kinematic inputs only, a diagnostic of the missing inputs, has one run and no input missing, and is evaluated on top tagging only. *: a probe that did not converge.

JetClass-II, AUC training jets	1,000	10,000	100,000	1,000,000
188 classes	0.9497 ± 0.0001	0.9701 ± 0.0001	0.9783 ± 0.0001	$0.9801 \pm <0.0001$
162 classes	0.9466 ± 0.0006	0.9692 ± 0.0002	0.9778 ± 0.0001	0.9796 ± 0.0001
43 classes	0.9324 ± 0.0029	0.9633 ± 0.0011	0.9735 ± 0.0003	0.9756 ± 0.0002
17 classes	0.9046 ± 0.0021	0.9508 ± 0.0006	0.9660 ± 0.0004	0.9691 ± 0.0004
162 classes, mass regression	0.9445, 0.9435	0.9684, 0.9684	0.9772, 0.9775	0.9791, 0.9794
17 classes, mass regression	0.8999, 0.9044	0.9483, 0.9518	0.9646, 0.9665	0.9679, 0.9696
random partition, no flavor boundary	0.9227, 0.9206	0.9562, 0.9544	0.9691, 0.9682	0.9717, 0.9710
random partition, b-quark boundary	0.9373, 0.9375	0.9647, 0.9648	0.9735, 0.9737	0.9758, 0.9759
random partition, random flavor boundary	0.9346, 0.9338	0.9642, 0.9645	0.9737, 0.9737	0.9760, 0.9760
random partitions (all five)	0.9401 ± 0.0020	0.9669 ± 0.0003	0.9757 ± 0.0001	0.9777 ± 0.0002
randomly initialized network	0.618 ± 0.004	0.718 ± 0.004	0.7946 ± 0.0023	0.8121 ± 0.0019
randomly initialized network, BatchNorm recomputed	0.67 ± 0.04	0.800 ± 0.013	0.855 ± 0.004	0.866 ± 0.004
JetClass, accuracy training jets	1,000	10,000	100,000	1,000,000
188 classes	0.711 ± 0.006	0.7529 ± 0.0008	0.7686 ± 0.0022	0.7709 ± 0.0023
162 classes	0.714 ± 0.008	0.7537 ± 0.0018	0.7687 ± 0.0002	0.7709 ± 0.0005
43 classes	0.7109 ± 0.0005	0.7521 ± 0.0004	0.7679 ± 0.0012	0.7695 ± 0.0014
17 classes	0.689 ± 0.006	0.742 ± 0.004	0.7586 ± 0.0026	0.7605 ± 0.0027
162 classes, mass regression	0.7083, 0.7093	0.7543, 0.7574	0.7694, 0.7741	0.7718, 0.7760
17 classes, mass regression	0.6863, 0.6947	0.7447, 0.7426	0.7606, 0.7595	0.7624, 0.7612
random partition, no flavor boundary	0.6795, 0.6738	0.7339, 0.7303	0.7517, 0.7470	0.7535, 0.7486
random partition, b-quark boundary	0.6995, 0.7020	0.7443, 0.7455	0.7597, 0.7628	0.7615, 0.7644
random partition, random flavor boundary	0.6920, 0.6930	0.7404, 0.7391	0.7575, 0.7546	0.7593, 0.7564
random partitions (all five)	0.6960 ± 0.0027	0.7380 ± 0.0006	0.7524 ± 0.0010	0.7541 ± 0.0009
randomly initialized network	0.306 ± 0.006	0.3854 ± 0.0016	0.4049 ± 0.0016	0.4077 ± 0.0013
randomly initialized network, BatchNorm recomputed	0.363 ± 0.009	0.450 ± 0.005	0.472 ± 0.006	0.475 ± 0.005
top, R_{30} (missing inputs: random draws) training jets	1,000	10,000	100,000	1,200,000
188 classes	162 ± 9	227 ± 18	228 ± 23	228 ± 25
162 classes	130 ± 8	199 ± 21	203 ± 26	199 ± 25
43 classes	146 ± 32	233 ± 18	236 ± 10	231 ± 8
17 classes	154 ± 5	252 ± 19	259 ± 20	255 ± 17
162 classes, mass regression	199, 186	248, 306	266, 312	260, 306
17 classes, mass regression	193.0, 191.0	286, 341	306, 379	306, 378
random partition, no flavor boundary	158, 138	265, 230	262, 248	262, 243
random partition, b-quark boundary	139.3, 139.6	209.2, 218.0	204, 217	202.5, 211.6
random partition, random flavor boundary	138.7, 148.5	204, 222	218, 242	214, 233
random partitions (all five)	101 ± 10	165 ± 11	165 ± 12	163 ± 11
randomly initialized network	43.1 ± 1.1	55.3 ± 3.1	57.5 ± 3.0	56.8 ± 2.7
randomly initialized network, BatchNorm recomputed	36.9 ± 2.6	49.0 ± 3.3	49.2 ± 3.1	48.6 ± 3.0
188 classes, kinematic inputs only (diagnostic)	133	208	222	231

continued

quark/gluon, R_{30} (missing inputs: the training mean)				
training jets	1,000	10,000	100,000	1,600,000
188 classes	66.5 ± 1.7	70.5 ± 1.8	73.4 ± 2.0	72.7 ± 2.4
162 classes	60.2 ± 1.6	68.9 ± 0.9	71.5 ± 0.8	70.99 ± 0.31
43 classes	63.4 ± 2.7	66.9 ± 0.6	70.8 ± 1.4	70.8 ± 1.8
17 classes	62 ± 4	72.0 ± 1.8	74.6 ± 1.5	74.3 ± 1.9
162 classes, mass regression	67.8, 60.6	70.1, 66.4	74.3, 71.9	73.9, 70.7
17 classes, mass regression	66.8, 60.7	64.6, 69.3	68.9, 70.9	68.4, 72.2
random partition, no flavor boundary	57.2, 54.7	70.2, 69.1	72.52, 72.15	73.9, 71.7
random partition, b-quark boundary	64.6, 56.4	68.6, 67.2	73.4, 70.8	72.52, 72.20
random partition, random flavor boundary	59.3, 64.2	68.6, 70.4	71.5, 72.6	71.2, 72.8
random partitions (all five)	55 ± 5	64.8 ± 2.3	68.3 ± 2.2	68.3 ± 2.1
randomly initialized network	43.0 ± 2.5	55.6 ± 1.2	56.7 ± 0.5	56.4 ± 1.0
randomly initialized network, BatchNorm recomputed	42.0 ± 2.3	51.1 ± 1.4	52.2 ± 1.3	52.2 ± 0.8

quark/gluon, Herwig, R_{30} (missing inputs: the training mean)				
training jets	1,000	10,000	100,000	1,600,000
188 classes	32.99 ± 0.13	34.5 ± 1.2	34.5 ± 2.4	35.0 ± 2.6
162 classes	31.3 ± 2.5	35.0 ± 2.2	35.8 ± 1.9	35.9 ± 1.7
43 classes	31.8 ± 0.8	32.4 ± 1.7	34.6 ± 1.8	34.4 ± 1.7
17 classes	31.0 ± 2.8	34.5 ± 1.1	35.3 ± 1.0	35.5 ± 1.0
162 classes, mass regression	34.7, 31.7	33.98, 33.85	34.55, 35.47	34.93, 35.60
17 classes, mass regression	33.7, 32.4	33.46, 33.94	34.5, 35.6	34.7, 36.3
random partition, no flavor boundary	26.39, 27.23	34.19, 34.20	34.01, 35.00	34.32, 34.93
random partition, b-quark boundary	32.6, 28.3	33.68, 32.86	34.82, 34.11	35.27, 34.60
random partition, random flavor boundary	29.8, 31.6	33.3, 34.8	34.79, 35.16	34.99, 35.51
random partitions (all five)	28.9 ± 1.8	33.0 ± 1.4	33.7 ± 1.5	34.0 ± 1.4
randomly initialized network	25.2 ± 1.0	30.8 ± 0.4	31.9 ± 1.2	31.6 ± 0.8
randomly initialized network, BatchNorm recomputed	24.7 ± 2.0	29.7 ± 0.5	30.5 ± 0.6	30.7 ± 0.5

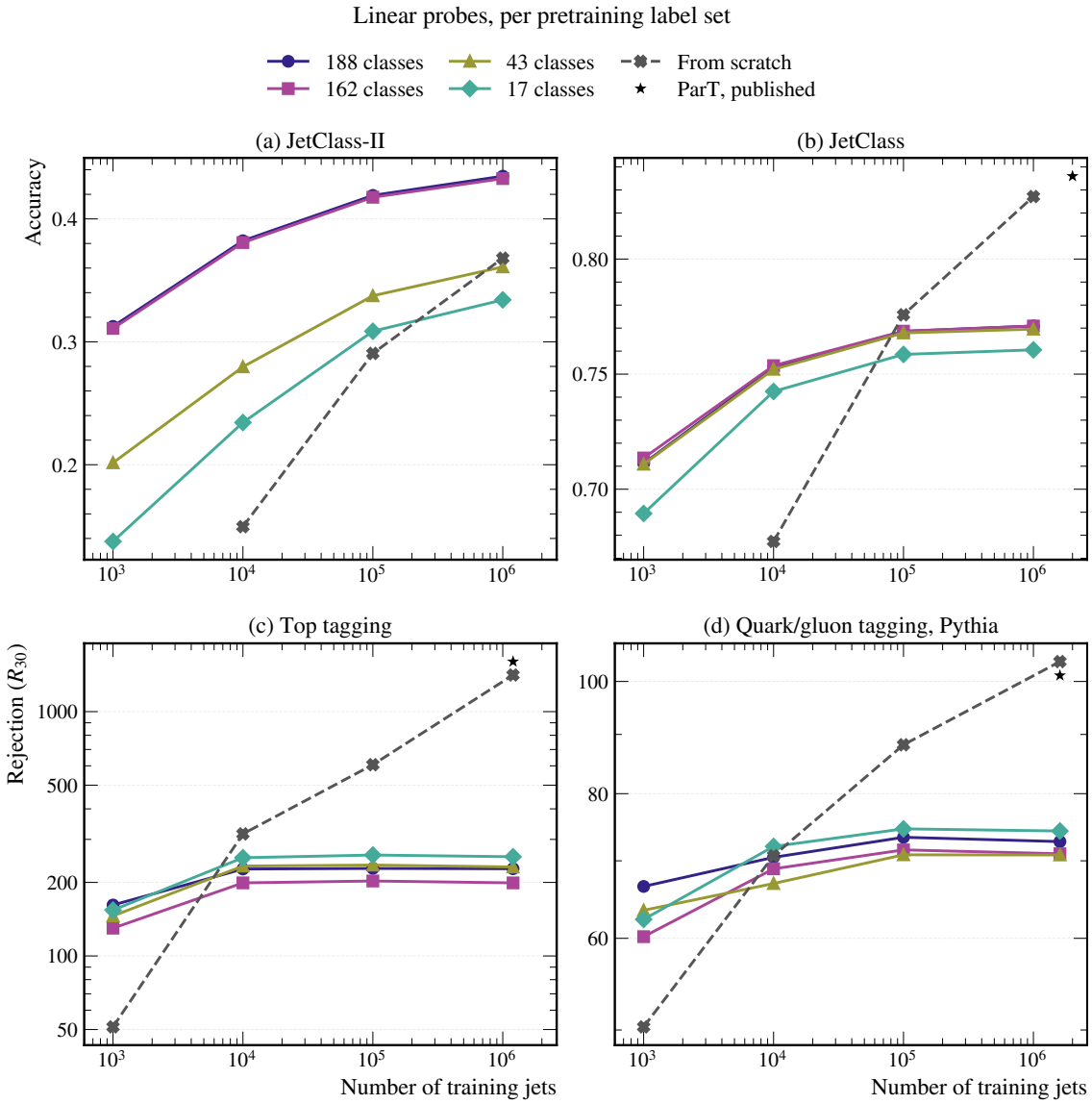


Figure 16: Linear probes of the four label sets against the number of training jets: accuracy on class-balanced JetClass-II and on JetClass, and rejection R_{30} at 30% signal efficiency on top and quark/gluon tagging, with a network trained from scratch dashed. Points are means over three runs; overlapping markers agree within their size. Stars are published ParT: same inputs on JetClass, kinematic only on top, kinematic and particle type on quark/gluon.

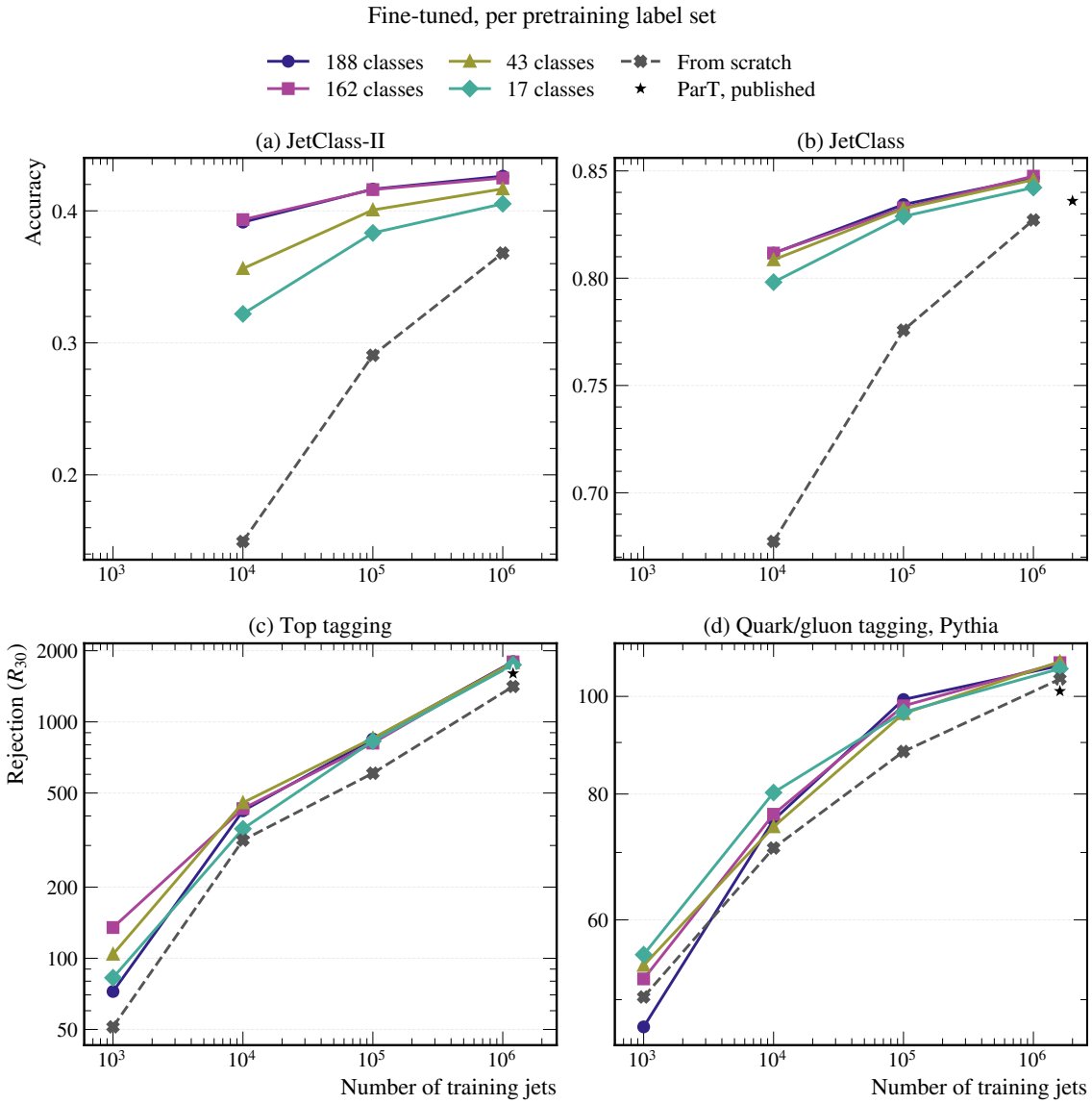


Figure 17: Fine-tuned models of the four label sets against the number of training jets: accuracy on class-balanced JetClass-II and on JetClass, and rejection R_{30} at 30% signal efficiency on top and quark/gluon tagging, with a network trained from scratch dashed. Points are means over three runs; overlapping markers agree within their size. Fine-tuning on JetClass-II and JetClass starts at 10^4 jets. Stars are published ParT: same inputs on JetClass, kinematic only on top, kinematic and particle type on quark/gluon.

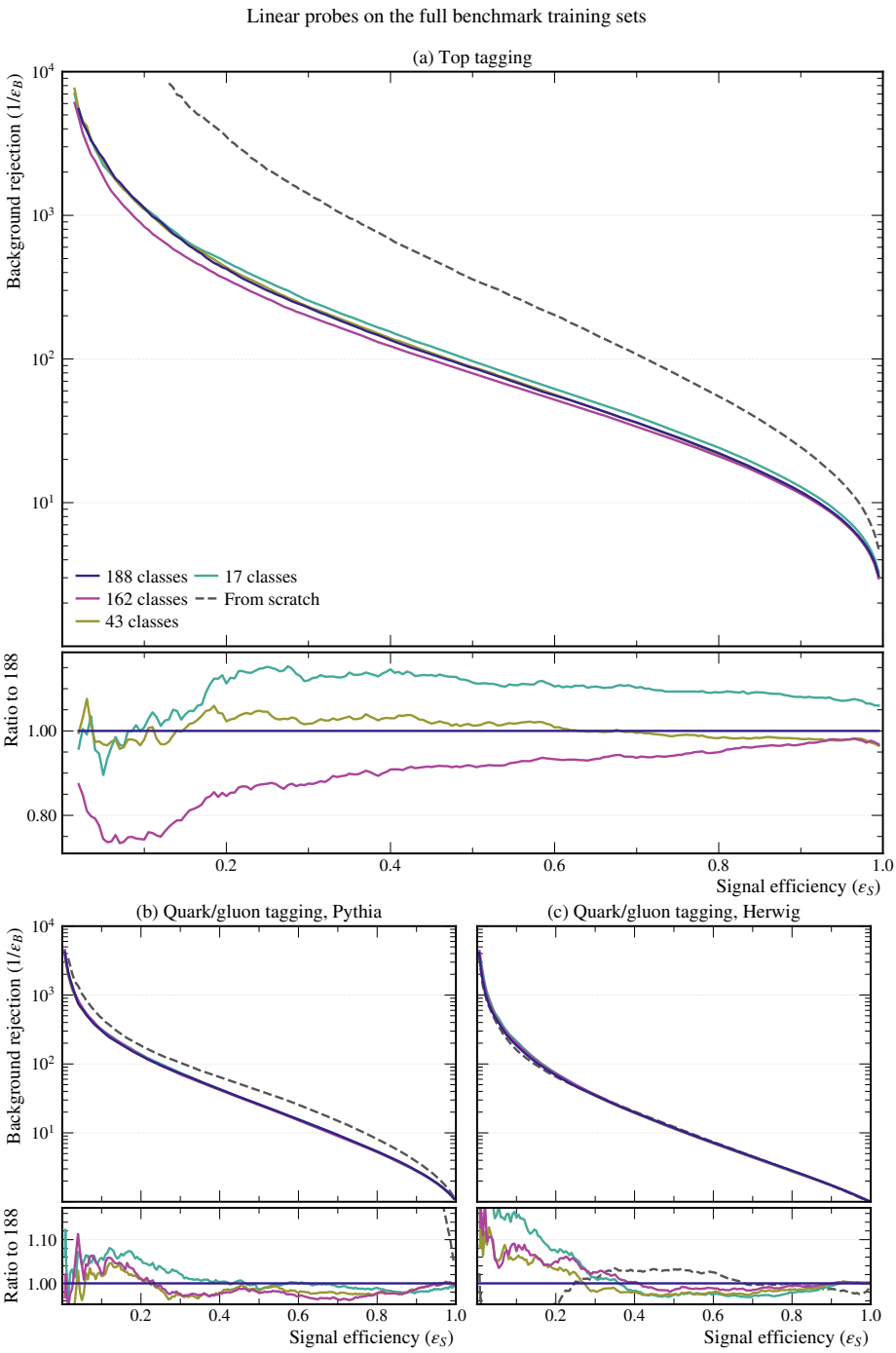


Figure 18: Background rejection against signal efficiency from linear probes on the full training sets (missing inputs filled by random draws on top tagging, by the training mean on quark/gluon), on networks pretrained with 188, 162, 43 and 17 classes, and for reference the network fully trained from scratch (three runs each). Curves end where fewer than 20 background jets remain. Lower panels: ratio to the 188-class probe; from scratch falls outside it in (a), (b) and partly (c).

Table 16: Fine-tuning against the number of training jets (column headers), in each task's metric: AUC averaged over the classes on class-balanced JetClass-II (all classes and the QCD classes), accuracy on JetClass, and R_{30} on top, quark/gluon and Herwig jets. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. Each pretrained run is fine-tuned once; from scratch: three training seeds, on the same jets for top and quark/gluon and for JetClass below 10^6 jets, and on 10^6 JetClass jets each on its own training subset. The last column of the JetClass-II block is ρ of equation (**??**) at 10^6 training jets, $1 - \text{AUC}$ over that of the 188-class run of the same seed. The 188-class model with kinematic inputs only, a diagnostic of the missing inputs, has one run and no input missing, and is evaluated on top tagging only.

JetClass-II, AUC training jets	10,000	100,000	1,000,000	ρ, 10^6 jets
188 classes	0.9722 ± 0.0002	0.9778 ± 0.0002	0.9797 ± 0.0001	1
162 classes	0.9725 ± 0.0001	0.9775 ± 0.0001	0.9794 ± 0.0002	1.014 ± 0.011
43 classes	0.9669 ± 0.0006	0.9763 ± 0.0001	0.9788 ± 0.0001	1.043 ± 0.002
17 classes	0.9623 ± 0.0003	0.9743 ± 0.0002	0.9772 ± 0.0001	1.123 ± 0.003
162 classes, mass regression	0.9721, 0.9724	0.9769, 0.9771	0.9790, 0.9792	1.036, 1.024
17 classes, mass regression	0.9625, 0.9627	0.9737, 0.9737	0.9766, 0.9769	1.152, 1.139
random partition 1, seed 1	**[no run yet]**	**[no run yet]**	0.978	1.07
random partition 2, seed 2	**[no run yet]**	**[no run yet]**	0.978	1.07
random partition 3, seed 1	**[no run yet]**	**[no run yet]**	0.979	1.06
random partition 4, seed 2	**[no run yet]**	**[no run yet]**	0.979	1.05
random partition 5, seed 2	**[no run yet]**	**[no run yet]**	0.978	1.07
random partition, no flavor boundary	**[no run yet]**	**[no run yet]**	0.9776, 0.9775	1.105, 1.108
random partition, b-quark boundary	**[no run yet]**	**[no run yet]**	0.9778, 0.9782	1.094, 1.072
random partition, random flavor boundary	**[no run yet]**	**[no run yet]**	0.9779, 0.9781	1.089, 1.076
from scratch	0.9196 ± 0.0005	0.9611 ± 0.0011	0.9732 ± 0.0004	–

JetClass-II, QCD classes, AUC over the QCD classes training jets	10,000	100,000	1,000,000
188 classes	0.9498 ± 0.0003	0.9598 ± 0.0002	$0.9631 \pm <0.0001$
162 classes	0.9479 ± 0.0001	0.9574 ± 0.0002	0.9618 ± 0.0004
43 classes	0.9416 ± 0.0011	0.9573 ± 0.0001	0.9613 ± 0.0002
17 classes	0.9368 ± 0.0014	0.9543 ± 0.0003	0.9586 ± 0.0001
162 classes, mass regression	0.9463, 0.9481	0.9566, 0.9571	0.9610, 0.9612
17 classes, mass regression	0.9335, 0.9349	0.9533, 0.9531	0.9580, 0.9581
random partition 1, seed 1	**[no run yet]**	**[no run yet]**	0.960
random partition 2, seed 2	**[no run yet]**	**[no run yet]**	0.960
random partition 3, seed 1	**[no run yet]**	**[no run yet]**	0.961
random partition 4, seed 2	**[no run yet]**	**[no run yet]**	0.961
random partition 5, seed 2	**[no run yet]**	**[no run yet]**	0.960
random partition, no flavor boundary	**[no run yet]**	**[no run yet]**	0.9592, 0.9589
random partition, b-quark boundary	**[no run yet]**	**[no run yet]**	0.9593, 0.9607
random partition, random flavor boundary	**[no run yet]**	**[no run yet]**	0.9595, 0.9603
from scratch	0.8873 ± 0.0010	0.9354 ± 0.0019	0.9530 ± 0.0006

JetClass, accuracy training jets	10,000	100,000	1,000,000
188 classes	0.8117 ± 0.0031	0.8343 ± 0.0035	0.8468 ± 0.0006
162 classes	0.8117 ± 0.0013	0.8328 ± 0.0005	0.8475 ± 0.0009
43 classes	0.8086 ± 0.0031	0.8324 ± 0.0004	0.8459 ± 0.0002
17 classes	0.7982 ± 0.0023	0.8288 ± 0.0005	0.8422 ± 0.0005
162 classes, mass regression	0.8107, 0.8098	0.8326, 0.8348	0.8459, 0.8471
17 classes, mass regression	0.7944, 0.7990	0.8253, 0.8262	0.8400, 0.8398
random partition 1, seed 1	0.801	0.830	0.843
random partition 2, seed 2	0.806	0.831	0.844
random partition 3, seed 1	0.799	0.831	0.845
random partition 4, seed 2	0.802	0.829	0.845
random partition 5, seed 2	0.800	0.831	0.845
random partition, no flavor boundary	0.8000, 0.7993	0.8283, 0.8294	0.8434, 0.8431
random partition, b-quark boundary	0.8058, 0.8032	0.8317, 0.8301	0.8434, 0.8443

continued

random partition, random flavor boundary	0.8021, 0.8015	0.8300, 0.8319	0.8432, 0.8436	
from scratch	0.677 ± 0.004	0.776 ± 0.005	0.8271 ± 0.0012	
top, R_{30}				
training jets	1,000	10,000	100,000	1,200,000
188 classes	72 ± 19	420 ± 9	850 ± 90	$1{,}800 \pm 60$
162 classes	135 ± 26	430 ± 60	810 ± 60	$1{,}790 \pm 60$
43 classes	104 ± 20	455 ± 30	850 ± 80	$1{,}780 \pm 150$
17 classes	83 ± 25	350 ± 40	830 ± 60	$1{,}740 \pm 140$
162 classes, mass regression	105.9, 112.7	373.2, 376.7	848, 848	1,941, 1,590
17 classes, mass regression	59.1, 63.9	365.8, 359.9	856, 774	1,683, 1,628
random partition 1, seed 1	102	330	711	1,660
random partition 2, seed 2	82.4	394	935	1,900
random partition 3, seed 1	80.1	370	789	1,640
random partition 4, seed 2	80.3	370	742	1,820
random partition 5, seed 2	62.1	291	952	1,740
random partition, no flavor boundary	107, 167	401, 380	841, 777	1,726, 1,852
random partition, b-quark boundary	116, 100	454, 316	878, 926	1,590, 2,060
random partition, random flavor boundary	56, 85	388, 347	783, 756	1,565, 1,655
from scratch	51 ± 5	320 ± 50	610 ± 80	$1{,}412 \pm 20$
188 classes, kinematic inputs only (diagnostic)	92.4	397	1,070	1,800
q/g, R_{30}				
training jets	1,000	10,000	100,000	1,600,000
188 classes	47 ± 10	75 ± 7	99.3 ± 1.4	107.3 ± 2.2
162 classes	52.4 ± 3.1	76.4 ± 0.6	97.9 ± 3.3	108.0 ± 1.4
43 classes	54 ± 6	74 ± 5	96.1 ± 1.8	108.3 ± 1.0
17 classes	55 ± 5	80.3 ± 1.1	96.4 ± 1.0	106.5 ± 0.8
162 classes, mass regression	59.3, 57.1	80.1, 72.6	95.6, 97.6	107.0, 106.6
17 classes, mass regression	54.2, 57.7	79.5, 77.9	97.85, 97.94	104.9, 105.6
random partition 1, seed 1	45.1	80.4	98.9	107
random partition 2, seed 2	52.0	81.4	99.6	109
random partition 3, seed 1	43.8	76.6	96.5	106
random partition 4, seed 2	52.5	79.5	99.1	107
random partition 5, seed 2	44.7	58.6	95.2	106
random partition, no flavor boundary	50.9, 58.5	79.2, 86.1	98.91, 99.90	108.9, 109.2
random partition, b-quark boundary	56.4, 55.3	84.2, 81.8	100.0, 96.8	106.5, 106.8
random partition, random flavor boundary	46.4, 51.9	68, 79	97.3, 98.7	108.2, 104.4
from scratch	50.3 ± 2.1	70.7 ± 2.8	88.2 ± 1.1	104.0 ± 1.0
q/g, Herwig, R_{30}				
training jets	1,000	10,000	100,000	1,600,000
188 classes	21 ± 6	27.9 ± 2.8	34.3 ± 1.0	35.2 ± 2.1
162 classes	26.2 ± 0.9	27.4 ± 0.5	35.5 ± 0.5	34.6 ± 2.0
43 classes	26.9 ± 2.7	28.4 ± 2.1	35.5 ± 2.1	36.1 ± 2.7
17 classes	25.2 ± 2.5	32.7 ± 1.8	34.9 ± 1.1	36.9 ± 0.9
162 classes, mass regression	28.0, 29.0	34.3, 28.5	34.0, 36.5	37.8, 34.4
17 classes, mass regression	25.8, 27.4	34.4, 31.2	35.9, 39.0	37.7, 32.9
random partition 1, seed 1	23.9	30.5	34.2	33.4
random partition 2, seed 2	25.2	34.1	35.2	35.2
random partition 3, seed 1	23.5	30.2	34.0	34.8
random partition 4, seed 2	24.6	34.4	33.9	37.5
random partition 5, seed 2	19.7	23.2	36.3	35.3
random partition, no flavor boundary	24.2, 27.7	31.6, 33.3	37.0, 35.1	34.1, 39.3
random partition, b-quark boundary	27.61, 26.89	35.2, 33.5	34.9, 37.0	38.1, 34.7
random partition, random flavor boundary	21.2, 25.3	24.7, 34.4	32.9, 35.1	34.9, 37.4
from scratch	27.3 ± 1.2	32.6 ± 2.9	37.3 ± 1.7	35.5 ± 1.2

Table 17: Fine-tuning on the full training sets of top and quark/gluon (also scored on Herwig jets), every configuration: accuracy, AUC and $1/\varepsilon_B$ at 50% and 30% signal efficiency. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. Every rejection rests on at least 98 background jets passing the cut.

pretraining	top				quark/gluon				quark/gluon, Herwig			
	accuracy	AUC	R_{50}	R_{30}	accuracy	AUC	R_{50}	R_{30}	accuracy	AUC	R_{50}	R_{30}
188 classes	0.9400 ± 0.0002	0.9861 ± 0.0001	461 ± 4	$1{,}800 \pm 60$	0.8433 ± 0.0002	0.9148 ± 0.0001	42.56 ± 0.10	107.3 ± 2.2	0.712 ± 0.005	0.8078 ± 0.0023	12.3 ± 0.4	35.2 ± 2.1
162 classes	0.9401 ± 0.0002	0.9861 ± 0.0001	443 ± 17	$1{,}790 \pm 60$	$0.8432 \pm <0.0001$	0.9148 ± 0.0001	42.39 ± 0.22	108.0 ± 1.4	0.7100 ± 0.0017	0.8074 ± 0.0010	12.33 ± 0.28	34.6 ± 2.0
43 classes	0.9402 ± 0.0004	0.9861 ± 0.0001	450 ± 13	$1{,}780 \pm 150$	0.8430 ± 0.0005	0.9146 ± 0.0003	42.2 ± 0.8	108.3 ± 1.0	0.713 ± 0.006	0.809 ± 0.005	12.5 ± 0.6	36.1 ± 2.7
17 classes	0.9398 ± 0.0001	0.9859 ± 0.0001	429 ± 10	$1{,}740 \pm 140$	0.8427 ± 0.0003	$0.9144 \pm <0.0001$	41.97 ± 0.25	106.5 ± 0.8	0.7175 ± 0.0033	0.8103 ± 0.0018	12.79 ± 0.20	36.9 ± 0.9
162 classes, mass regression	0.9406, 0.9402	0.9860, 0.9860	420, 436	1,941, 1,590	0.8425, 0.8427	0.9147, 0.9145	42.46, 42.21	107.0, 106.6	0.7178, 0.7121	0.8109, 0.8074	12.99, 12.21	37.8, 34.4
17 classes, mass regression	0.9397, 0.9399	0.9858, 0.9859	435, 421	1,683, 1,628	0.8425, 0.8424	0.9142, 0.9140	41.77, 41.39	104.9, 105.6	0.7160, 0.7107	0.8101, 0.8052	12.58, 11.96	37.7, 32.9
random partition 1, seed 1	0.940	0.986	432	1,660	0.843	0.915	42.1	107	0.712	0.807	12.2	33.4
random partition 2, seed 2	0.940	0.986	430	1,900	0.843	0.915	42.5	109	0.712	0.808	12.3	35.2
random partition 3, seed 1	0.940	0.986	421	1,640	0.843	0.914	42.1	106	0.711	0.807	12.2	34.8
random partition 4, seed 2	0.940	0.986	458	1,820	0.843	0.915	41.8	107	0.714	0.809	12.7	37.5
random partition 5, seed 2	0.940	0.986	431	1,740	0.843	0.915	42.1	106	0.713	0.808	12.2	35.3
random partition, no flavor boundary	0.9402, 0.9401	0.9860, 0.9861	427, 463	1,726, 1,852	0.8423, 0.8434	0.9144, 0.9146	41.67, 41.95	108.9, 109.2	0.7091, 0.7121	0.8045, 0.8094	12.07, 12.92	34.1, 39.3
random partition, b-quark boundary	0.9401, 0.9405	0.9860, 0.9861	432, 444	1,590, 2,060	0.8430, 0.8431	0.9147, 0.9147	42.43, 42.77	106.5, 106.8	0.7153, 0.7110	0.8109, 0.8066	12.93, 12.19	38.1, 34.7
random partition, random flavor boundary	0.9397, 0.9406	0.9858, 0.9861	394, 447	1,565, 1,655	0.8427, 0.8428	0.9145, 0.9147	42.12, 42.09	108.2, 104.4	0.7114, 0.7171	0.8047, 0.8106	12.18, 12.76	34.9, 37.4
from scratch	0.9387 ± 0.0002	0.9852 ± 0.0001	358 ± 11	$1{,}412 \pm 20$	0.8397 ± 0.0007	0.9121 ± 0.0002	41.47 ± 0.29	104.0 ± 1.0	0.7153 ± 0.0024	0.8061 ± 0.0007	12.18 ± 0.11	35.5 ± 1.2
188 classes, kinematic inputs only	0.941	0.986	443	1,800	–	–	–	–	–	–	–	–

Table 18: Every pretraining run's values at the largest training set: linear probe and fine-tuning. Linear probes on top and quark/gluon with the missing inputs substituted as chosen on validation (random draws for top, the training mean for quark/gluon); fine-tuning with them set to zero.

run	linear probe				fine-tuning				
	JetClass-II AUC	JetClass AUC	top R_{30}	q/g R_{30}	JetClass-II AUC	JetClass accuracy	top R_{30}	q/g R_{30}	Herwig R_{30}
188 classes, seed 1	0.9801	0.9683	203	75.5	0.9797	0.8475	1,740	108	33.7
188 classes, seed 2	0.9801	0.9687	226	71.7	0.9797	0.8463	1,850	109	37.5
188 classes, seed 3	0.9801	0.9675	253	71.0	0.9796	0.8467	1,800	105	34.2
162 classes, seed 1	0.9795	0.9682	177	71.3	0.9792	0.8479	1,740	110	33.3
162 classes, seed 2	0.9797	0.9683	194	70.9	0.9795	0.8482	1,790	107	36.9
162 classes, seed 3	0.9796	0.9685	226	70.7	0.9794	0.8465	1,850	108	33.5
43 classes, seed 1	0.9758	0.9678	223	72.0	0.9788	0.8459	1,760	107	33.1
43 classes, seed 2	0.9755	0.9678	235	68.8	0.9789	0.8461	1,940	108	38.2
43 classes, seed 3	0.9755	0.9685	237	71.7	0.9787	0.8457	1,640	109	37.1
17 classes, seed 1	0.9695	0.9658	236	72.1	0.9772	0.8418	1,870	106	37.5
17 classes, seed 2	0.9689	0.9643	260	75.8	0.9772	0.8421	1,770	106	37.4
17 classes, seed 3	0.9688	0.9651	268	74.9	0.9771	0.8428	1,590	107	35.8
162 classes, mass regression, seed 1	0.9791	0.9684	260	73.9	0.9790	0.8459	1,940	107	37.8
162 classes, mass regression, seed 2	0.9794	0.9697	306	70.7	0.9792	0.8471	1,590	107	34.4
17 classes, mass regression, seed 1	0.9679	0.9655	306	68.4	0.9766	0.8400	1,680	105	37.7
17 classes, mass regression, seed 2	0.9696	0.9651	378	72.2	0.9769	0.8398	1,630	106	32.9
random partition 1, seed 1	0.9776	0.9636	174	69.6	0.9783	0.8435	1,660	107	33.4
random partition 2, seed 2	0.9776	0.9641	151	69.3	0.9783	0.8436	1,900	109	35.2
random partition 3, seed 1	0.9779	0.9638	164	68.4	0.9786	0.8448	1,640	106	34.8
random partition 4, seed 2	0.9778	0.9636	152	64.7	0.9786	0.8449	1,820	107	37.5
random partition 5, seed 2	0.9775	0.9639	173	69.6	0.9783	0.8448	1,740	106	35.3
random partition, no flavor boundary, seed 1	0.9717	0.9633	262	73.9	0.9776	0.8434	1,730	109	34.1
random partition, no flavor boundary, seed 2	0.9710	0.9622	243	71.7	0.9775	0.8431	1,850	109	39.3
random partition, *b*-quark boundary, seed 1	0.9758	0.9661	203	72.5	0.9778	0.8434	1,590	106	38.1
random partition, *b*-quark boundary, seed 2	0.9759	0.9665	212	72.2	0.9782	0.8443	2,060	107	34.7
random partition, random flavor boundary, seed 1	0.9760	0.9654	214	71.2	0.9779	0.8432	1,570	108	34.9
random partition, random flavor boundary, seed 2	0.9760	0.9645	233	72.8	0.9781	0.8436	1,660	104	37.4
188 classes, kinematic inputs only	–	–	231	–	–	–	1,800	–	–
from scratch, seed 1	–	–	–	–	0.9736	0.8277	1,410	103	34.9
from scratch, seed 2	–	–	–	–	0.9729	0.8280	1,430	105	36.9
from scratch, seed 3	–	–	–	–	0.9731	0.8258	1,390	104	34.7

Table 19: JetClass fine-tuning as table 2, every configuration. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. Bold and † as there.

				rejection of QCD								
	jets	accuracy	AUC	$H \to b\bar{b}$ R_{50}	$H \to c\bar{c}$ R_{50}	$H \to gg$ R_{50}	$H \to 4q$ R_{50}	$H \to \ell\nu qq'$ R_{99}	$t \to bqq'$ R_{50}	$t \to b\ell\nu$ $R_{99.5}$	$W \to qq'$ R_{50}	$Z \to q\bar{q}$ R_{50}
from scratch	1 M	0.8271 ± 0.0012	0.9819 ± 0.0002	$4{,}920 \pm 190$	$1{,}580 \pm 60$	88.3 ± 3.2	670 ± 40	$1{,}270 \pm 100$	$6{,}100 \pm 600$	$3{,}940 \pm 320$	272 ± 6	215 ± 8
188 classes	1 M	0.8468 ± 0.0006	0.9854 ± 0.0001	$6{,}600 \pm 330$	$\mathbf{2{,}650 \pm 260}$	$\mathbf{104 \pm 4}$	$1{,}091 \pm 15$	$\mathbf{3{,}100 \pm 400}$	$12{,}900 \pm 1{,}600^\dagger$	$9{,}400 \pm 1{,}500^\dagger$	374 ± 8	$\mathbf{312 \pm 6}$
162 classes	1 M	$\mathbf{0.8475 \pm 0.0009}$	$\mathbf{0.9856 \pm <0.0001}$	6,060	$\mathbf{2{,}820 \pm 40}$	101.8 ± 1.5	$\mathbf{1{,}150 \pm 70}$	$\mathbf{3{,}070 \pm 330}$	$\mathbf{14{,}100 \pm 3{,}500}^\dagger$	$7{,}800 \pm 340$	$\mathbf{389 \pm 9}$	307.7 ± 1.2
43 classes	1 M	0.8459 ± 0.0002	0.9853 ± 0.0001	$6{,}500 \pm 500$	$2{,}610 \pm 160$	103.5 ± 1.3	$1{,}030 \pm 40$	$\mathbf{3{,}160 \pm 280}$	$14{,}200 \pm 2{,}100^\dagger$	$8{,}450 \pm 210$	376 ± 6	$\mathbf{307 \pm 7}$
17 classes	1 M	0.8422 ± 0.0005	0.9847 ± 0.0001	$6{,}670 \pm 220$	$2{,}390 \pm 40$	101 ± 6	$1{,}049 \pm 17$	$2{,}800 \pm 200$	$10{,}600 \pm 2{,}700^\dagger$	$8{,}600 \pm 1{,}300$	$\mathbf{378 \pm 22}$	285 ± 11
162 classes, mass regression	1 M	0.8459, 0.8471	0.9852, 0.9854	6,452, 6,667	2,532, 2,299	105.3, 101.9	1,099, 1,130	2,778, 2,941	11,111, $14{,}286^\dagger$	8,333, 6,897	**392, 380**	302.6, 300.8
17 classes, mass regression	1 M	0.8400, 0.8398	0.9844, 0.9843	6,452, 5,556	2,299, 2,020	105.5, 102.7	966, 935	2,941, 2,899	10,526, $14{,}286^\dagger$	**8,000, 11,111**†	363, 353	289, 276
random partitions (all five)	1 M	0.8443 ± 0.0007	0.9851 ± 0.0001	$6{,}100 \pm 900$	$2{,}580 \pm 160$	102.6 ± 1.6	$1{,}006 \pm 31$	$2{,}690 \pm 300$	$10{,}900 \pm 2{,}000^\dagger$	$7{,}700 \pm 1{,}300$	$\mathbf{379 \pm 18}$	299 ± 7
random partition, no flavor boundary	1 M	0.8434, 0.8431	0.9849, 0.9848	6,060, 6,060	2,151, 2,174	104.7, 106.3	**1,176, 1,036**	2,820, 2,820	**13,333, 16,667**†	6,667, 7,143	351, 351	291.5, 292.0
random partition, b-quark boundary	1 M	0.8434, 0.8443	0.9849, 0.9850	**5,714, 7,407**	2,353, 2,410	**104.9, 108.0**	1,042, 1,031	2,564, 2,597	12,500, $14{,}286^\dagger$	7,140, 7,140	366, 347	292.4, 297.6
random partition, random flavor boundary	1 M	0.8432, 0.8436	0.9849, 0.9849	6,452, 5,882	**2,857, 2,632**	106.0, 96.8	1,000, 1,047	2,817, 2,667	11,765, $13{,}333^\dagger$	7,143, 7,407	371.7, 371.1	295.9, 301.2

Table 20: JetClass linear probes on the frozen features, largest training set, in the columns of table 2 (AUC one against one where the fit records it). Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. † as there.

			rejection of QCD								
pretraining	accuracy	AUC	$H \to b\bar{b}$	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$	$H \to \ell\nu qq'$	$t \to bqq'$	$t \to b\ell\nu$	$W \to qq'$	$Z \to q\bar{q}$
188 classes	0.7709 ± 0.0023	0.9682 ± 0.0006	840 ± 90	382 ± 34	20.0 ± 1.0	200 ± 15	920 ± 40	$1{,}100 \pm 80$	$2{,}690 \pm 310$	76.1 ± 1.7	63.1 ± 2.5
162 classes	0.7709 ± 0.0005	0.9683 ± 0.0002	800 ± 130	410 ± 50	20.7 ± 0.8	207 ± 15	930 ± 50	$1{,}220 \pm 150$	$3{,}400 \pm 600$	81.6 ± 2.1	65.8 ± 3.2
43 classes	0.7695 ± 0.0014	0.9680 ± 0.0004	880 ± 180	440 ± 31	20.8 ± 0.7	207 ± 13	920 ± 70	$1{,}300 \pm 200$	$3{,}700 \pm 500$	79.1 ± 1.6	69.5 ± 0.9
17 classes	0.7605 ± 0.0027	0.9650 ± 0.0008	831 ± 16	294 ± 20	21.6 ± 0.7	183 ± 12	810 ± 110	$1{,}180 \pm 310$	$3{,}090 \pm 280$	71 ± 5	68.9 ± 2.8
162 classes, mass regression	0.7718, 0.7760	0.9684, 0.9697	823, 806	377, 413	19.9, 22.5	210, 243	901, 957	980, 1,183	5,000, 4,878	77.9, 72.9	69.1, 67.6
17 classes, mass regression	0.7624, 0.7612	0.9655, 0.9651	637, 926	287, 261	21.90, 22.58	169.6, 171.2	781, 800	873, 813	1,923, 2,381	69.7, 65.8	70.9, 68.1
random partition 1, seed 1	0.753	0.964	712	323	18.3	161	557	847	2,250	67.2	56.1
random partition 2, seed 2	0.755	0.964	658	357	18.7	186	667	664	1,550	71.3	57.7
random partition 3, seed 1	0.755	0.964	877	382	18.5	180	617	1,120	2,020	72.4	58.7
random partition 4, seed 2	0.754	0.964	797	348	17.7	154	662	858	2,500	70.1	56.5
random partition 5, seed 2	0.754	0.964	749	353	16.5	164	610	840	2,330	69.0	56.0
random partition, no flavor boundary	0.7535, 0.7486	0.9633, 0.9622	775, 508	233, 244	21.5, 19.2	157.4, 154.2	683, 712	889, 909	1,449, 2,667	65.5, 69.2	69.2, 68.2
random partition, b-quark boundary	0.7615, 0.7644	0.9661, 0.9665	990, 1,099	398, 301	20.6, 18.0	205, 195	722, 690	837, 1,058	1,724, 2,222	75.0, 70.6	67.8, 66.1
random partition, random flavor boundary	0.7593, 0.7564	0.9654, 0.9645	758, 662	332, 318	18.86, 18.80	184.7, 174.7	660, 685	1,117, 758	2,083, 1,818	72.3, 75.0	66.29, 65.64
randomly initialized network	0.4077 ± 0.0013	0.8280 ± 0.0011	25.3 ± 1.1	20.4 ± 1.6	17.9 ± 0.8	15.8 ± 1.2	1.452 ± 0.035	20.0 ± 1.4	1.204 ± 0.015	17.3 ± 1.3	14.3 ± 1.0
randomly initialized network, BatchNorm recomputed	0.475 ± 0.005	0.866 ± 0.004	46.3 ± 2.3	13.7 ± 1.2	15.4 ± 0.5	13.6 ± 0.8	2.1 ± 0.5	23.0 ± 1.5	2.3 ± 0.4	13.47 ± 0.28	11.0 ± 0.6

Table 21: Area under the curve of each JetClass class against the other nine at one million training jets, for every configuration. Upper part: after fine-tuning, with the network trained from scratch. Lower part: linear probes on the frozen network, with the randomly initialized network. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run.

pretraining	QCD	$H \to b\bar{b}$	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$	$H \to \ell\nu qq'$	$Z \to q\bar{q}$	$W \to qq'$	$t \to bqq'$	$t \to b\ell\nu$
188 classes	0.9753 ± 0.0001	0.9964 ± 0.0001	0.9853 ± 0.0001	0.9774 ± 0.0001	0.9865 ± 0.0001	0.9993 ± <0.0001	0.9603 ± 0.0003	0.9762 ± 0.0003	0.9977 ± 0.0001	0.9998 ± <0.0001
162 classes	0.9753 ± <0.0001	0.9964 ± 0.0001	0.9855 ± 0.0002	0.9776 ± 0.0001	0.9867 ± 0.0001	0.9993 ± <0.0001	0.9610 ± 0.0002	0.9766 ± 0.0001	0.9978 ± <0.0001	0.9998 ± <0.0001
43 classes	0.9752 ± <0.0001	0.9962 ± 0.0001	0.9850 ± 0.0001	0.9773 ± 0.0001	0.9863 ± 0.0001	0.9993 ± <0.0001	0.9604 ± 0.0002	0.9762 ± 0.0002	0.9977 ± <0.0001	0.9998 ± <0.0001
17 classes	0.9745 ± 0.0004	0.9955 ± 0.0001	0.9831 ± 0.0001	0.9764 ± 0.0002	0.9859 ± <0.0001	0.9992 ± <0.0001	0.9595 ± 0.0005	0.9758 ± 0.0002	0.9975 ± <0.0001	0.9998 ± <0.0001
162 classes, mass regression	0.9751, 0.9753	0.9961, 0.9963	0.9845, 0.9851	0.9773, 0.9776	0.9863, 0.9866	0.9993, 0.9992	0.9608, 0.9603	0.9764, 0.9764	0.9977, 0.9978	0.9997, 0.9998
17 classes, mass regression	0.9744, 0.9745	0.9951, 0.9954	0.9824, 0.9826	0.9759, 0.9761	0.9854, 0.9854	0.9992, 0.9992	0.9592, 0.9578	0.9754, 0.9751	0.9972, 0.9974	0.9997, 0.9997
random partition 1, seed 1	0.975	0.996	0.984	0.977	0.986	0.999	0.960	0.976	0.997	1.00
random partition 2, seed 2	0.975	0.996	0.984	0.977	0.986	0.999	0.960	0.976	0.998	1.00
random partition 3, seed 1	0.975	0.996	0.984	0.977	0.986	0.999	0.959	0.976	0.998	1.00
random partition 4, seed 2	0.975	0.996	0.984	0.977	0.986	0.999	0.960	0.976	0.998	1.00
random partition 5, seed 2	0.975	0.996	0.984	0.977	0.986	0.999	0.960	0.976	0.998	1.00
random partition, no flavor boundary	0.9749, 0.9751	0.9957, 0.9957	0.9835, 0.9835	0.9772, 0.9768	0.9863, 0.9859	0.9992, 0.9993	0.9593, 0.9594	0.9755, 0.9755	0.9975, 0.9976	0.9997, 0.9997
random partition, b-quark boundary	0.9750, 0.9752	0.9959, 0.9961	0.9840, 0.9842	0.9770, 0.9769	0.9861, 0.9863	0.9993, 0.9993	0.9591, 0.9592	0.9756, 0.9756	0.9976, 0.9977	0.9998, 0.9998
random partition, random flavor boundary	0.9750, 0.9746	0.9957, 0.9957	0.9837, 0.9836	0.9768, 0.9766	0.9861, 0.9860	0.9992, 0.9992	0.9600, 0.9602	0.9758, 0.9761	0.9974, 0.9975	0.9998, 0.9998
from scratch	0.9714 ± 0.0002	0.9939 ± 0.0001	0.9780 ± 0.0001	0.9727 ± 0.0003	0.9822 ± 0.0003	0.9990 ± <0.0001	0.9541 ± 0.0011	0.9726 ± 0.0005	0.9965 ± 0.0001	0.9996 ± <0.0001

pretraining	QCD	$H \to b\bar{b}$	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$	$H \to \ell\nu qq'$	$Z \to q\bar{q}$	$W \to qq'$	$t \to bqq'$	$t \to b\ell\nu$
188 classes	0.9474 ± 0.0005	0.9912 ± 0.0003	0.9719 ± 0.0006	0.9640 ± 0.0003	0.9776 ± 0.0002	0.9988 ± 0.0001	0.8958 ± 0.0029	0.9428 ± 0.0012	0.9925 ± 0.0003	0.9995 ± <0.0001
162 classes	0.9482 ± 0.0010	0.9912 ± 0.0001	0.9720 ± 0.0005	0.9642 ± 0.0002	0.9775 ± 0.0002	0.9988 ± <0.0001	0.8962 ± 0.0002	0.9431 ± 0.0005	0.9925 ± 0.0001	0.9995 ± <0.0001
43 classes	0.9484 ± 0.0004	0.9909 ± 0.0002	0.9719 ± 0.0003	0.9632 ± 0.0004	0.9769 ± 0.0002	0.9988 ± <0.0001	0.8959 ± 0.0023	0.9424 ± 0.0014	0.9924 ± 0.0002	0.9995 ± <0.0001
17 classes	0.9467 ± 0.0010	0.9873 ± 0.0003	0.9568 ± 0.0020	0.9622 ± 0.0007	0.9746 ± 0.0009	0.9985 ± <0.0001	0.8937 ± 0.0025	0.9409 ± 0.0006	0.9905 ± 0.0008	0.9993 ± 0.0001
162 classes, mass regression	0.9473, 0.9492	0.9912, 0.9912	0.9720, 0.9732	0.9638, 0.9651	0.9775, 0.9782	0.9988, 0.9988	0.8987, 0.9021	0.9428, 0.9461	0.9925, 0.9929	0.9995, 0.9995
17 classes, mass regression	0.9476, 0.9460	0.9865, 0.9873	0.9573, 0.9559	0.9622, 0.9621	0.9753, 0.9746	0.9983, 0.9984	0.8954, 0.8958	0.9423, 0.9421	0.9909, 0.9900	0.9992, 0.9991
random partition 1, seed 1	0.945	0.989	0.965	0.961	0.973	0.998	0.882	0.933	0.990	0.999
random partition 2, seed 2	0.945	0.989	0.966	0.961	0.974	0.998	0.883	0.935	0.990	0.999
random partition 3, seed 1	0.946	0.989	0.967	0.960	0.975	0.998	0.880	0.935	0.990	0.999
random partition 4, seed 2	0.943	0.989	0.967	0.961	0.974	0.999	0.880	0.933	0.991	0.999
random partition 5, seed 2	0.943	0.989	0.966	0.961	0.975	0.998	0.883	0.935	0.989	0.999
random partition, no flavor boundary	0.9462, 0.9453	0.9850, 0.9812	0.9564, 0.9495	0.9619, 0.9624	0.9739, 0.9736	0.9982, 0.9981	0.8867, 0.8867	0.9363, 0.9367	0.9894, 0.9896	0.9991, 0.9990
random partition, b-quark boundary	0.9476, 0.9445	0.9889, 0.9898	0.9649, 0.9657	0.9633, 0.9635	0.9766, 0.9771	0.9985, 0.9985	0.8907, 0.8948	0.9403, 0.9404	0.9907, 0.9914	0.9993, 0.9993
random partition, random flavor boundary	0.9450, 0.9448	0.9882, 0.9876	0.9644, 0.9630	0.9629, 0.9626	0.9755, 0.9749	0.9983, 0.9981	0.8918, 0.8884	0.9386, 0.9366	0.9901, 0.9897	0.9992, 0.9992
randomly initialized network, BatchNorm recomputed	0.850 ± 0.007	0.855 ± 0.009	0.7793 ± 0.0018	0.883 ± 0.004	0.837 ± 0.005	0.906 ± 0.017	0.791 ± 0.008	0.8703 ± 0.0031	0.9298 ± 0.0009	0.954 ± 0.006

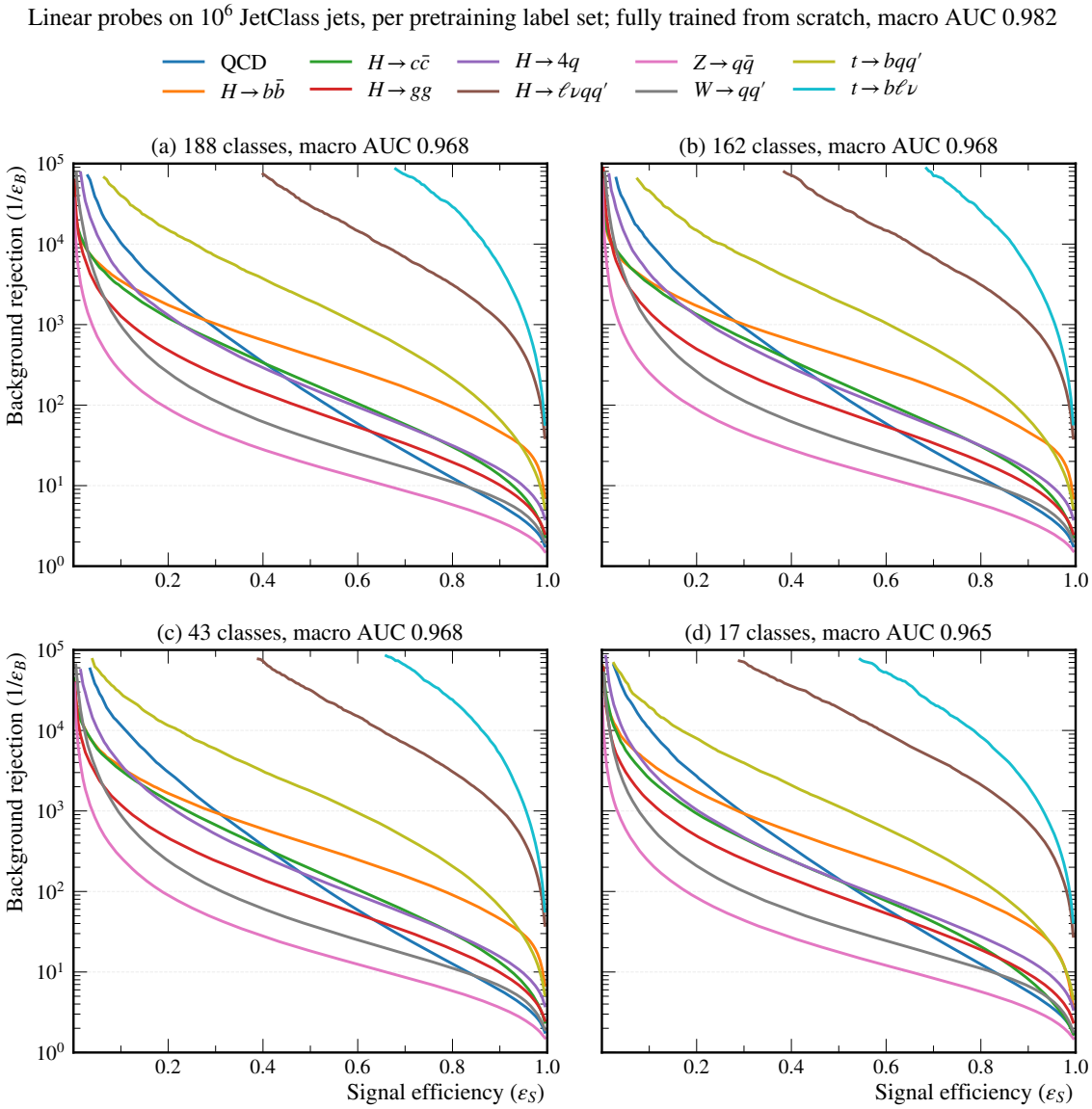


Figure 19: Background rejection against signal efficiency on JetClass from linear probes, trained on one million jets, on frozen pretrained networks. Each curve takes one of the ten classes as signal and the other nine as background, and is the mean over three pretraining runs. Curves begin where 20 background jets remain. Titles give the mean area under the curve; the heading gives the fully trained network from scratch for reference.

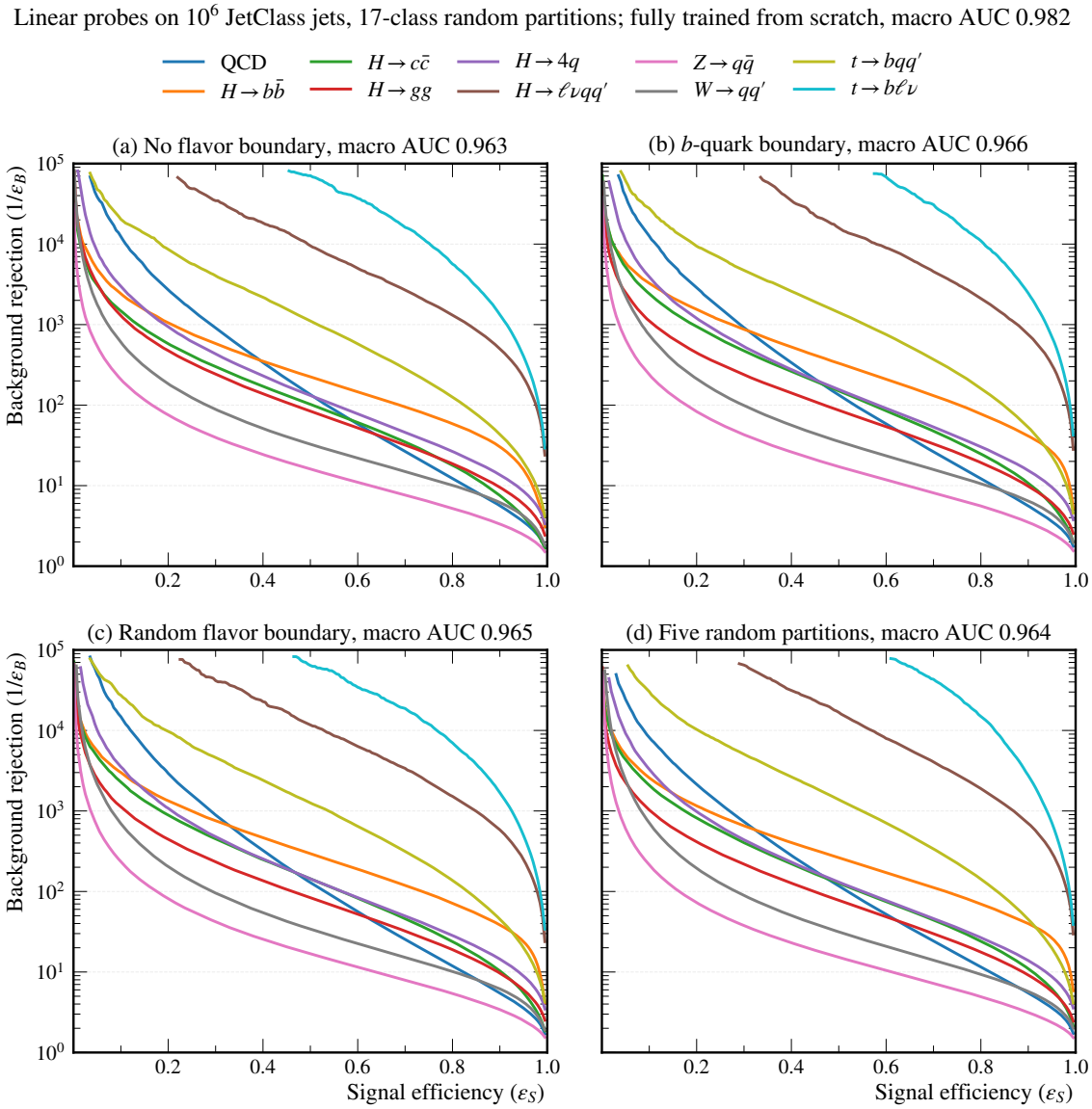


Figure 20: Background rejection against signal efficiency on JetClass from linear probes trained on one million jets, on frozen networks pretrained on 17-class random partitions: with no flavor boundary, a b-quark boundary or a random flavor boundary (two runs each), and five plain ones (averaged). Each curve takes one class as signal against the other nine and begins where 20 background jets remain. Titles give the mean area under the curve; the heading, from scratch.

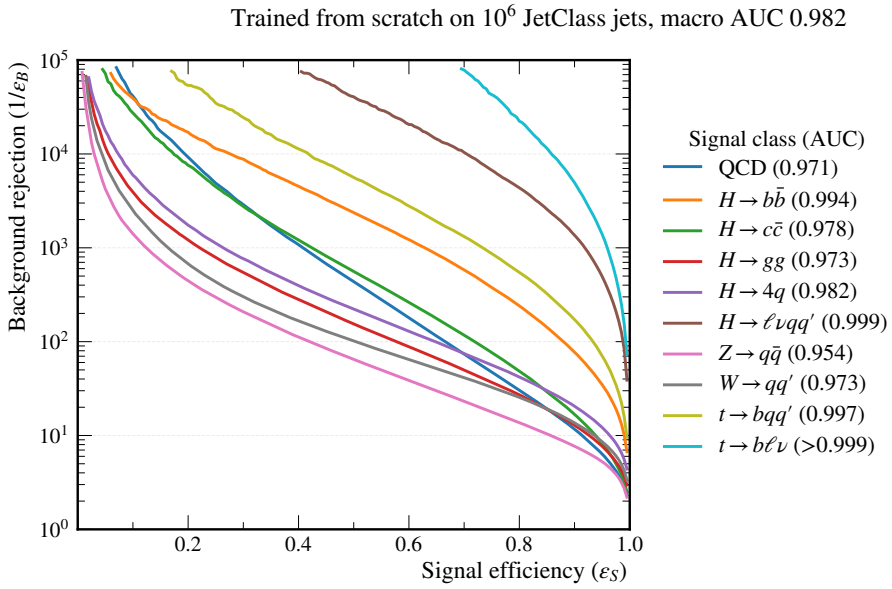


Figure 21: Background rejection against signal efficiency on JetClass for the network trained from scratch on one million jets, with no pretraining. Each curve takes one of the ten JetClass classes as signal and the other nine as background, and is the mean over three trainings. Curves begin where 20 background jets remain. The legend gives each class's area under the curve; the heading gives their mean.

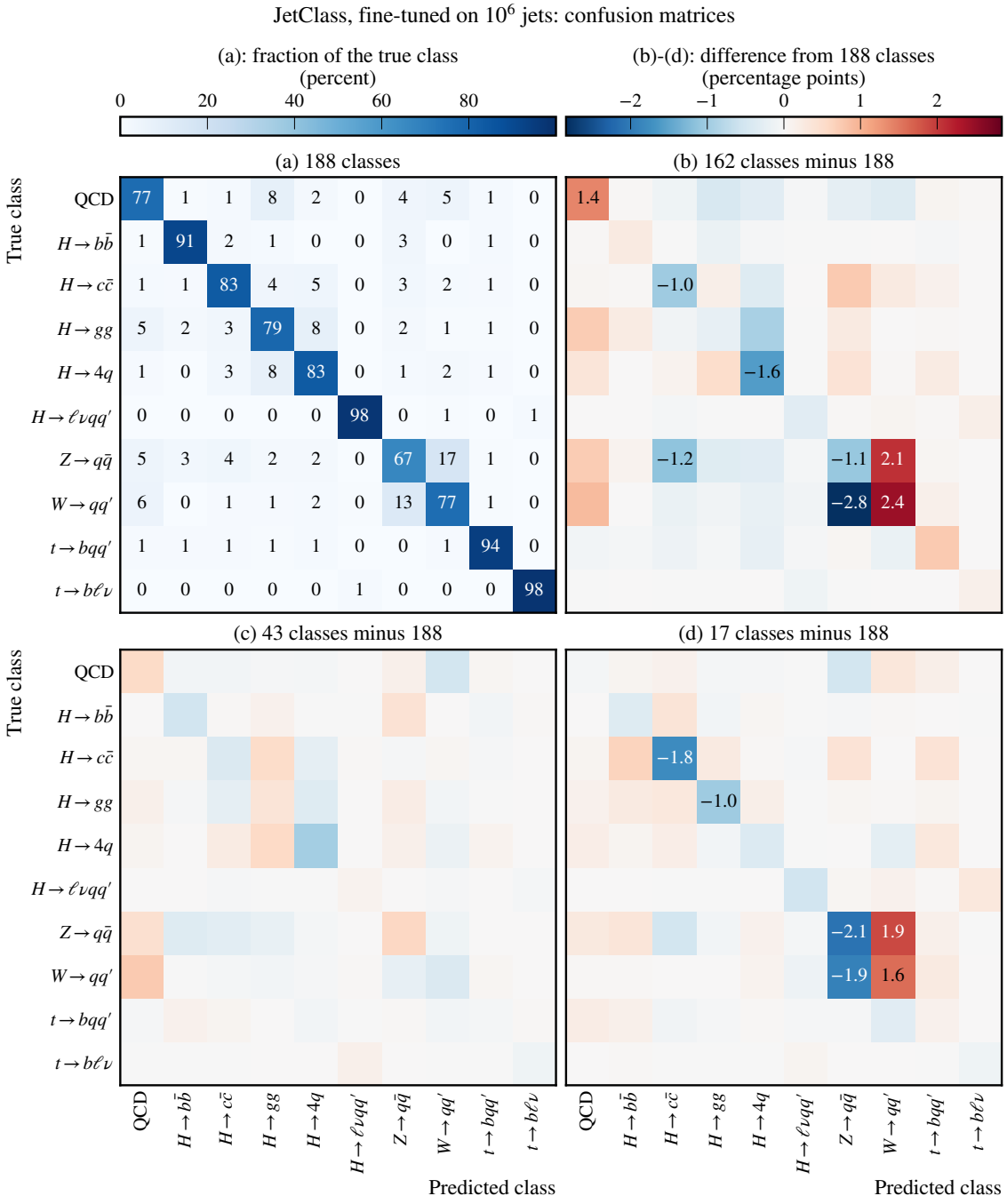


Figure 22: Confusion matrices on JetClass after fine-tuning on one million jets: (a) the percentage of each true class assigned to each predicted class by the model pretrained on 188 classes; (b)–(d) the change in that percentage for the models pretrained on 162, 43 and 17 classes, each against the 188-class model of the same seed, printed where it reaches one percentage point, which no change in (c) does. Values are means over three runs.

Table 22: QCD jets passing each signal class's working point after fine-tuning, the fewest over the runs of each configuration: the counts behind the rejections of table 2.

fine-tuned from	$H \to b\bar{b}$	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$	$H \to \ell\nu qq'$	$t \to bqq'$	$t \to b\ell\nu$	$W \to qq'$	$Z \to q\bar{q}$
from scratch	39	123	2216	286	145	30	48	724	897
188 classes	29	68	1884	181	57	14	19	524	628
162 classes	33	70	1933	166	58	11	25	508	648
43 classes	29	72	1914	187	58	13	23	525	636
17 classes	29	82	1873	188	66	15	21	503	679
162 classes, mass regression	30	79	1899	177	68	14	24	510	661
17 classes, mass regression	31	87	1896	207	68	14	18	551	691
random partitions (all five)	26	71	1921	194	64	15	22	492	649
random partition, no flavor boundary	33	92	1882	170	71	12	28	569	685
random partition, b-quark boundary	27	83	1851	192	77	14	28	546	672
random partition, random flavor boundary	31	70	1886	191	71	15	27	538	664

Table 23: The random partitions on JetClass: which prong-count boundaries each keeps (•: no group holds classes of both prong counts), the groups that mix prong counts, and its fine-tuned rejection of QCD per signal class over that of the 188-class run of its seed. †: fewer than $n_{\min}$ QCD jets pass in one of the two runs (equation (**??**)).

	prong-count boundary kept				fine-tuned rejection over the 188-class model								
partition	2/3	2/4	3/4	mixed groups	$H \to b\bar{b}$	$H \to c\bar{c}$	$H \to gg$	$H \to 4q$	$H \to \ell\nu qq'$	$t \to bqq'$	$t \to b\ell\nu$	$W \to qq'$	$Z \to q\bar{q}$
random partition 1, seed 1	—	—	—	12	0.842	1.04	0.984	0.862	1.12	$0.652^\dagger$	1.13	0.991	0.929
random partition 2, seed 2	—	—	—	12	0.853	1.10	0.944	0.935	0.958	$0.875^\dagger$	0.606	1.03	0.929
random partition 3, seed 1	—	—	—	13	0.889	1.09	0.966	0.928	0.935	$1.00^\dagger$	1.18	0.934	0.939
random partition 4, seed 2	—	—	—	12	0.906	0.940	0.966	0.959	0.908	$0.636^\dagger$	0.667	1.11	0.971
random partition 5, seed 2	—	—	—	11	1.12	0.963	0.981	0.944	0.793	$0.778^\dagger$	0.800	1.03	0.991

Table 24: Binary linear probes between pairs of JetClass-II classes: the AUC and the ratio ρ of $1 - \text{AUC}$, equation (**??**), to the 188-class run of the same seed. †: the label set merges the two sides into one class; ‡: it merges the first side with part of the second. Mean ± standard deviation over the three pretraining runs of each level of the class hierarchy; configurations with two runs give both values, and a random partition has a single run. The b-quark boundary and the random flavor boundary lie only in the four-quark decays; every label set but the 188-class one merges all QCD classes. The randomly initialized network, as initialized and with its BatchNorm statistics recomputed, has no label set and no ρ. The $X \to ee$ vs $\mu\mu$ probe uses the jets of the two classes in the class-balanced sets (9,571 training jets, 128,833–135,245 test jets per side).

pretraining	$X \to bc$ vs bq, cs		$X \to b\bar{b}$ vs $c\bar{c}$		$X \to YY \to b\bar{b}qq$ vs $c\bar{c}qq$		$X \to ee$ vs $\mu\mu$	
	AUC	ρ	AUC	ρ	AUC	ρ	AUC	ρ
188 classes	0.9409 ± 0.0003	1.00	$0.9977 \pm <0.0001$	1.00	$0.9939 \pm <0.0001$	1.00	$1.0000 \pm <0.0001$	1.00
162 classes	0.9410 ± 0.0002	0.998 ± 0.007	0.9976 ± 0.0001	1.015 ± 0.027	0.9939 ± 0.0002	1.005 ± 0.032	$1.0000 \pm <0.0001$	0.4 ± 0.4
43 classes	$0.9319 \pm 0.0014^\ddagger$	1.151 ± 0.029	$0.9974 \pm <0.0001$	1.102 ± 0.013	$0.9932 \pm <0.0001$	1.117 ± 0.008	$0.9610 \pm 0.0030^\dagger$	$4{,}000{,}000 \pm 8{,}000{,}000$
17 classes	$0.885 \pm 0.004^\dagger$	1.95 ± 0.06	$0.9902 \pm 0.0007^\dagger$	4.22 ± 0.29	$0.9716 \pm 0.0033^\dagger$	4.7 ± 0.5	$0.960 \pm 0.020^\dagger$	$7{,}000{,}000 \pm 11{,}000{,}000$
162 classes, mass regression	0.9375, 0.9400	1.057, 1.010	0.9973, 0.9975	1.184, 1.087	0.9930, 0.9935	1.164, 1.070	1.0000, 1.0000	0.002, 0.030
17 classes, mass regression	0.8808, $0.8833^\dagger$	2.018, 1.964	0.9891, $0.9904^\dagger$	4.70, 4.13	0.9676, $0.9706^\dagger$	5.36, 4.82	0.969, $0.998^\dagger$	30,560, 2,829
random partition 1, seed 1	$0.931^\dagger$	1.17	$0.996^\dagger$	1.51	$0.991^\dagger$	1.43	$1.00^\dagger$	9.99
random partition 2, seed 2	$0.930^\ddagger$	1.18	0.996	1.54	0.991	1.46	$1.00^\dagger$	14.5
random partition 3, seed 1	0.937	1.07	$0.996^\dagger$	1.53	$0.991^\dagger$	1.43	1.00	0.214
random partition 4, seed 2	0.936	1.08	0.997	1.37	0.992	1.32	$1.00^\dagger$	9.62
random partition 5, seed 2	$0.927^\dagger$	1.24	$0.996^\dagger$	1.67	0.991	1.48	1.00	0.00523
random partition, no flavor boundary	0.882, $0.870^\dagger$	2.00, 2.19	0.986, $0.969^\dagger$	5.9, 13.4	0.961, $0.943^\dagger$	6.4, 9.4	1.0000, 1.0000	0.097, 0.010
random partition, b-quark boundary	0.9103, $0.9091^\dagger$	1.518, 1.530	0.9957, $0.9959^\dagger$	1.837, 1.748	0.9909, 0.9912	1.510, 1.447	1.0000, 1.0000	0.0, 1.1
random partition, random flavor boundary	0.9096, $0.9100^\dagger$	1.530, 1.515	0.9940, $0.9924^\dagger$	2.58, 3.25	0.9863, $0.9860^\dagger$	2.272, 2.281	1.0000, 1.0000	0.0024, 0.0049
randomly initialized network	0.6011 ± 0.0028	–	0.701 ± 0.012	–	0.680 ± 0.007	–	0.9995 ± 0.0003	–
randomly initialized network, BatchNorm recomputed	0.653 ± 0.009	–	0.817 ± 0.020	–	0.782 ± 0.019	–	$1.0000 \pm <0.0001$	–

pretraining	QCD $b\bar{b}$ vs $c\bar{c}$		QCD with b vs light		QCD with c, no b, vs light	
	AUC	ρ	AUC	ρ	AUC	ρ
188 classes	0.9867 ± 0.0001	1.00	0.9880 ± 0.0001	1.00	0.9324 ± 0.0003	1.00
162 classes	$0.9848 \pm 0.0004^\dagger$	1.15 ± 0.04	$0.9870 \pm 0.0001^\dagger$	1.085 ± 0.004	$0.9282 \pm 0.0001^\dagger$	1.061 ± 0.005
43 classes	$0.9846 \pm 0.0001^\dagger$	1.163 ± 0.016	$0.9870 \pm 0.0001^\dagger$	1.081 ± 0.008	$0.9275 \pm 0.0003^\dagger$	1.072 ± 0.008
17 classes	$0.9563 \pm 0.0003^\dagger$	3.294 ± 0.033	$0.9815 \pm 0.0007^\dagger$	1.54 ± 0.07	$0.9048 \pm 0.0010^\dagger$	1.407 ± 0.021
162 classes, mass regression	0.9829, $0.9842^\dagger$	1.30, 1.19	0.9868, $0.9870^\dagger$	1.107, 1.078	0.9262, $0.9273^\dagger$	1.093, 1.070
17 classes, mass regression	0.9481, $0.9531^\dagger$	3.95, 3.51	0.9804, $0.9800^\dagger$	1.638, 1.662	0.9046, $0.9002^\dagger$	1.414, 1.470
random partition 1, seed 1	$0.980^\dagger$	1.56	$0.986^\dagger$	1.20	$0.922^\dagger$	1.16
random partition 2, seed 2	$0.979^\dagger$	1.58	$0.985^\dagger$	1.21	$0.921^\dagger$	1.16
random partition 3, seed 1	$0.980^\dagger$	1.51	$0.986^\dagger$	1.21	$0.923^\dagger$	1.14
random partition 4, seed 2	$0.981^\dagger$	1.43	$0.986^\dagger$	1.16	$0.924^\dagger$	1.12
random partition 5, seed 2	$0.979^\dagger$	1.60	$0.986^\dagger$	1.19	$0.922^\dagger$	1.14
random partition, no flavor boundary	0.944, $0.930^\dagger$	4.24, 5.21	0.9814, $0.9825^\dagger$	1.56, 1.45	0.9088, $0.9122^\dagger$	1.352, 1.293
random partition, b-quark boundary	0.9795, $0.9795^\dagger$	1.561, 1.532	0.9855, $0.9854^\dagger$	1.212, 1.213	0.9166, $0.9171^\dagger$	1.237, 1.220
random partition, random flavor boundary	0.9721, $0.9707^\dagger$	2.124, 2.189	0.9848, $0.9845^\dagger$	1.277, 1.286	0.9204, $0.9211^\dagger$	1.179, 1.162
randomly initialized network	0.655 ± 0.005	–	0.742 ± 0.008	–	0.656 ± 0.005	–
randomly initialized network, BatchNorm recomputed	0.765 ± 0.016	–	0.829 ± 0.012	–	0.6821 ± 0.0003	–

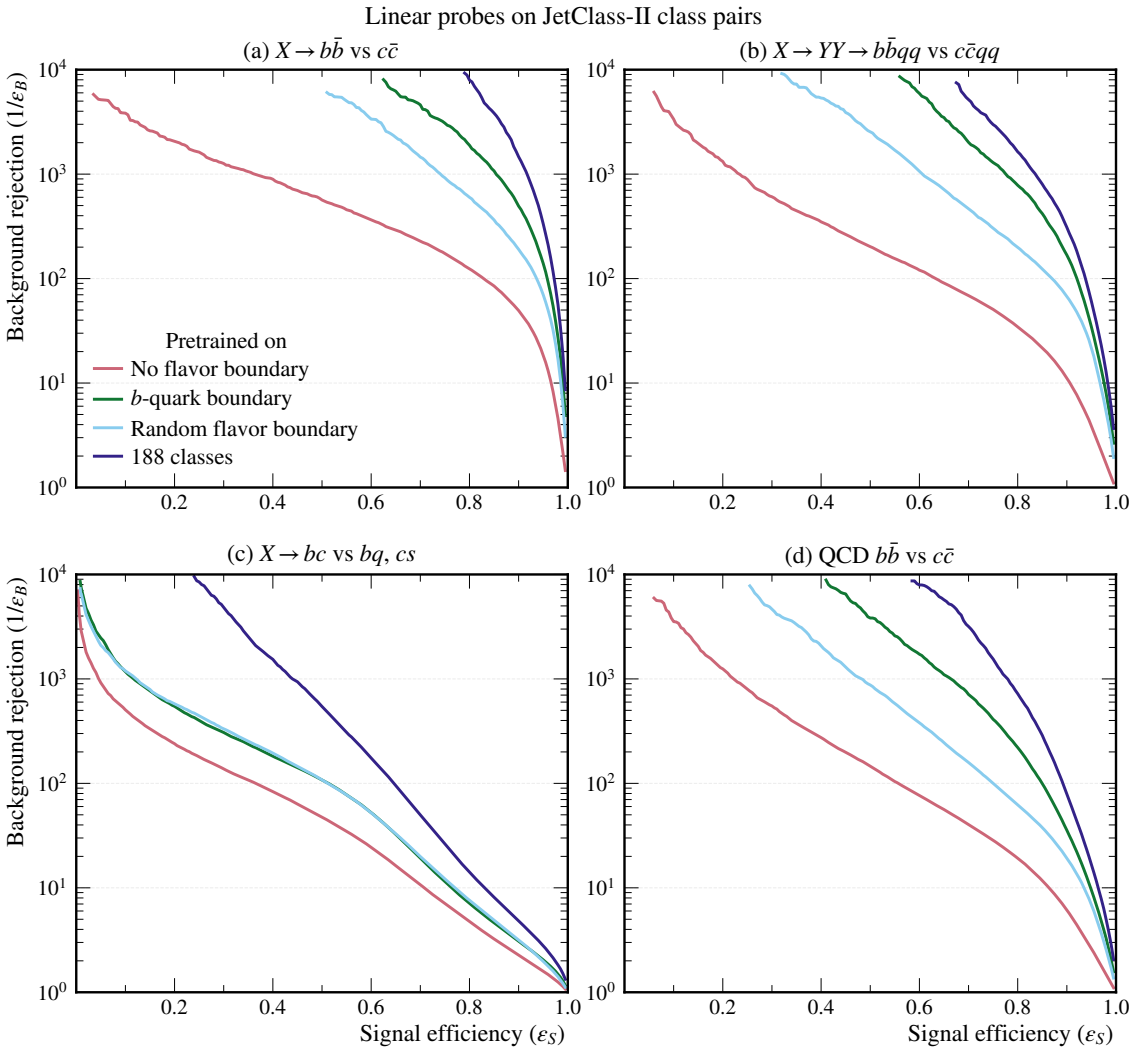


Figure 23: Background rejection against signal efficiency for four pairs of JetClass-II classes that differ only in flavor, (d) of QCD jets, from linear probes on networks pretrained on 17-class random partitions with no flavor boundary, a b-quark boundary or a random flavor boundary, and on 188 classes for reference. The first class named is the signal. Curves are means over two runs (188 classes: three) and end where fewer than 20 background jets remain.

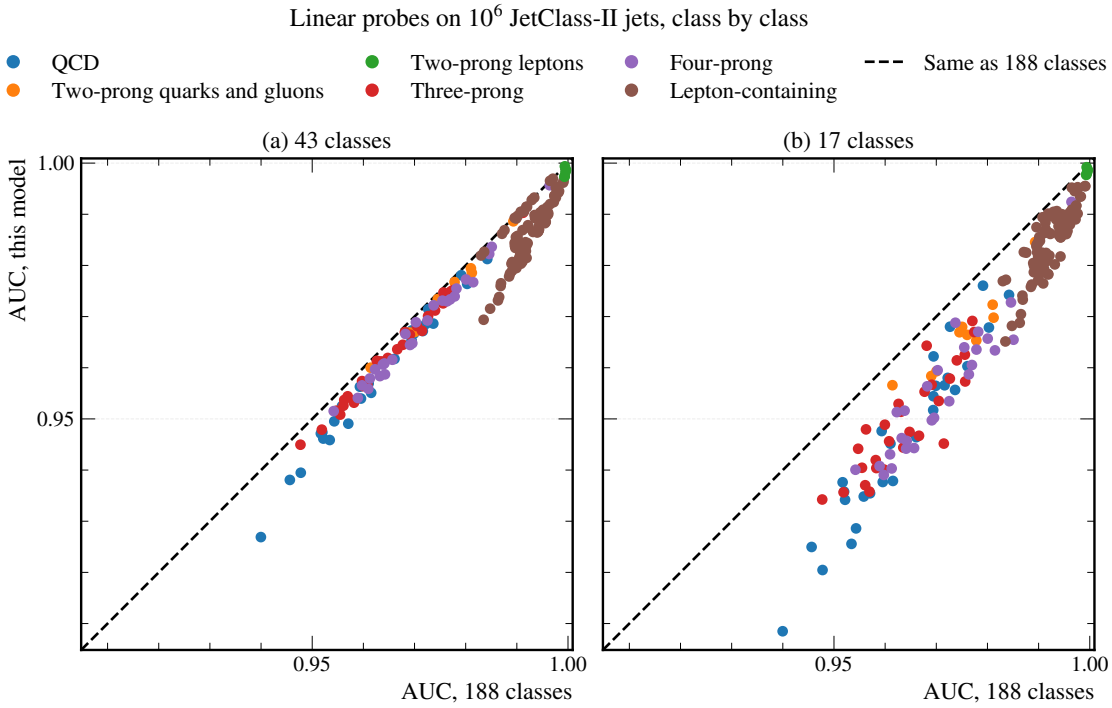


Figure 24: Each of the 188 classes of class-balanced JetClass-II, from linear probes trained on one million jets: the class's area under the curve for the 43- and 17-class label sets, against the same for the 188-class model. On the dashed line the two agree; below it the class does worse than with 188 classes. Colors are decay families. Values are means over runs.

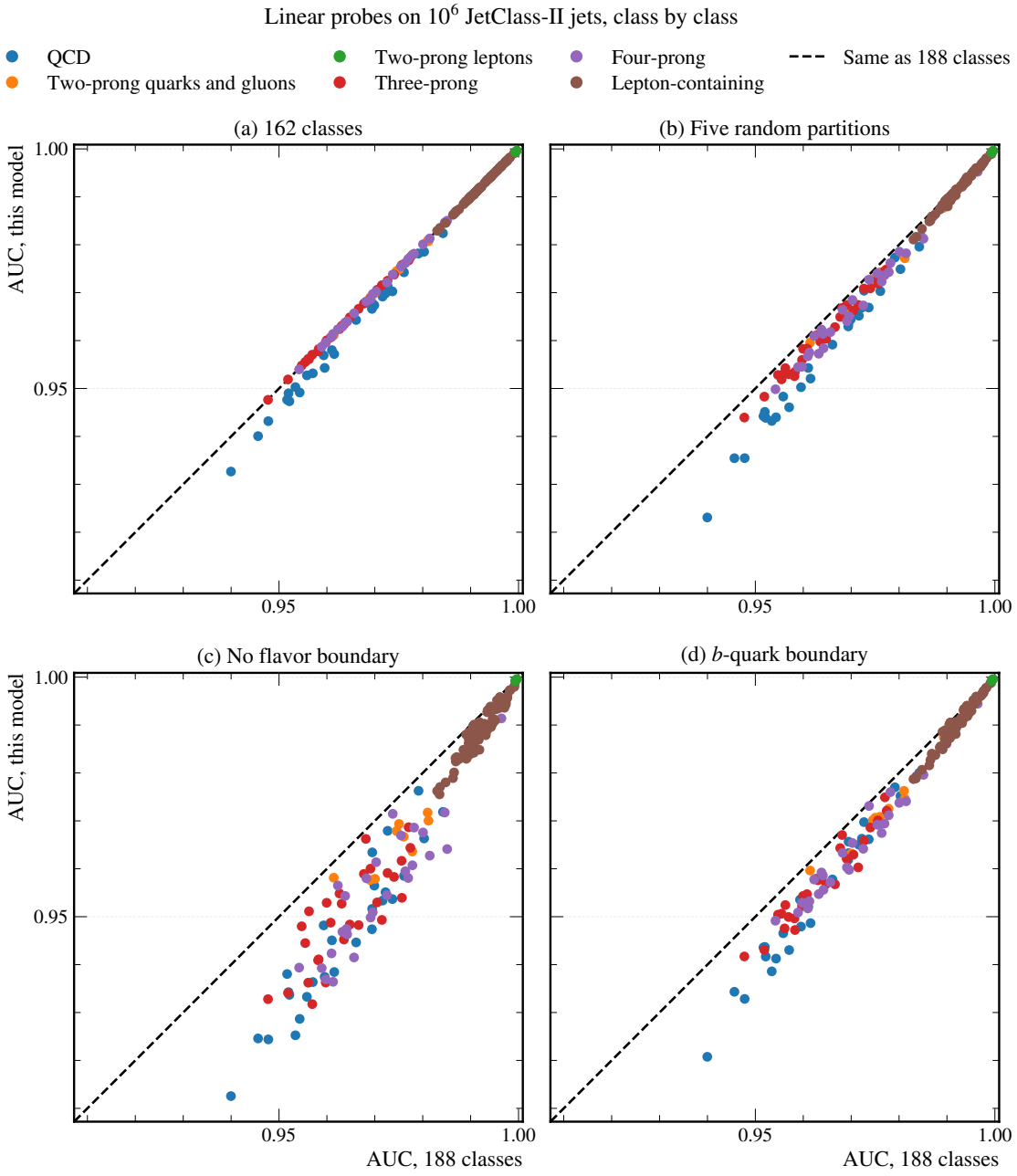


Figure 25: Each class of class-balanced JetClass-II, from linear probes trained on one million jets: its area under the curve for (a) the 162-class label set, (b) the five plain random partitions averaged, (c) the no-flavor-boundary and (d) the b-quark-boundary partitions, against the same for the 188-class model. Below the dashed line the class does worse than with 188 classes. Colors are decay families; values are means over runs.

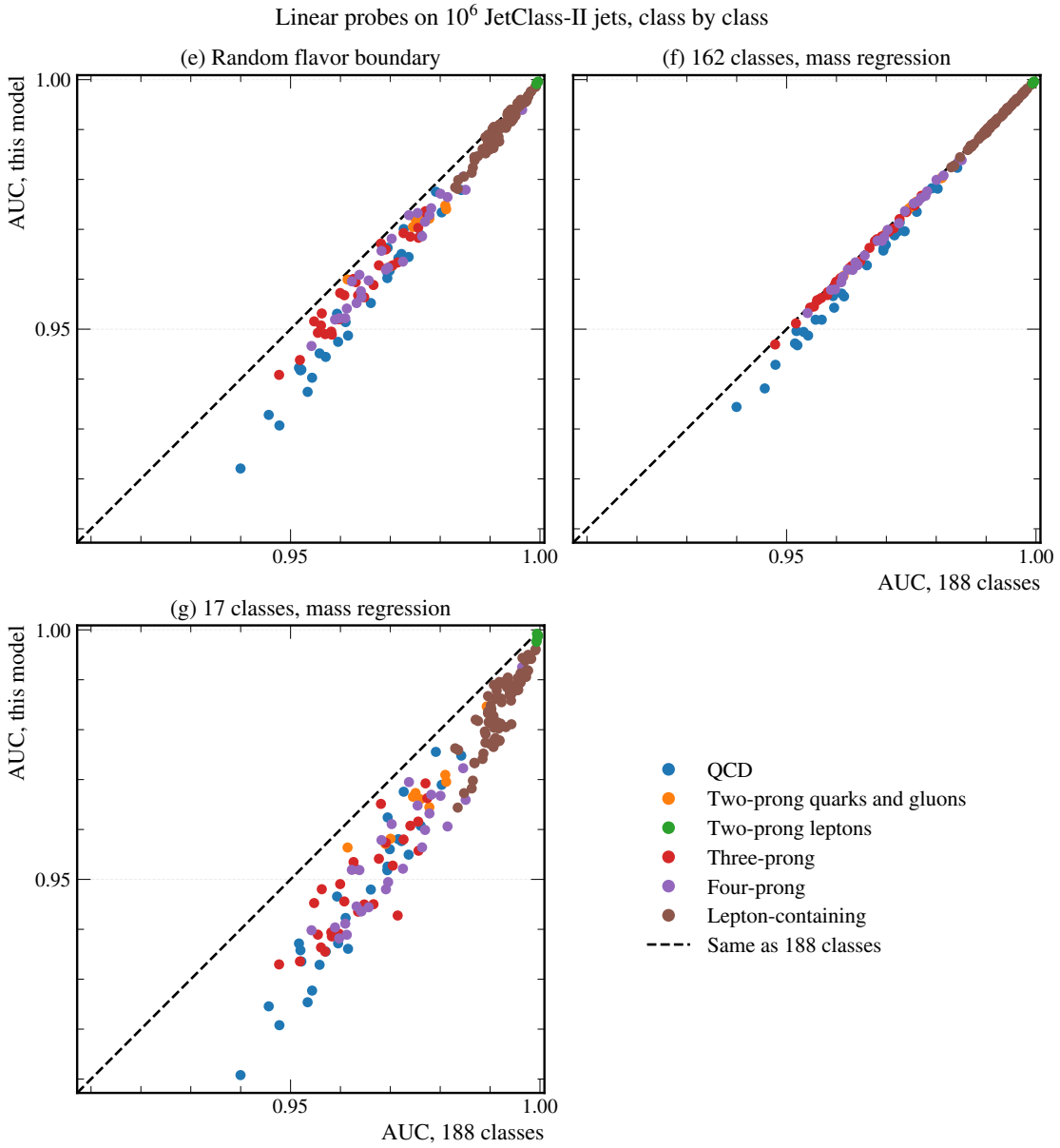


Figure 26: Continued from the previous figure: each class's area under the curve for (e) the random-flavor-boundary partition and the (f) 162- and (g) 17-class label sets pretrained with an auxiliary regression of the jet mass, against the same for the 188-class model, from linear probes trained on one million jets on class-balanced JetClass-II. Colors are decay families; values are means over runs.

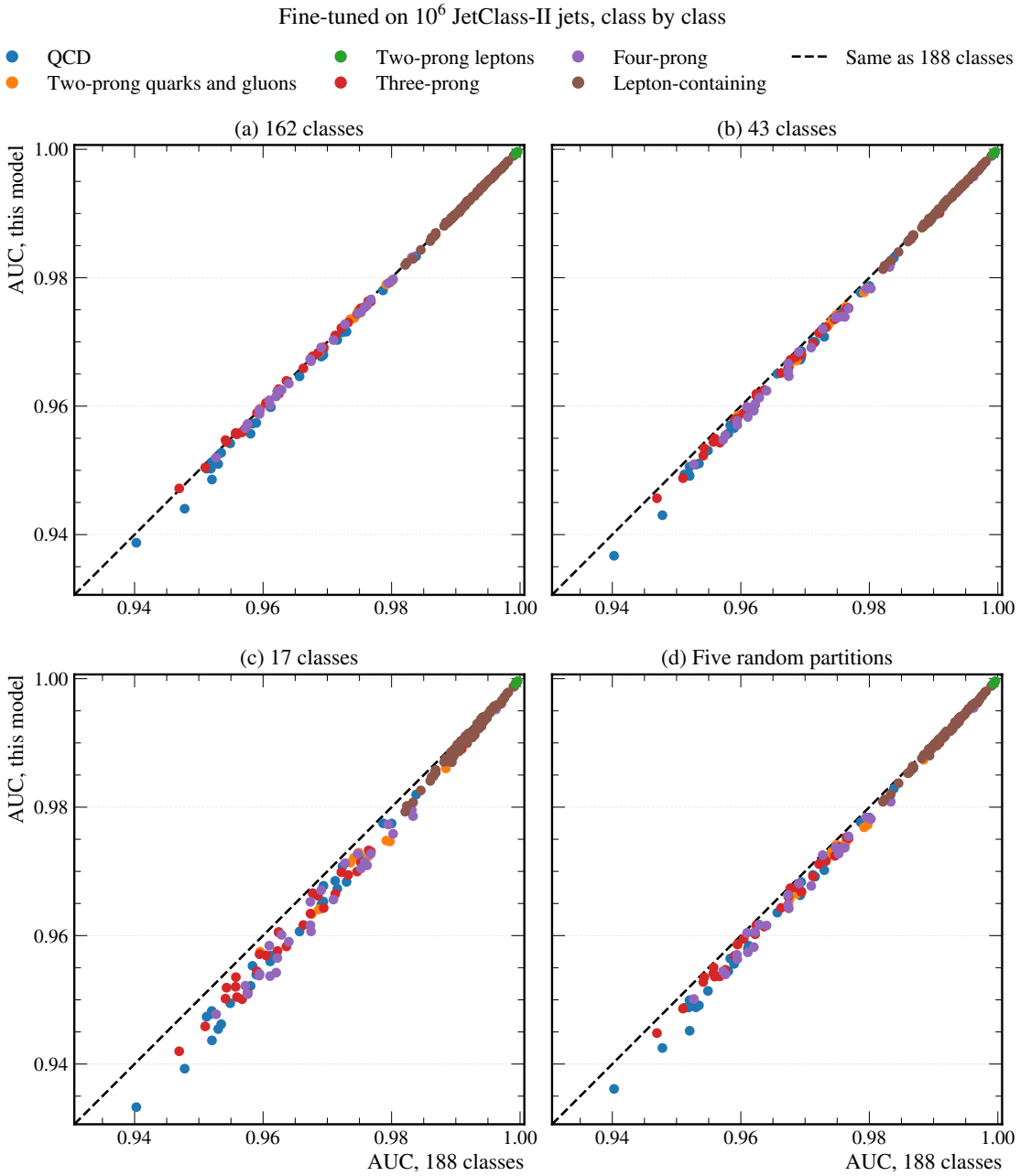


Figure 27: Each class of class-balanced JetClass-II after fine-tuning on one million jets: its area under the curve for the (a) 162-, (b) 43- and (c) 17-class label sets and (d) the five plain random partitions averaged, against the same for the 188-class model. Below the dashed line the class does worse than with 188 classes. Colors are decay families; values are means over three runs, in (d) over the five partitions, one run each.

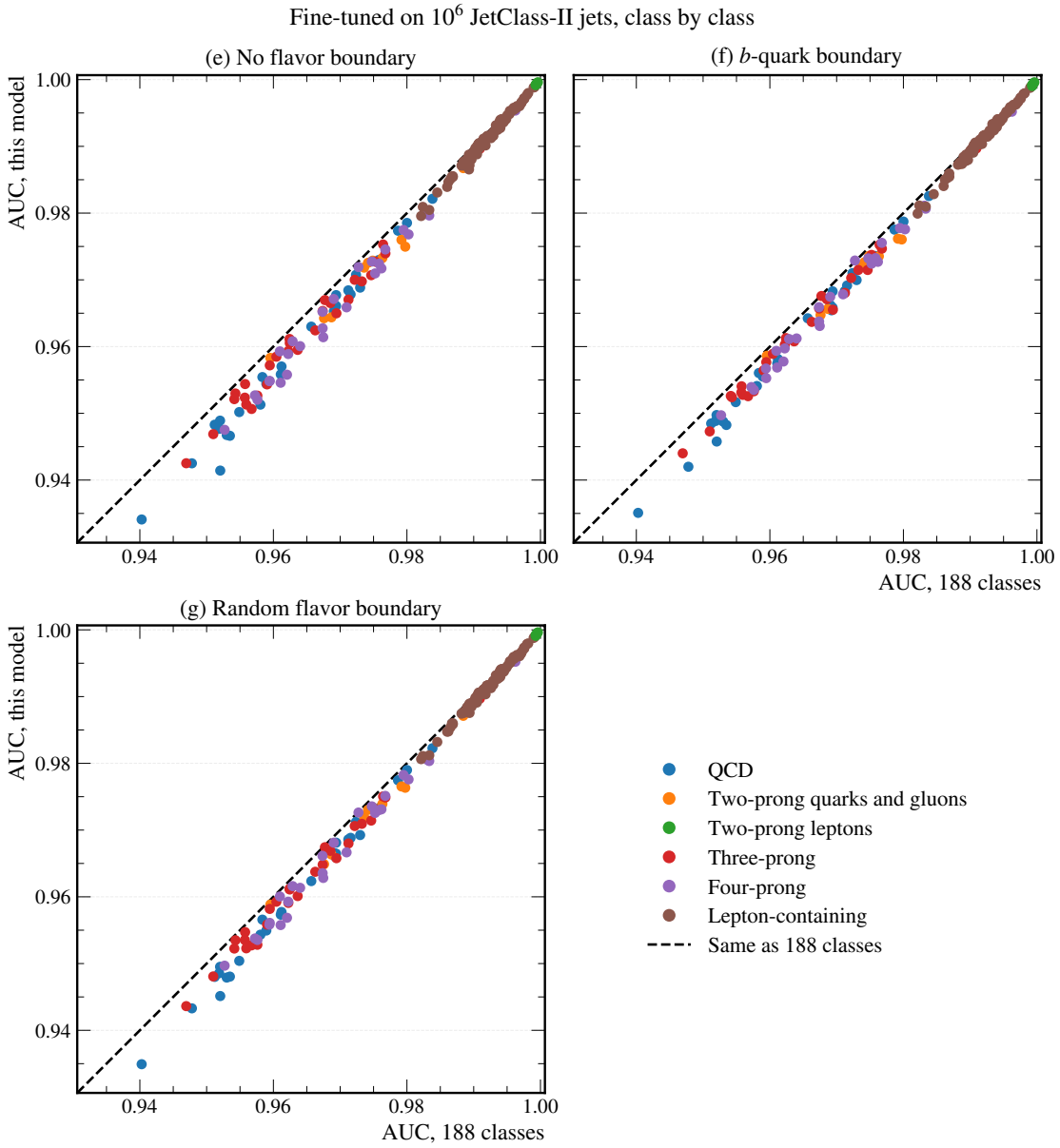


Figure 28: Continued from the previous figure: each class's area under the curve after fine-tuning on one million class-balanced JetClass-II jets, for the partitions with (e) no flavor boundary, (f) a b-quark boundary and (g) a random flavor boundary, against the same for the 188-class model. Colors are decay families; values are means over two runs.

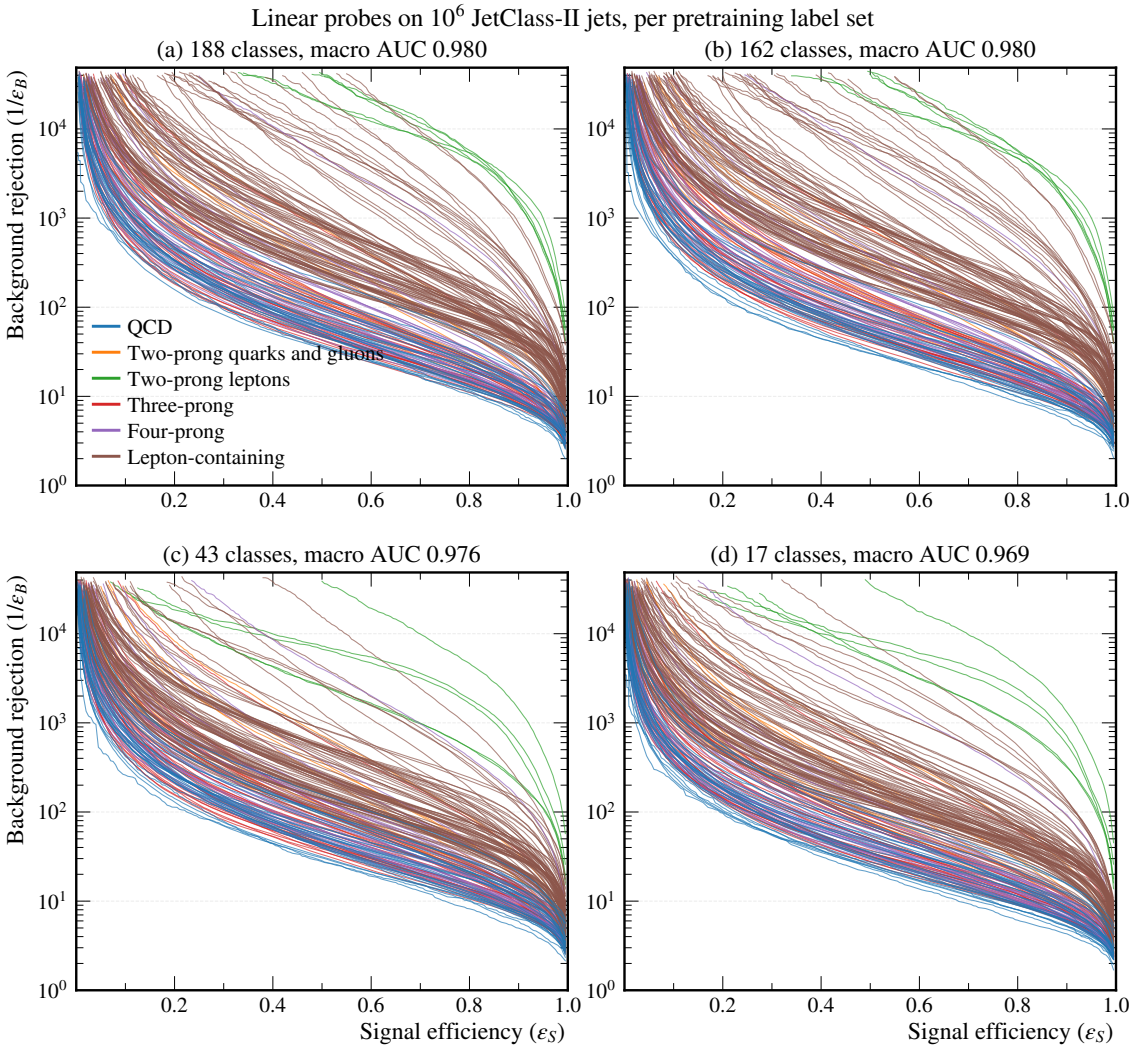


Figure 29: Background rejection against signal efficiency for each of the 188 JetClass-II classes against the rest of the class-balanced test set, from linear probes on frozen networks pretrained with 188, 162, 43 and 17 classes. Thin curves are means over three pretraining runs, colored by decay family, and end where fewer than 20 background jets remain. Each title gives the mean area under the curve over all classes.

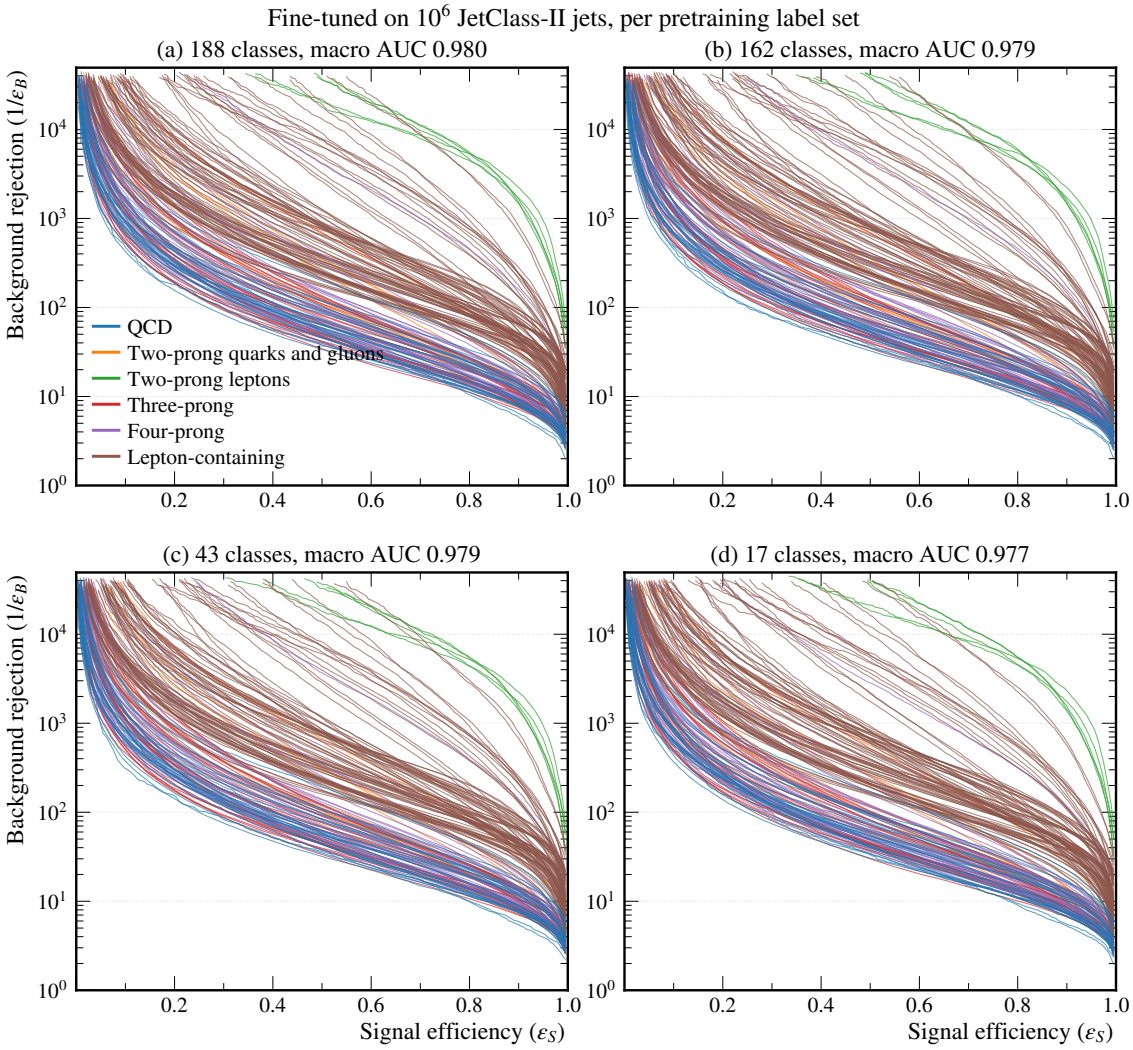


Figure 30: Background rejection against signal efficiency for each of the 188 JetClass-II classes against the rest of the class-balanced test set, after fine-tuning on one million jets, for networks pretrained with 188, 162, 43 and 17 classes. Thin curves are means over runs, colored by decay family, and end where fewer than 20 background jets remain. Each title gives the mean area under the curve over all classes.

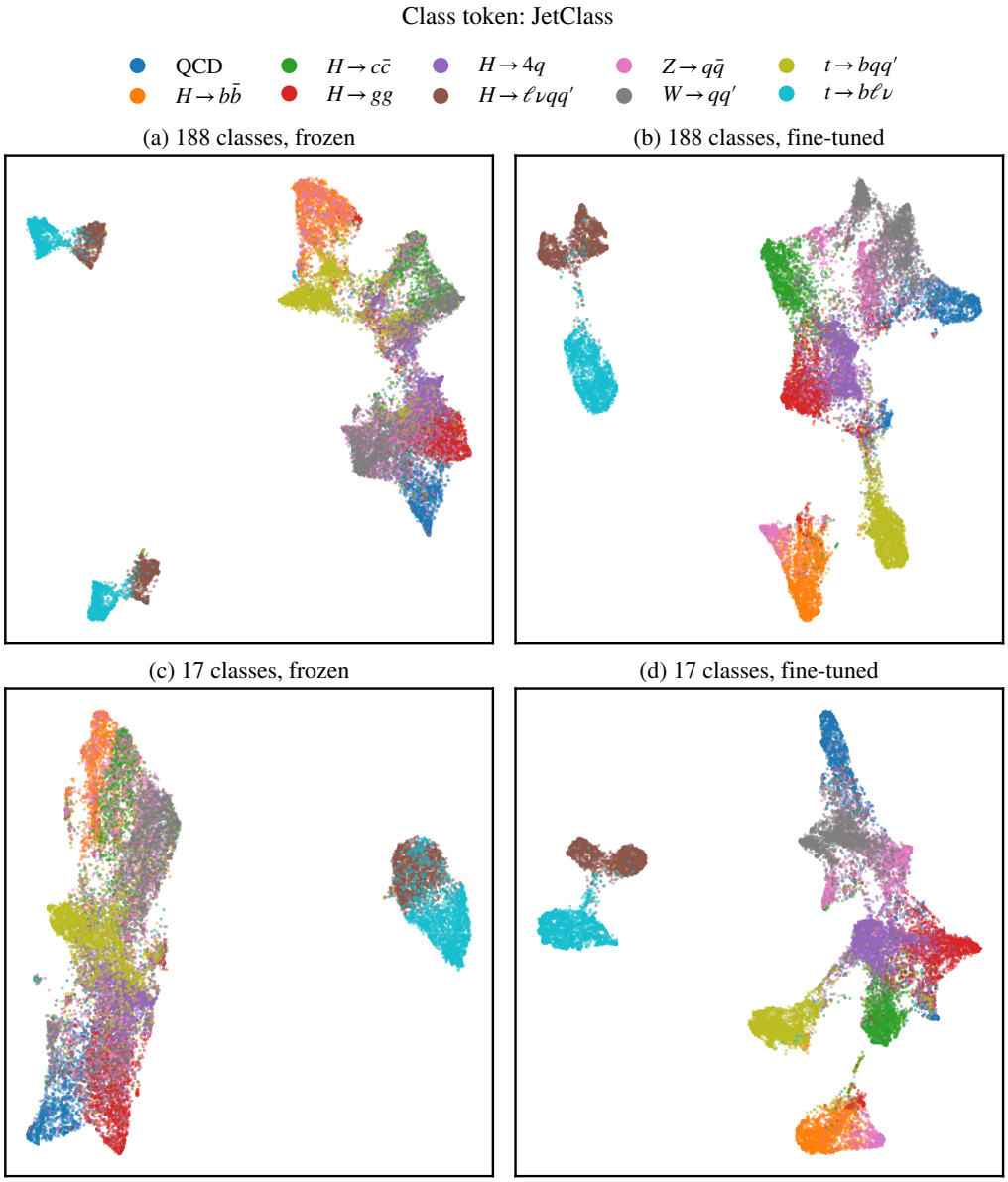


Figure 31: UMAP of the class token of a network pretrained on 188 classes (upper) and one on 17 (lower), frozen (left) and fine-tuned on one million JetClass jets (right), for 2,000 test jets of each JetClass class in every panel; the axes carry no scale. Both models' frozen features set the leptonic decays apart, in two islands for 188 classes; fine-tuning separates most hadronic classes, $Z \to q\bar{q}$ partly mixed with $H \to b\bar{b}$ and $W \to qq'$.

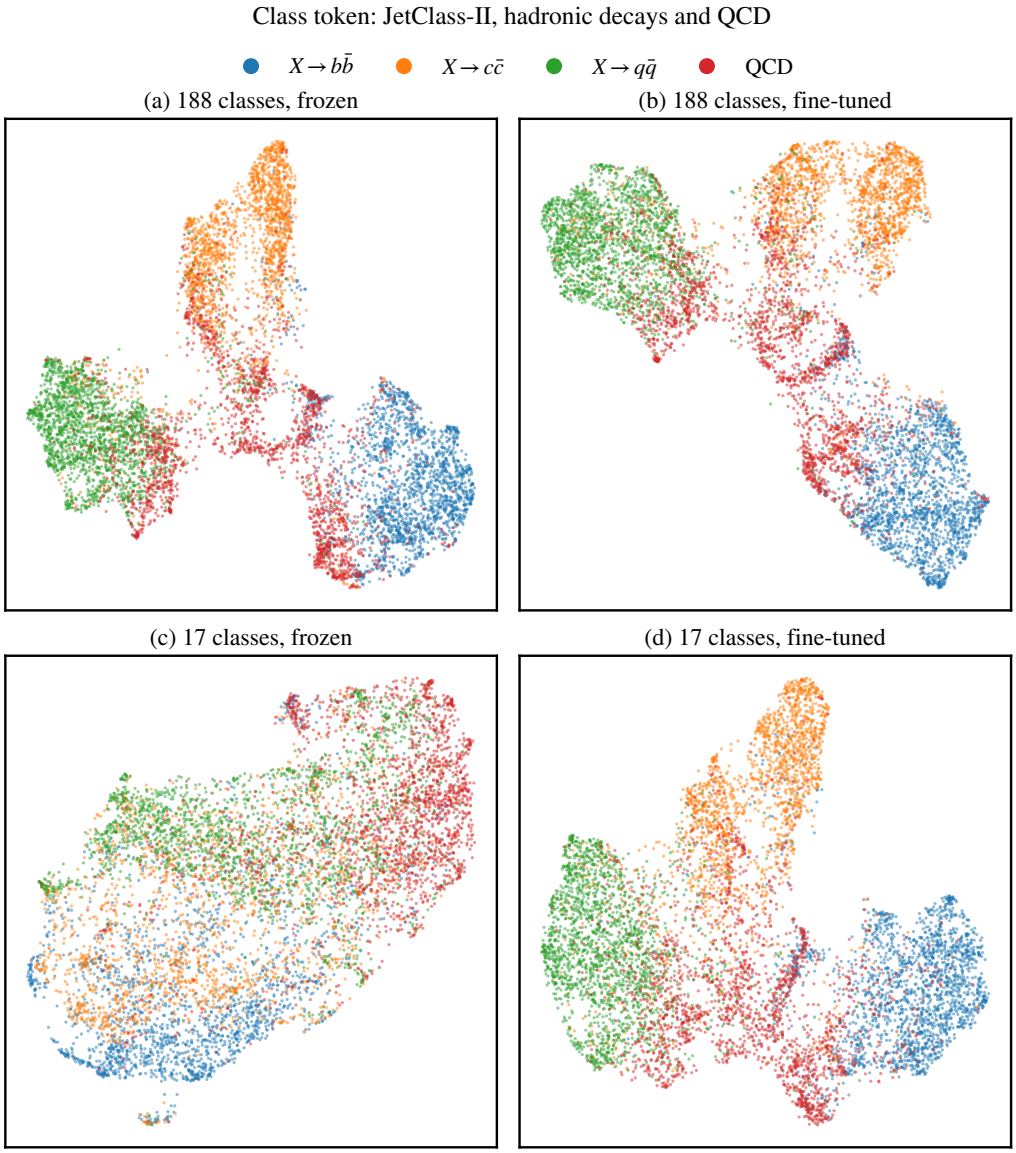


Figure 32: UMAP of the class token of a network pretrained on 188 classes (upper) and one on 17 (lower), frozen (left) and fine-tuned on one million class-balanced JetClass-II jets (right), for 2,000 test jets each of $X \to b\bar{b}$, $X \to c\bar{c}$, $X \to q\bar{q}$ and QCD in every panel; the axes carry no scale. The 188-class frozen features separate the three decays; the 17-class ones only order them, $X \to b\bar{b}$ at one edge and $X \to q\bar{q}$ near QCD at the other; fine-tuning separates them.

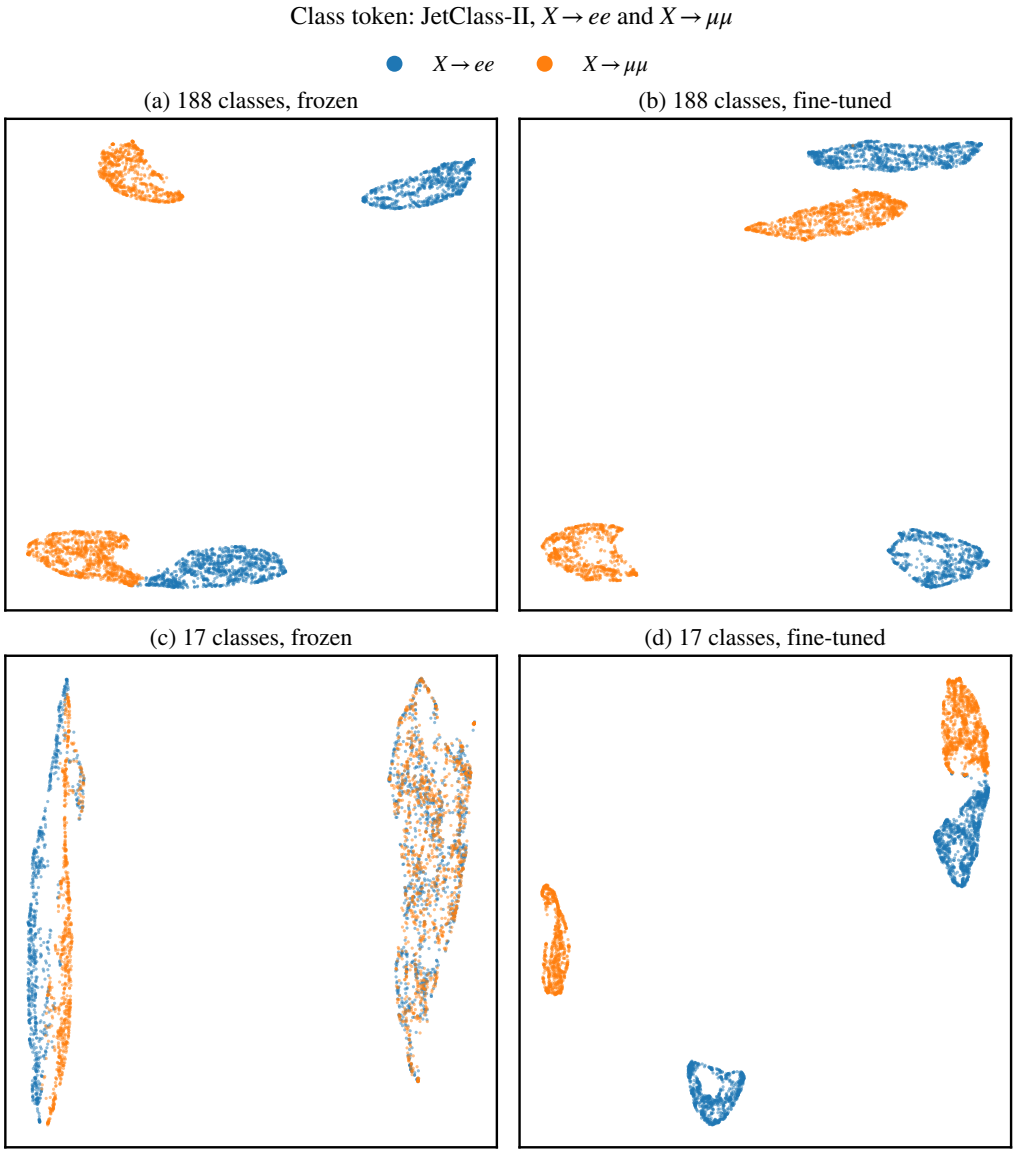


Figure 33: UMAP of the class token of a network pretrained on 188 classes (upper) and one on 17 (lower), frozen (left) and fine-tuned on one million class-balanced JetClass-II jets (right), for 2,000 test jets each of $X \to ee$ and $X \to \mu\mu$ in every panel; the axes carry no scale. Each class falls in two islands in every panel; the 188-class frozen features keep electrons and muons apart, the 17-class ones mix them, and fine-tuning separates them.

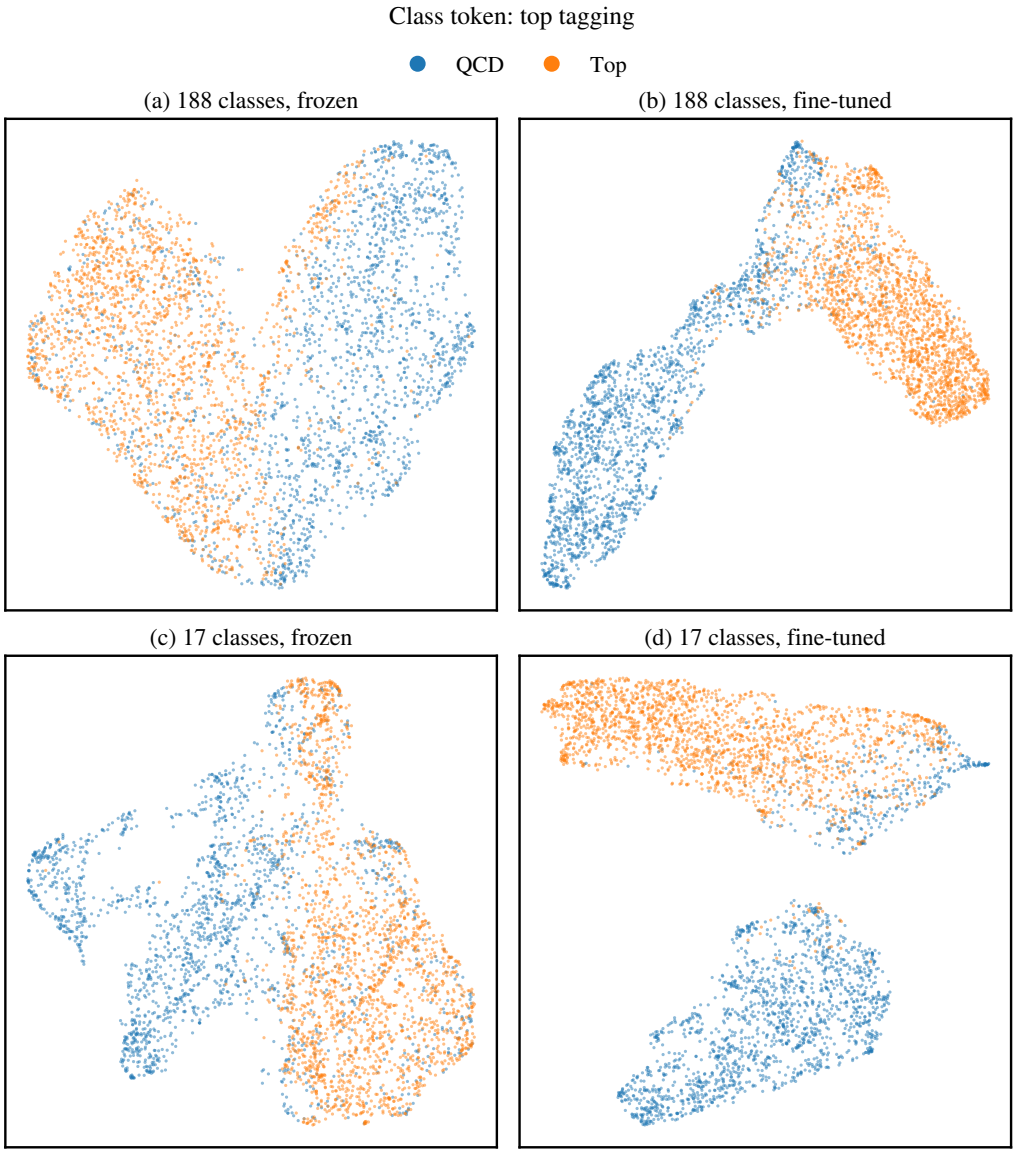


Figure 34: UMAP of the class token of a network pretrained on 188 classes (upper) and one on 17 (lower), frozen (left) and fine-tuned on the full top-tagging training set (right), for 2,000 top and 2,000 QCD test jets in every panel; the axes carry no scale. Top and QCD jets lie mostly apart in both models' frozen features; fine-tuned, the 17-class model gives two islands, some QCD jets in the top one; the 188-class model joins them by mixed jets.

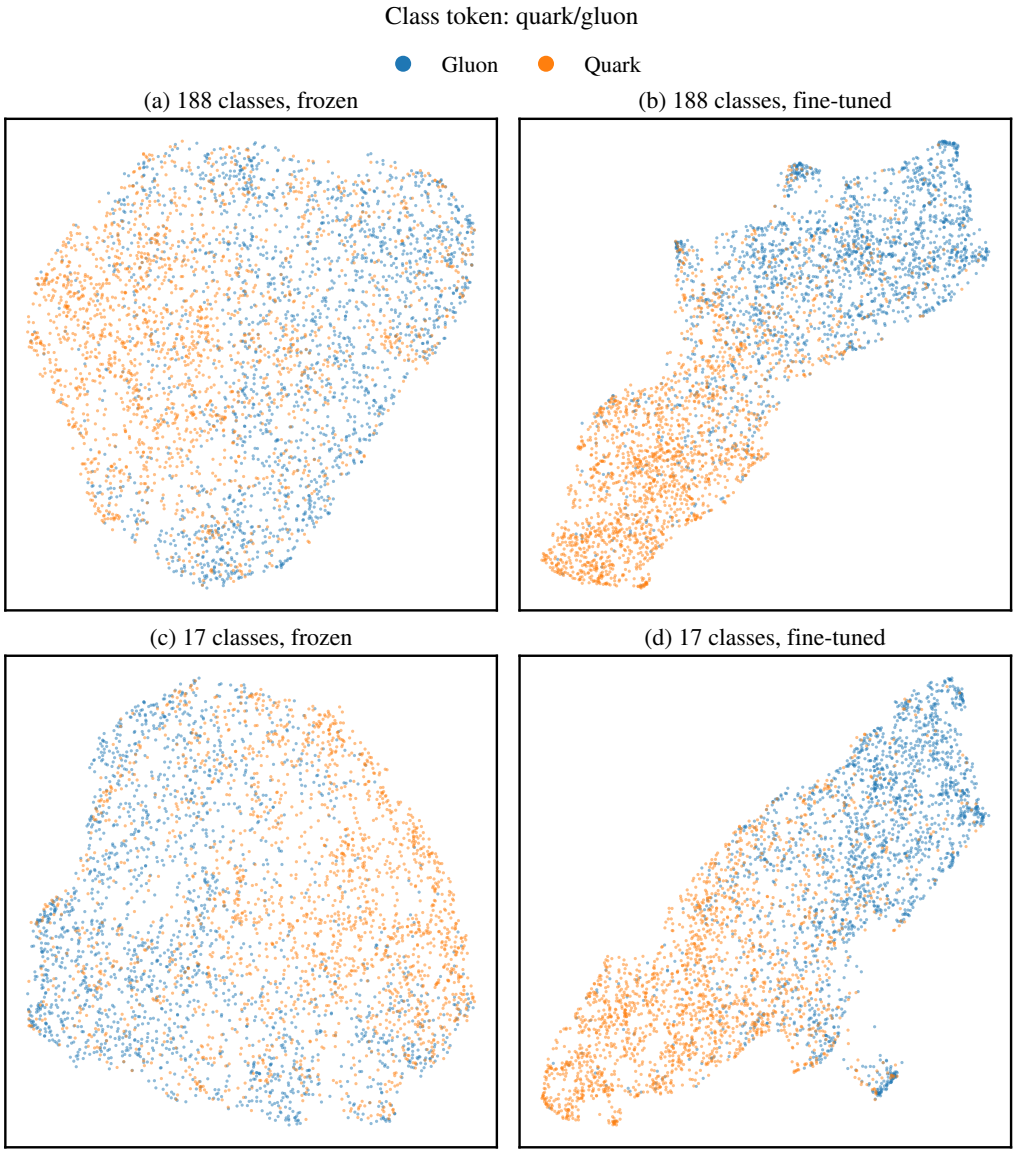


Figure 35: UMAP of the class token of a network pretrained on 188 classes (upper) and one on 17 (lower), frozen (left) and fine-tuned on the full quark/gluon training set (right), for 2,000 quark and 2,000 gluon test jets (Pythia) in every panel; the axes carry no scale. Quark and gluon jets overlap in the frozen features of both models, with only a gradient between them, and separate partly after fine-tuning.

Table 25: Jet-mass regression in detail, over all test jets and over resonant and QCD jets. Upper part: resolution and median response of the regression output of the models pretrained with the mass regression, and of ridge and multilayer perceptron regressions on frozen features. Lower parts: resolution and median response of the groomed mass and of the regression output against jet p_{T} and against the generator soft-drop mass; bins with fewer than 100 jets are blank. Mean ± standard deviation over the three pretraining runs; the models with the mass regression have two runs, and both values are given.

		resolution			median response		
estimator	pretraining	all	resonant	QCD	all	resonant	QCD
regression output	162 classes, mass regression	0.1471, 0.1463	0.1332, 0.1320	0.2263, 0.2256	0.995, 1.003	0.996, 1.004	0.9897, 0.9996
regression output	17 classes, mass regression	0.1450, 0.1461	0.1312, 0.1320	0.2239, 0.2240	1.005, 1.005	1.006, 1.006	1.001, 1.000
ridge, frozen features	randomly initialized network	0.312 ± 0.007	0.286 ± 0.006	0.913 ± 0.031	0.9406 ± 0.0035	0.936 ± 0.004	0.9649 ± 0.0035
ridge, frozen features	188 classes	0.3211 ± 0.0034	0.2921 ± 0.0026	0.559 ± 0.013	0.9641 ± 0.0012	0.9754 ± 0.0006	0.905 ± 0.006
ridge, frozen features	162 classes	0.316 ± 0.009	0.288 ± 0.008	0.548 ± 0.021	0.9655 ± 0.0005	0.9768 ± 0.0008	0.909 ± 0.007
ridge, frozen features	43 classes	0.304 ± 0.016	0.280 ± 0.013	0.49 ± 0.06	0.968 ± 0.005	0.977 ± 0.004	0.922 ± 0.012
ridge, frozen features	17 classes	0.293 ± 0.012	0.271 ± 0.012	0.448 ± 0.030	0.973 ± 0.005	0.980 ± 0.004	0.937 ± 0.010
ridge, frozen features	162 classes, mass regression	0.1885, 0.1922	0.1751, 0.1785	0.2674, 0.2701	0.9998, 0.9998	1.003, 1.003	0.9826, 0.9830
ridge, frozen features	17 classes, mass regression	0.1842, 0.1824	0.1696, 0.1687	0.2630, 0.2601	1.002, 1.002	1.005, 1.003	0.9888, 0.9935
MLP, frozen features	randomly initialized network	0.2421 ± 0.0018	0.2214 ± 0.0017	0.502 ± 0.014	0.983 ± 0.005	0.984 ± 0.005	0.980 ± 0.007
MLP, frozen features	188 classes	0.2200 ± 0.0014	0.1990 ± 0.0016	0.3487 ± 0.0029	0.991 ± 0.005	0.994 ± 0.004	0.975 ± 0.007
MLP, frozen features	162 classes	0.2195 ± 0.0017	0.1989 ± 0.0020	0.3425 ± 0.0031	0.9876 ± 0.0014	0.9900 ± 0.0024	0.974 ± 0.006
MLP, frozen features	43 classes	0.2158 ± 0.0029	0.1949 ± 0.0023	0.338 ± 0.007	0.991 ± 0.005	0.993 ± 0.004	0.976 ± 0.010
MLP, frozen features	17 classes	0.2098 ± 0.0031	0.1894 ± 0.0035	0.327 ± 0.006	0.991 ± 0.004	0.993 ± 0.004	0.977 ± 0.008
MLP, frozen features	162 classes, mass regression	0.1591, 0.1586	0.1436, 0.1430	0.2473, 0.2452	1.004, 1.006	1.006, 1.008	0.9883, 0.9930
MLP, frozen features	17 classes, mass regression	0.1583, 0.1588	0.1420, 0.1431	0.2465, 0.2463	1.009, 1.006	1.012, 1.008	0.9942, 0.9932

resolution	jet p_{T} [GeV]	200–400	400–700	700–1000	1000–2500
groomed mass	all	0.242	0.225	0.231	0.327
162 classes, mass regression	all	0.1720, 0.1703	0.1483, 0.1468	0.1426, 0.1411	0.1396, 0.1395
17 classes, mass regression	all	0.1691, 0.1702	0.1455, 0.1470	0.1403, 0.1414	0.1383, 0.1391
groomed mass	resonant	0.229	0.215	0.216	0.211
162 classes, mass regression	resonant	0.1642, 0.1621	0.1412, 0.1393	0.1337, 0.1315	0.1201, 0.1201
17 classes, mass regression	resonant	0.1610, 0.1620	0.1382, 0.1396	0.1313, 0.1320	0.1188, 0.1200
groomed mass	QCD	0.279	0.333	1.10	3.61
162 classes, mass regression	QCD	0.1945, 0.1928	0.1854, 0.1846	0.2146, 0.2110	0.2801, 0.2831
17 classes, mass regression	QCD	0.1918, 0.1947	0.1816, 0.1830	0.2089, 0.2090	0.2800, 0.2775

median response	jet p_{T} [GeV]	200–400	400–700	700–1000	1000–2500
groomed mass	all	0.893	0.914	0.917	0.953
162 classes, mass regression	all	0.986, 0.997	0.994, 1.003	0.997, 1.005	0.997, 1.004
17 classes, mass regression	all	0.9930, 0.9974	1.001, 1.003	1.006, 1.007	1.010, 1.007
groomed mass	resonant	0.904	0.917	0.915	0.947
162 classes, mass regression	resonant	0.987, 0.998	0.995, 1.004	0.998, 1.005	0.998, 1.004
17 classes, mass regression	resonant	0.9933, 0.9980	1.002, 1.004	1.006, 1.007	1.010, 1.008
groomed mass	QCD	0.863	0.895	0.931	1.02
162 classes, mass regression	QCD	0.984, 0.995	0.986, 0.998	0.994, 1.003	0.993, 1.001
17 classes, mass regression	QCD	0.9921, 0.9955	0.9965, 0.9973	1.004, 1.004	1.008, 1.002

resolution	generator soft-drop mass [GeV]	20–50	50–100	100–150	150–250	250–500
groomed mass	all	0.281	0.206	0.204	0.210	0.225
162 classes, mass regression	all	0.1751, 0.1756	0.1425, 0.1416	0.1302, 0.1291	0.1233, 0.1213	0.1163, 0.1135
17 classes, mass regression	all	0.1755, 0.1740	0.1418, 0.1411	0.1284, 0.1288	0.1210, 0.1220	0.1134, 0.1145
groomed mass	resonant	0.275	0.194	0.192	0.201	0.216
162 classes, mass regression	resonant	0.1632, 0.1634	0.1366, 0.1359	0.1261, 0.1252	0.1198, 0.1175	0.1133, 0.1107
17 classes, mass regression	resonant	0.1631, 0.1621	0.1357, 0.1351	0.1246, 0.1250	0.1174, 0.1183	0.1107, 0.1118
groomed mass	QCD	0.292	0.239	0.246	0.265	0.278
162 classes, mass regression	QCD	0.2033, 0.2030	0.1653, 0.1639	0.1508, 0.1515	0.1515, 0.1509	0.1511, 0.1470
17 classes, mass regression	QCD	0.2032, 0.2006	0.1630, 0.1631	0.1505, 0.1514	0.1483, 0.1515	0.1434, 0.1482

median response	generator soft-drop mass [GeV]	20–50	50–100	100–150	150–250	250–500
groomed mass	all	1.00	0.936	0.908	0.890	0.852
162 classes, mass regression	all	1.025, 1.037	1.006, 1.014	0.993, 1.001	0.9881, 0.9954	0.9710, 0.9772
17 classes, mass regression	all	1.032, 1.041	1.014, 1.016	1.002, 1.002	0.9986, 0.9970	0.9837, 0.9792
groomed mass	resonant	1.01	0.953	0.922	0.902	0.862
162 classes, mass regression	resonant	1.032, 1.043	1.012, 1.019	0.997, 1.004	0.9905, 0.9977	0.9722, 0.9783
17 classes, mass regression	resonant	1.038, 1.047	1.019, 1.021	1.006, 1.006	1.001, 0.999	0.9847, 0.9804
groomed mass	QCD	0.971	0.864	0.816	0.781	0.738
162 classes, mass regression	QCD	1.006, 1.019	0.9802, 0.9899	0.9670, 0.9766	0.9647, 0.9720	0.9565, 0.9631
17 classes, mass regression	QCD	1.015, 1.021	0.9906, 0.9901	0.9791, 0.9759	0.9787, 0.9743	0.9704, 0.9641

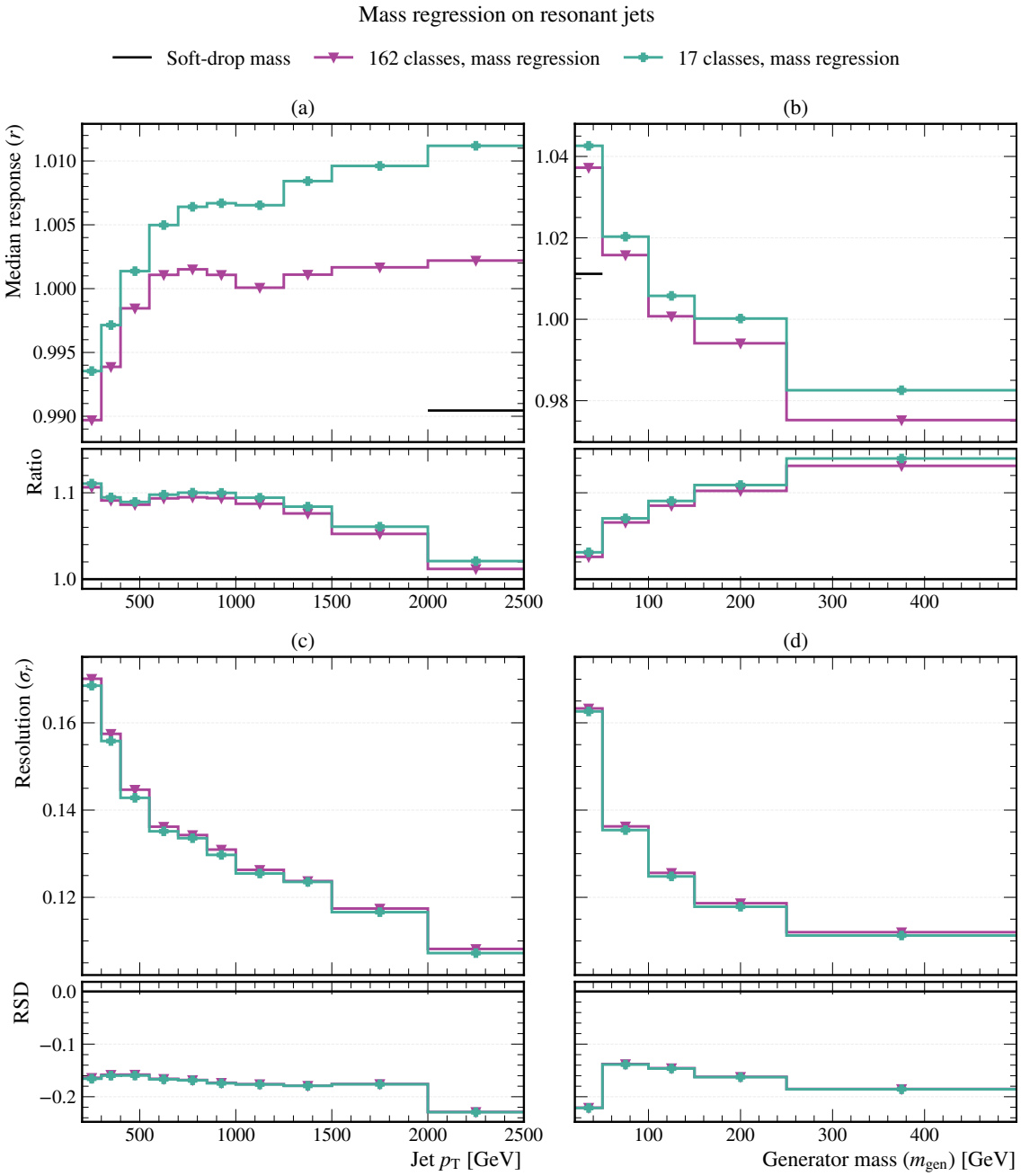


Figure 36: Median mass response r (a, b) and resolution σ_r (c, d), as in the main-text figure, of resonant JetClass-II test jets against transverse momentum (a, c; all jets) and generator mass (b, d; 20–500 GeV): mass regression of the models pretrained on 162 and 17 classes (two runs). Lower panels: ratio to the soft-drop mass (black where in range) and signed root square difference (RSD), negative where better. Unlabelled axes share the left ones.

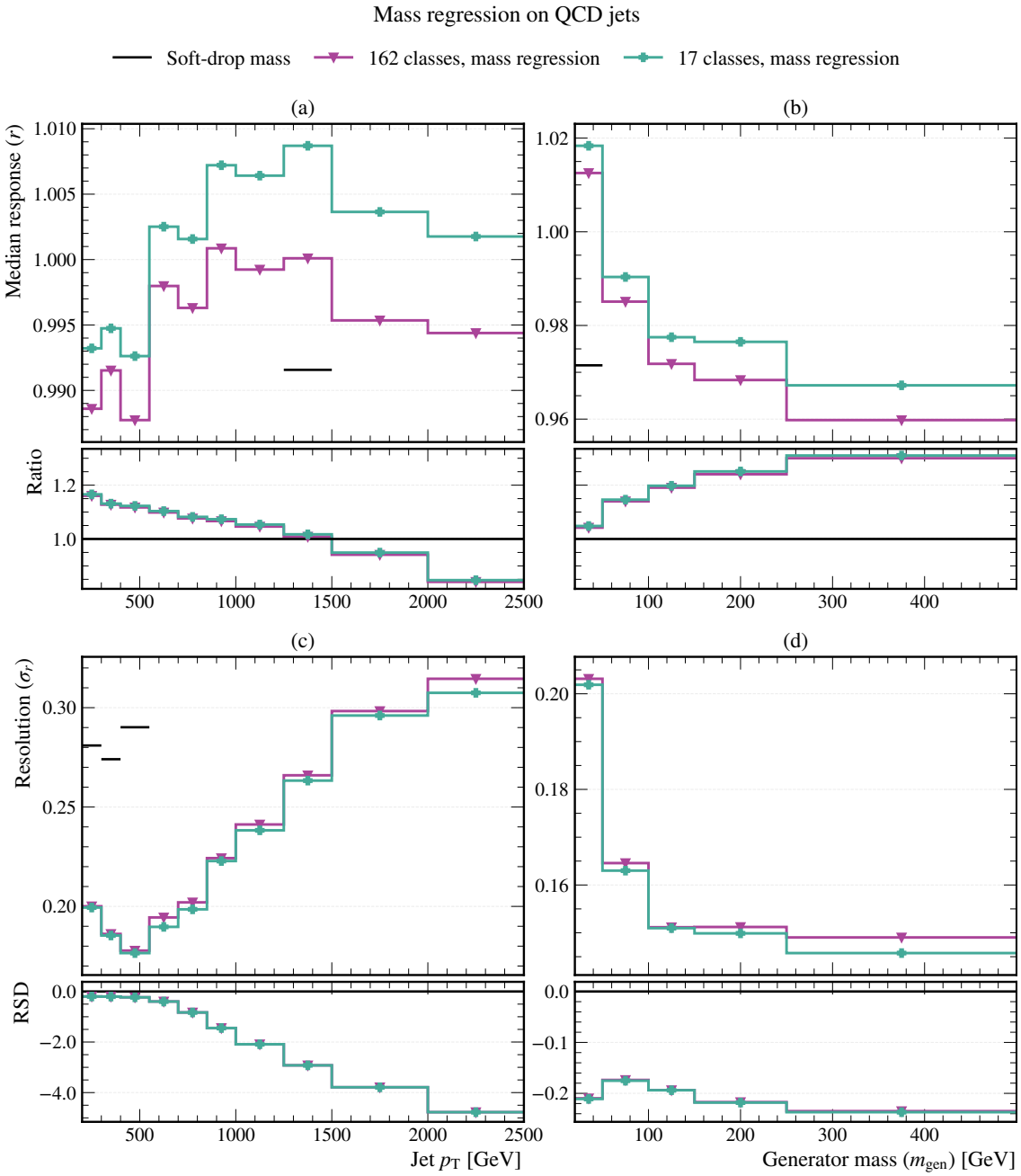


Figure 37: Median mass response r (a, b) and resolution σ_r (c, d), as in the main-text figure, of QCD JetClass-II test jets against transverse momentum (a, c; all jets) and generator mass (b, d; 20–500 GeV): mass regression of the models pretrained on 162 and 17 classes (two runs). Lower panels: ratio to the soft-drop mass (black where in range) and signed root square difference (RSD), negative where better. Unlabelled axes share the left ones.

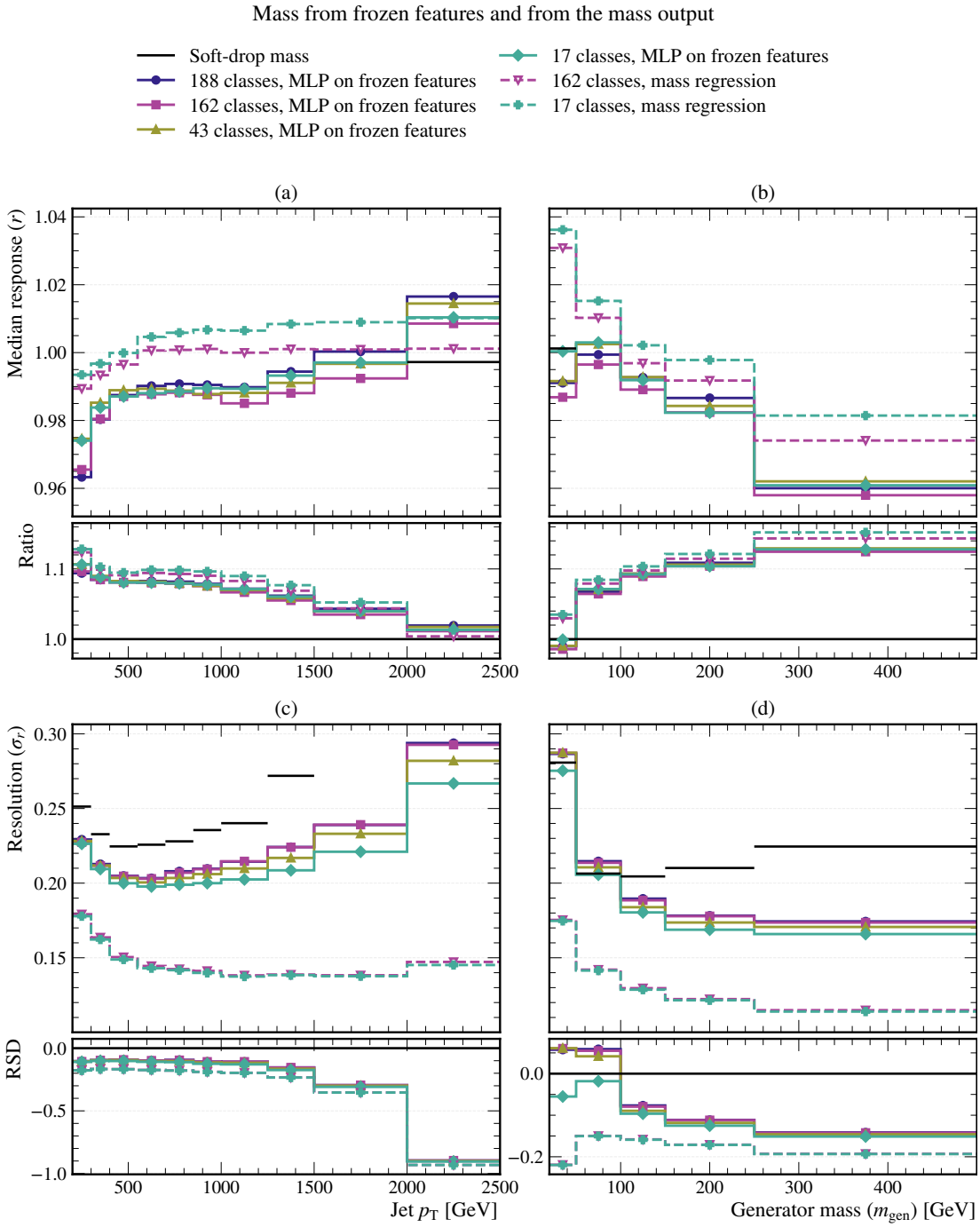


Figure 38: Median mass response r (a, b) and resolution σ_r (c, d) of JetClass-II test jets against transverse momentum (a, c; all jets) and generator mass (b, d; 20–500 GeV): an MLP regressing the mass from the frozen features of the models pretrained on 188, 162, 43 and 17 classes (solid, three runs), and the mass regression of the 162- and 17-class models (dashed, two runs). Sub-panels as in the main-text mass figure; overlapping markers agree within their size.

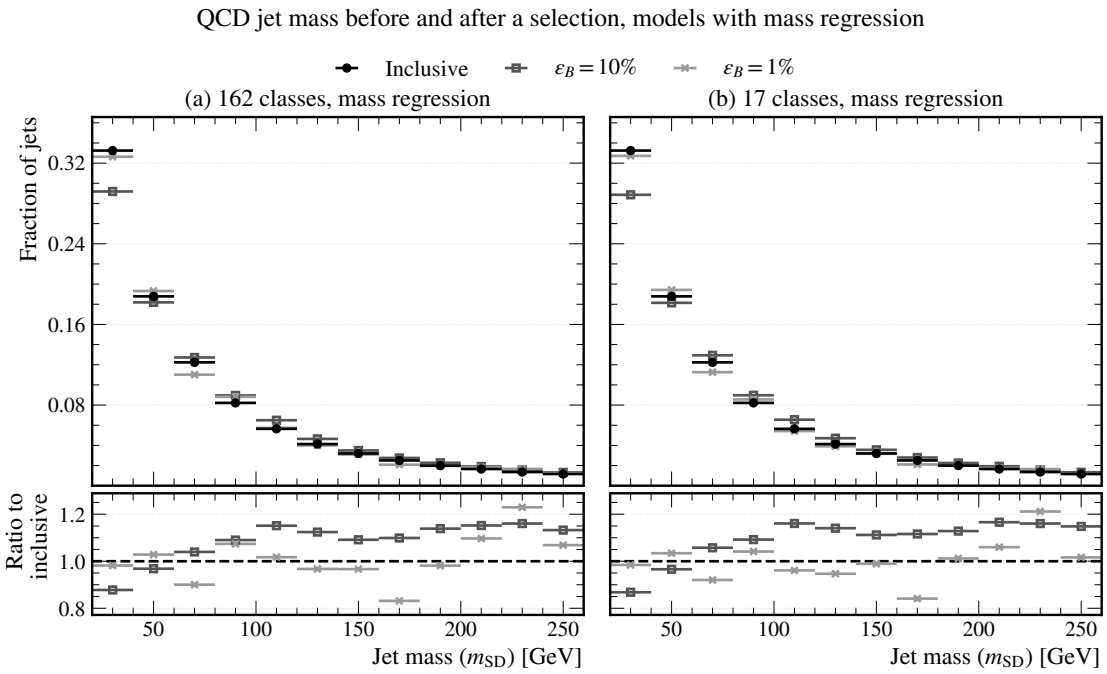


Figure 39: The mass sculpting figure for the models pretrained on 162 and 17 classes with an auxiliary regression of the jet mass: soft-drop mass distribution of QCD test jets of JetClass-II before any selection and after selections on $1 - g_{\mathrm{QCD}}$ at QCD efficiencies ε_B of 10% and 1% (mean of two runs), with the ratio to the inclusive distribution below.

Table 26: Mass sculpting: Jensen–Shannon distance (base 2) $\times 10^2$ between the m_{SD} distributions (24 bins of 20 GeV from 20 to 500 GeV) of all QCD test jets and of those passing a cut on $1 - g_{\mathrm{QCD}}$ at the QCD efficiency ε_B given; lower is less sculpting. The last row is the distance expected for a random selection of as many of the QCD test jets (drawn without replacement). Mean ± standard deviation over the three pretraining runs; the models with the mass regression have two runs, and both values are given.

pretraining	$\varepsilon_B = 10\%$	$\varepsilon_B = 1\%$
188 classes	5.24 ± 0.34	4.39 ± 0.24
162 classes	5.32 ± 0.09	4.29 ± 0.29
43 classes	5.60 ± 0.27	5.1 ± 0.7
17 classes	5.43 ± 0.10	5.61 ± 0.17
162 classes, mass regression	4.22, 4.64	4.24, 4.42
17 classes, mass regression	5.22, 4.23	4.05, 4.52
random selection of as many QCD jets	0.944	3.13